4 Random Variable and Noise

Random Experiment

The basic concept of random variable is the probability theory. We use random variables as a tool to understand signals in communication system. Probability theory can be (explicitly or implicitly) modelled by an experiment whose outcome is based on 'chance'. Also, if experiment is repeated multiple times the outcome can differ each time because of an underlying random or chance phenomenon. Such an experiment is called random experiment.

There are three important features of a random experiment:

1. The experiment is repeatable only under identical conditions of the previous one.
2. The outcome is unpredictable on any trial of the experiment.
3. For a fairly large number of trials of the experiment, the outcome tends to exhibit 'statistical regularity' or it gives a definite average pattern of outcomes that can be observed.

Definitions

Random experiment: Experiment is any process of observation. Here in random experiment, the outcome is not certain or outcome cannot be predicated with 100 % certainty, e.g. tossing of fair coin, rolling of fair dice etc.

Random Variable

A sample space is a set of outcomes of any random experiment. So, we can pretend the outcome of the experiment is a variable that can take any value from the set of sample points and the values are determined by the experiment. If the outcome of an experiment is λ, the random variable is denoted as $X(\lambda)$ or simply X.

Random variable: A function with a sample space as its domain and its range is some set of real numbers.

Sample Point

For example:

An experiment of tossing of 3 coins simultaneously.

Outcome: {TTT, TTH, THT, HTT, HHT, HTH, THH, HHH} = 8 (sample space)

Note:

Every sample point represents an event, but an event does not represent a sample point (as it can be more than one).

Whenever a random experiment is performed, the occurrence of these sample points is always random which when it is mapped by random variable, the outcome of the random variable is always deterministic.

Deterministic

A random variable X(λ) has a domain which is random and a range which is deterministic. Number of sample points can be infinite.

Definitions

Deterministic: When the experiment is said to have no uncertainty in its outcome, tossing of fair coin is deterministic experiment.

Example 1:

An experiment is performed in which dice is rolled until 3 is observed for the first time. Let no. of times dice rolling is assumed as random variable. Perform the mapping.

Solution:

Sample space $= \{3, \overline{3}3, \overline{3}\overline{3}, \overline{3333}, \ldots\}$

Here, sample point are infinite ($\overline{3}$ means 'not' 3)

$X(\lambda) = \{1, 2, 3, 4, \ldots\}$

Where X is no. of times dice rolled.

Here, $X(L_1) = 1, X(L_2) = 2; X(L_3) = 3$ etc.

If random experiment is given, where two coins are tossed simultaneously.

Sample space $= \{TT, TH, HT, HH\}$

Here X is no. of heads (X is random variable)

$X(\lambda) = \{0,1,1,2\}$

In form of set $\{X \le 1.5\}$ is actually $\{X(\lambda) \le 1.5\}$. So, the answer is 3.

That is the no. of sample points: {TT, TH, HT}

And $\{X = -3\}$ = Null set {f}

$\{X = 1\} = \{TH, HT\}$

$\{X < \infty\} =$ All.

$\{X = 1.5\} =$ Null set {f }

However, the probability of the null set is zero.

Now, we can extend our discussion to probability.

$P\{X = 1.5\} = 0$

$P\{X < \infty\} = 1$

$P\{X \le 1.5\} = \frac{3}{4}$

For $\{X < -\infty\} =$ Null

$P\{X < -\infty\} = 0$

And $\{|X| < 1\}$ or from -1 to 1 range of X. So, $\{-1 < X < 1\}$ where both 1 and -1 are not included. So, the value is zero, i.e., {TT}.

Therefore, $P\{|X| < 1\} = \frac{1}{4}$

And $\{|X| \le 1\} = \{-1 \le X \le 1\} = \{TT, TH, HT\}$

$P\{|X| \le 1\} = \frac{3}{4} = 0.75$

- Mathematical tool to study random variable:
  - Cumulative Distributive Function (CDF).
  - Probability Density Function (PDF).

CDF of a random variable:

$\{X \le 2\} = \{TT, TH, HT, HH\}$

$= \{X = 0\} + \{X = 1\} + \{X = 2\}$

$\{X \le 1\} = \{TT, TH, HT\} = \{X = O\} + \{X = 1\}$

$\{X \le a\}$ can also be defined for any value of a. Also, $P\{X \le a\} = F_x\{a\}$ is the CDF or we can define CDF as $F_x(x)$ for a value of x; here, x is the values taken by random variable $(R_v)\, X$.

Therefore, $F_x(x) = P\{X \le x\} = 1 - P\{X > x\}$. If x value is fixed at 'a' then

$F_x(a) = P\{X \le a\} = 1 - P\{x > a\}$

Definitions

Continuous Random Variable (CRV): If a random variable X can take any value in the whole interval of observation, then X is continuous.

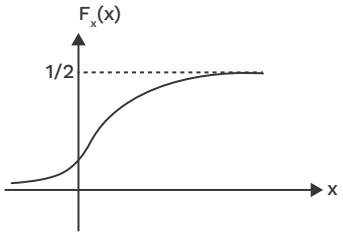
Example: Amplitude of noise voltage at any instant of time.

Discrete Random Variable (DRV)

When random variable takes only a discrete set of values that is random variable X takes only a finite number of values in any finite interval of observation.

Properties of CDF

1. $0 \le F_x(x) \le 1$
2. $F_X(\infty) = P\{X \le \infty\} = 1$
3. $F_x(-\infty) = P\{X \le -\infty\} = 0$
4. $F_x(\infty) + F_x\{-\infty\} = 1$
5. CDF is monotonically non-decreasing function of x.



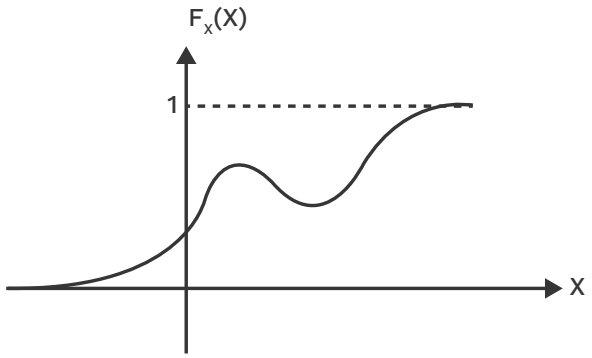
Definitions

Cumulative distribution function: $F_x(x)$ is called a CDF or simply the distributive function of random variable X and we should mean that it is a function of x and not of

$R \cdot V$. X. But X is written in subscript as it is the assigned random variable. For a value of x, the distribution function or CDF $F_x(x)$ expresses a probability.

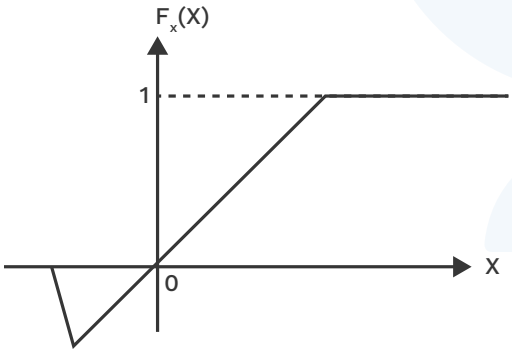
But $F_X(\infty) = 1$, so this graph of CDF is not valid.

$F_x(x)$

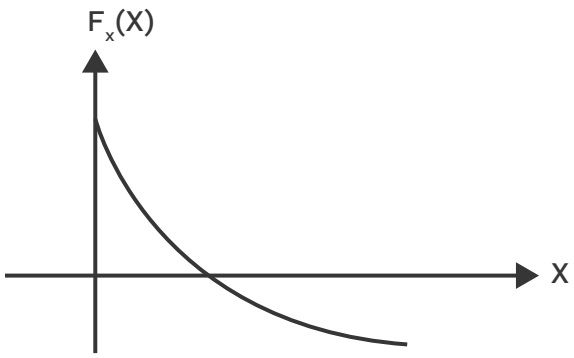


This CDF is not valid as this graph is not non-decreasing.

$F_x(x)$



This CDF is decreasing as well as negative which is not the property of CDF. Here, CDF is monotonically or strictly decreasing.



6. CDF is always continuous from right.

$$F_x\left(a^+\right) = F_x(a)$$

$$F_x\left(a^+\right) = F_x(a) = F_x\left(a^-\right)$$

Where 'a' is the discontinuity point.

So, for $F_x(x)$ is continuous at x = a, then

$$F_x\left(a^+\right) = F_x(a) = F_x\left(a^-\right)$$

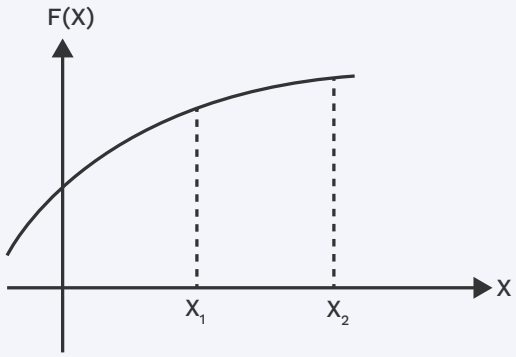
For all continuous graphs and for $F_x(X)$ is discontinuous at x = a, then

$$F_x\left(a^+\right) = F_x(a) \neq F_x\left(a^-\right)$$

Here on left side continuity would break, but on right side it should always be continuous.

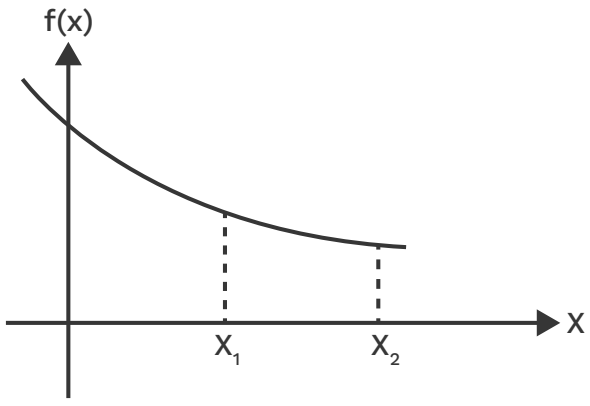
Grey Matter Alert!

1. Increasing function



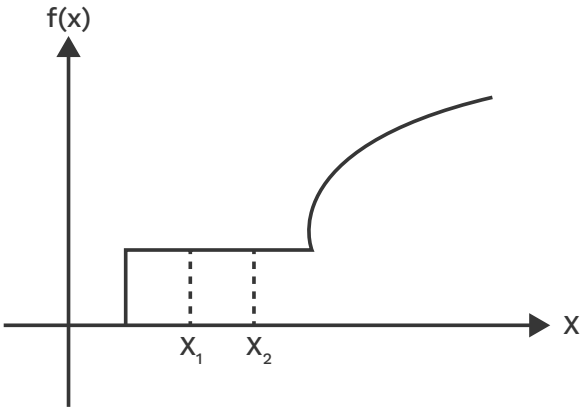
$$x_2 > x_1; f\left(x_2\right) > f\left(x_1\right)$$

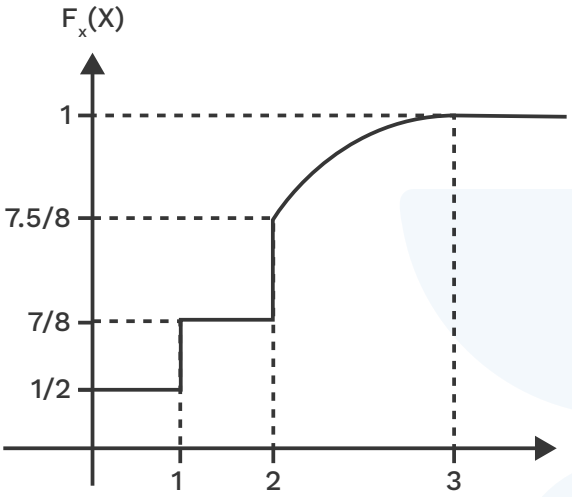
2. Decreasing function



$$x_2 < x_1; f\left(x_2\right) < f\left(x_1\right)$$

3. Non-decreasing function





For this CDF, which is valid, we have

a) Value of $F_x(x)$ at $x = 1$

$$F_x(1) = F_x\left(1^+\right) = \frac{7}{8}$$

b) Value of $F_x(x)$ at $x = 2$

$$F_x(2) = F_x\left(2^+\right) = \frac{7.5}{8}$$

c) Value of $F_x(x)$ at $x = 1.5$

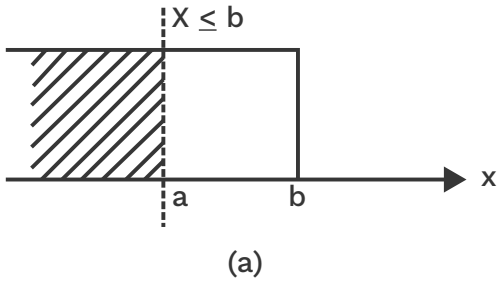
$$F_x(1.5) = F_x\left(1.5^+\right) = \frac{7}{8}$$

d) Value of $F_x(x)$ at $x = 3$

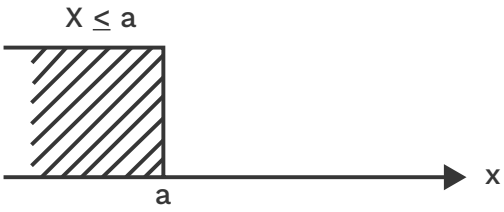
$$F_x(3) = F_x\left(3^+\right) = 1$$

Probability in a range

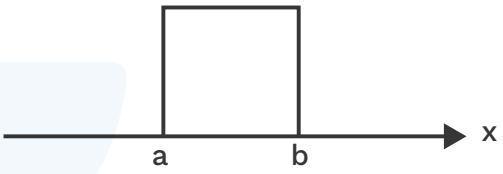
For $P\{a < X \le b\}$



(a)



(b)



(c)

So,

$$P\{a < X \le b\} = P\{X \le b\} - P\{X \le a\}$$

$$= F_x(b) - F_x(a)$$

$$\left(\because F_x(x) = P\{X \le x\}\right)$$

$$= F_x\left(b^+\right) - F_x\left(a^+\right)$$

Similarly, for a few ranges of probability given below, we can define CDF as

$$P\{a < X \le b\} = F_x\left(b^+\right) - F_x\left(a^+\right)$$

$$P\{a < X < b\} = F_x\left(b^-\right) - F_x\left(a^+\right)$$

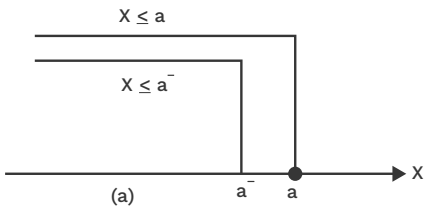
$$P\{a \le X < b\} = F_x\left(b^-\right) - F_x\left(a^-\right)$$

$$P\{a \le X \le b\} = F_x\left(b^+\right) - F_x\left(a^-\right)$$

Probability at a Point

With help of CDF, we can find

P{X = a}



Previous Years' Question

1. Let the random variable X represents the number of times a fair coin needs to be tossed till two consecutive heads appear for the first time. The expectation of X is

So, $(X \le a) - \left(X \le a^{-}\right) = \{X = a\}$

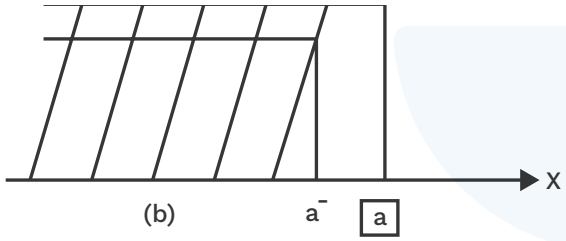


Fig. 4.1 (A) Lines Showing $X \le a$ and $X \le a^{-}$

(B) $(x \le a) - \left(x \le a^{-}\right)$ Gives Only a Point 'a'

$$P\{X = a\} = P\{X \le a\} - P\left\{X \le a^{-}\right\}$$
$$= F_x(a) - F_x\left(a^{-}\right)$$
$$= F_x\left(a^{+}\right) - F_x\left(a^{-}\right)$$

Probability Density Function (PDF):

For a random variable X, the PDF can be written as $f_x(x)$.

$f_x(x) = \frac{dF_x(x)}{dx}$ (Derivative of CDF with respect to x)

$\Rightarrow$ PDF of " X " $= \frac{d}{dx}$ [CDF of "X"]

PDF properties:

1. $f_x(x) = \frac{dF_x(x)}{dx}$ so, for CDF $F_x(x) = \int_{-\infty}^{x} f_x(x)dx$, here the integral is known as running integral.
2. As CDF is a non-decreasing function, its derivative, PDF will be either zero or positive. So, PDF is non-negative.
   $f_x(x) \ge 0$
3. PDF can be $0 \le f_x(x) < \infty$
4. $F_x(x) = P\{X \le x\} = \int_{-\infty}^{x} f_x(x)dx$ ...(i)

   $F_x(\infty) = P\{X \le \infty\} = \int_{-\infty}^{\infty} f_x(x)dx = 1$

   Or $\int_R f_x(x)dx = 1$, $R =$ Region in which PDF is defined.

The PDF functions can never be odd function as area under the curve of odd function is zero.

And $\int_{-\infty}^{\infty} f_x(x)dx = 2\int_{0}^{\infty} f_x(x)dx = 1$

Probability in a range:

From equation (i)

$$P(X \le x) = \int_{-\infty}^{x} f_x(x)dx$$

$$P\{-\infty \le X \le x\} = \int^{x} f_x(x)dx$$

= Area of PDF in the range random variable lies.

Now, for $P\{a < X \le b\} = \int_{a^+}^{b^+} f_x(x)dx$

And also,

$$P\{a < X < b\} = \int_{a^+}^{b^-} f_x(x)dx$$

$$P\{a \le X < b\} = \int_{a^-}^{b^-} f_x(x)dx$$

$$P\{a \le X \le b\} = \int_{a^-}^{b^+} f_x(x)dx$$

Grey Matter Alert!

The name probability density f arises from the fact that, we can define the probability of an event:

$$P\left(x_1 < x \le x_2\right) = P\left(X \le x_2\right) - P(x)$$
$$= F_x\left(x_2\right) - F_x\left(x_1\right)$$
$$= \int_{x_1}^{x_2} f_x(x)dx$$

Probability at a point:

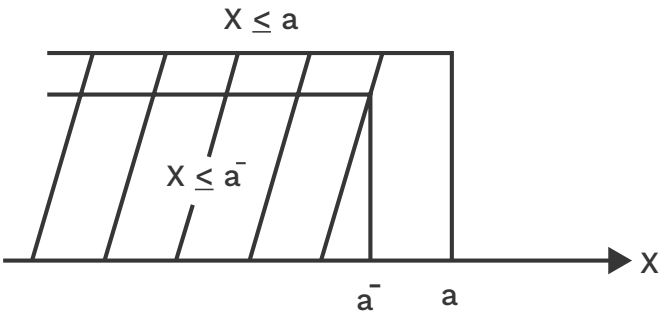


Fig. 4.2 Probability at a Point 'a'

$$P\{X = a\} = F_x\left(a^+\right) - F_x\left(a^-\right)$$

$$= \int_{-\infty}^{a^+} f_x(x)dx - \int_{-\infty}^{a^-} f_x(x)dx$$

$$\left(as F_x(a) = \int_{-\infty}^{a} f_x(x)dx\right)$$

$$= \int_{-\infty}^{a^+} f_x(x)dx + \int_{a^-}^{-\infty} f_x(x)dx$$

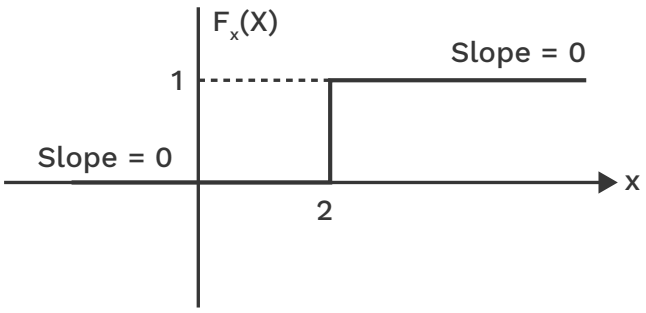
$$= \int_{a^-}^{a^+} f_x(x)dx$$

Chapter Summary

	CDF	PDF
$P\{a < X \le b\} =$	$F_x\left(b^+\right) - F_x\left(a^+\right)$	$\int_{a^+}^{b^+} f_x(x)dx$
$P\{X = a\} =$	$F_x\left(a^+\right) - F_x\left(a^-\right)$	$\int_{a^-}^{a^+} f_x(x)dx$

Table 6.1 Summary Table of CDF and PDF for Probability in a Range and at a Point

A CDF is given



So, now differentiate to get PDF

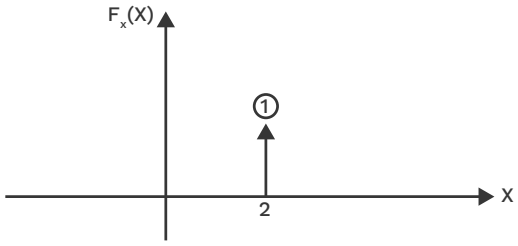


Fig. 4.3 (A) CDF Graph (B) PDF Graph: PDF is an Impulse with Area of Same Height (Amplitude) as of CDF

Now calculate,

(From CDF) $P\{X = 2\} = F_x\left(2^+\right) - F_x\left(2^-\right) = 1 - 0 = 1$

(From PDF) $P\{X = 2\} = \int_{2^-}^{2^+} f_x(x)dx = 1$

So, probability can be calculated from both CDF and PDF.

From above discussion we can see that if we are asked to find probability of a point, suppose, point 'a' and if CDF is continuous at point 'a'.

That is, $F_x\left(a^+\right) = F_x\left(a^-\right) = F_x(a)$

Then, $P\{X = a\} = F_x\left(a^+\right) - F_x\left(a^-\right)$

P{X = a} = 0

Therefore,

$$P\{X = a\} = F_x\left(a^+\right) - F_x\left(a^-\right)$$

1. Equal to 0 if CDF is continuous at point a (x = a)
2. Not equal to 0 if CDF is discontinuous at point a (x = a)

   Similarly,

   $$P\{X = a\} = \int_{a^-}^{a^+} f_x(x)dx$$

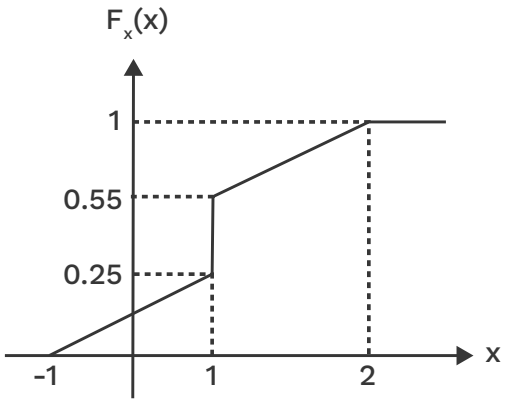
   a) = 0 if PDF does not contain impulse at x = a

   b) ≠ 0 if PDF contains impulse at x = a

{ impulse function gives non-zero area for width equal to zero, a for range a^- to a^+, the width is zero }

Example 2:

The distribution function $F_x(x)$ of a random variable X is shown. The probability that X = 1 is



Solution:

$P\{X = 1\} = F_x(1^+) - F_X(1^-)$

= 0.55-0.25

= 0.30

Also, $P\{X = -1\} = 0$

P{X = 3} = 0

Discrete Random Variable

Random variable here maps the points on sample space or sample points on the real line (discrete points).

Parameters of DRV:

1. Probability Mass Function (PMF)
2. PDF of DRV
3. CDF of DRV

Grey Matter Alert!

Random variable can be real random variable, which maps sample space on real line and complex random variable which maps sample space on the complex plane

1. PMF of DRV 'X': x_i is values of different X.

$$p_x(x_i) = P(X = x_i)$$

2. PDF of DRV 'X':

$$f_x(x) = \frac{dF_x(x)}{dx} = \sum_i \phi_x(x_i)\delta(x - x_i)$$

Which is defined in terms of PMF.

$$\Rightarrow f_x(x) = \frac{dF_x(x)}{dx} = \sum_i p(x = x_i)\delta(x - x_i)$$

Where $\delta(x) \rightarrow$ Impulse function.

PDF of discrete RV is always in the form of 'impulses'.

3. CDF of DRV 'X':

$$F_x(x) = \int_{-\infty}^{x} f_x(x)dx = \sum_i^P x\{x = x_i\}$$

CDF of DRV is in the form of 'staircase'.

Example 3:

Two coins are tossed simultaneously. Number of heads of random variable ' X ':

Solution:

Random experiments are 2 coins tossed

S = {TT, TH, HT, HH}

X = {0, 1, 1, 2} which is DRV.

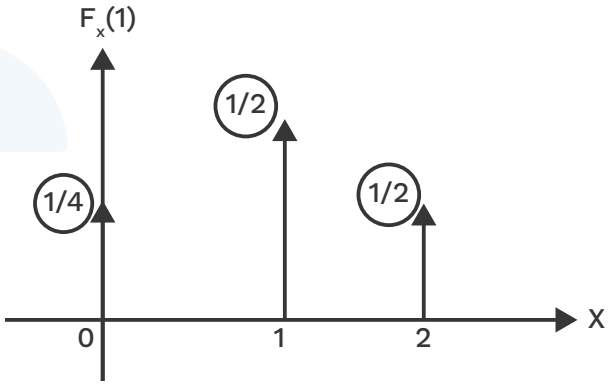
PDF: $f_x(x) = \sum_i P(x = x_i)\}\delta(x - x_i)$

$P\{X = 0\}\delta(x - 0) + P\{x = 1\}(x - 1) + P\{x = 2\}\delta(x - 2)$

$\Rightarrow$ Here $P\{X = 0\} = \frac{1}{4}; P\{x = 1\} = \frac{2}{4} = \frac{1}{2}$

And $P\{x = 2\} = \frac{1}{4}$

$$f_x(x) = \frac{1}{4}\delta(x) + \frac{1}{2}\delta(x - 1) + \frac{1}{4}\delta(x - 2)$$



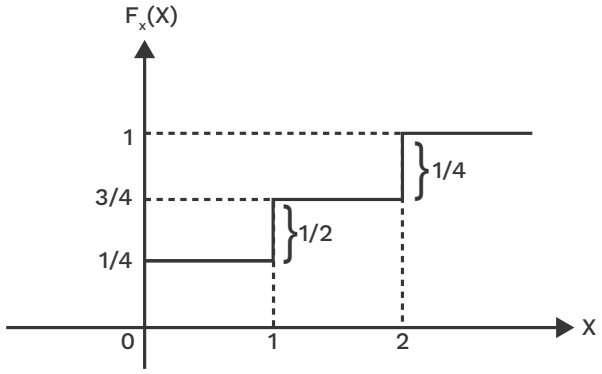
From here,

$$P\{x = 2\} = \int_{2^-}^{2^+} f_x(x)dx = \frac{1}{4}$$

$$P\{1 < x \le 2\} = \int_{1^+}^{2^+} f_x(x)dx = \frac{1}{4}$$

$$P\{1 \le x \le 2\} = \int_{1^-}^{2^+} f_x(x)dx = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

CDF: We can integrate PDF $f_x(x)$ to find CDF.



Here, $P\{x=2\} = F_x\left(2^+\right) - F_x\left(2^-\right)$

$= 1 - \frac{3}{4} = \frac{1}{4}$

(This is the jump in staircase)

So, $P\{x=1\} = \frac{1}{2}$ and $P\{x=0\} = \frac{1}{4}$

And,

$P\{x=0.5\} = 0 = F_x\left(0.5^+\right) - F_x\left(0.5^-\right)$

$= \frac{1}{4} - \frac{1}{4} = 0$

Example 4:

The probability cumulative distribution function must be monotone and

(A) Increasing

(B) Decreasing

(C) Non-increasing

(D) Non-decreasing

Solution:

From CDF properties,

The correct answer is (D).

That is non-decreasing.

Example 5:

Consider a discrete RV assuming finitely many values. The CDF of such a random variable is:

(A) Non-decreasing.

(B) Non-decreasing function with finitely many discontinuities and assuming values less than 1.

(C) Non-decreasing function without discontinuity.

(D) Non-decreasing function assuming value larger than 1.

Solution:

The correct answer is (B).

As CDF is non-decreasing and has finitely many values are present with values less than 1. (Or equal to 1), CDF never be more than 1;

(A) Can also be right, but we need to choose most appropriate.

(C) and (D) are wrong options.

Continuous Random Variable (CRV)

1. PDF of CRV
2. CDF of CRV

Here, PMF is not used.

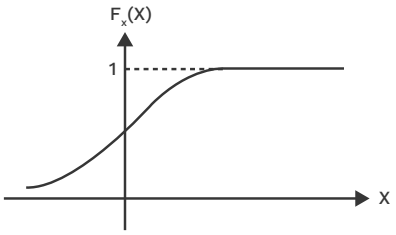
PDF

$$f_x(x) = \frac{d}{dx} F_x(x)$$

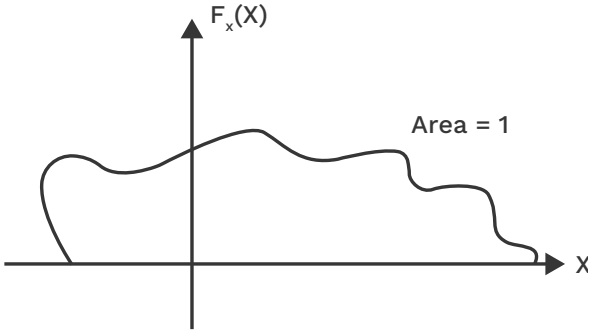
Should be continuous and should not contain impulses.

CDF

$F_x(x)$ here should be continuous and no step change anywhere.



(a)



(b)

Figure: (a) CDF of CRV (b) PDF of CRV

In above figure, there is no step change observed in the graph (A) and in (B) three is no impulse. [For CRV]

Now, we have a CDF group as follows:

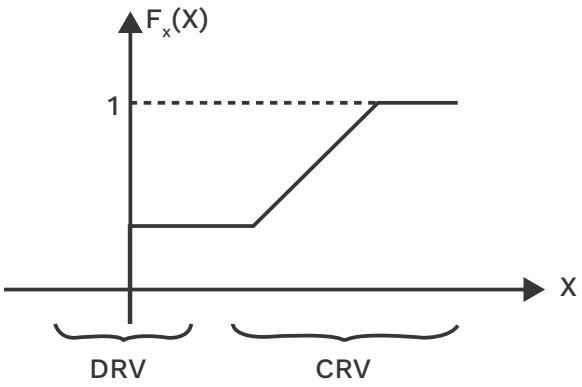
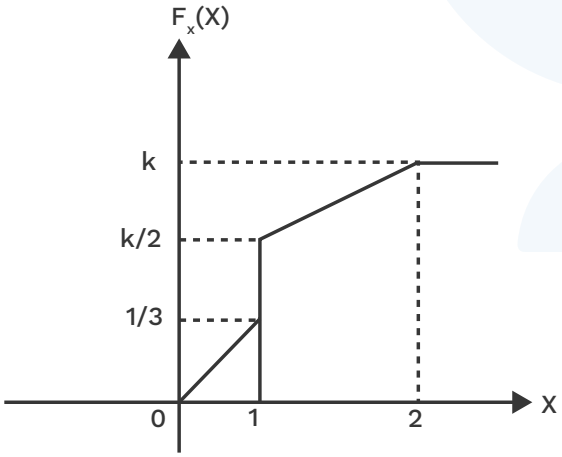


Figure: CDF of Mixed random variable

These types of random variables are called 'Mixed RV'.

Example 6:

CDF of a RV is given



Find

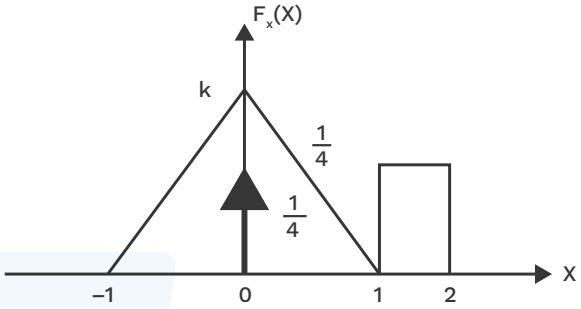
1. Value of k ?
2. $P\{x = 1\} = ?$
3. $P\{x = 2\} = ?$
4. $P\{|x| < 1\} = ?$
5. $P\{|x| \leq 1\} = ?$

Solution:

1. Value of k = 1
2. $P\{x = 1\} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$ (Jump-step change)
3. $P\{x = 2\} = 0$ (No jump)
4. $P\{|x| < 1\} = P\{-1 < x < 1\}$
   $= F_x(1^-) - F_x(1^+)$
   $= \frac{1}{3} - 0 = \frac{1}{3}$
5. $P\{|x| \leq 1\} == F_x(1^+) - F_x(-1^-)$
   $= \frac{1}{2} - 0 = \frac{1}{2}$

Example 7:

PDF of a RV is given by $f_x(x) =$



Find

1. k = ?
2. $P\{X = 0\} = ?$
3. $P\{X = 1.5\} = ?$

Rack Your Brain

For this example, also find

4. $P\{0 < X < 1\} = ?$
   $P\{1 < x < 2\}$ and $P\{1 \leq x \leq 2\}$.
5. $P\{0 \leq X \leq 1\} = ?$

Solution:

$\frac{1}{2}$ and $\frac{2}{3}$ respectively

This is a PDF which is of a mixed RV. To calculate K, we find the area

$$\int_{-\infty}^{\infty} f_x(x) = 1$$

$$\Rightarrow \frac{1}{4} + \frac{1}{4} + \frac{1}{2} \times k \times 2 = 1$$

$$\Rightarrow \frac{1}{4}+\frac{1}{4}+k=1$$

$$\Rightarrow k=1-\frac{1}{2}=\frac{1}{2}$$

$P\{x=0\}=\int_{0^-}^{0+} f_x(x)dx=\frac{1}{4}$ (as impulse present with area = $\frac{1}{4}$)

$P\{x=1.5\}=\int_{1.5^-}^{1.5^+} f_x(x)dx=0$ (as no impulse present)

$$P\{0<x<1\}=\int_{0^+}^{r} f_x(x)dx=\frac{1}{2}\times 1\times\frac{1}{2}=\frac{1}{4}$$

$$P\{0\le x\le 1\}=\int_{0^+}^{r} f_x(x)dx=\frac{1}{4}+\frac{1}{4}=\frac{1}{2}$$

Expectation operator:

- It acts only on random variable or a function of random variable.
- It is written as E[RV] or E[function of RV].
- So, for a constant C, E[C] = C [Can't process deterministic process]. $E[X]\neq X$ (For a random variable X)
  Also, $E[cx]=cE[X]$
  Similarly, $E[ax+b]=aE[x]+b$
- Suppose X : RV

$$E[X]=\begin{cases}\int_{-\infty}^{\infty} x\overbrace{f_x}^{\text{PDF of } x}(x)\,dx & X: CRV\\ \sum_i x_i\overbrace{\phi_x(x_i)}^{\text{PMF of } x}=\sum_i x_i P\{X=x_i\} & X: DRV\end{cases}$$

$$E\left[X^2\right]=\begin{cases}\int_{-\infty}^{\infty} x^2 f_x(x)dx & X: CRV\\ \sum_i \overbrace{x_i^2\phi_x(x_i)}^{x_i^2\phi_x}=\sum_i x_i^2 P\{X=x_i\} & X: DRV\end{cases}$$

- Now if X is CRV and we need to find
  $E[\sin x]=\int_{-\infty}^{\infty}\sin x f_x(x)dx$

Statistical Averages of Random Variable

1. Mean of

$$RV\ "X": E[X]=\begin{cases}\int_{-\infty}^{\infty} x f_x(x)dx & X: \text{CRV}\\ \sum_i x_i P\{X=x_i\} & X: \text{DRV}\end{cases}$$

It is also represented as 'μ_x'.

It can also be called as DC value of RV
Average value of RV
Expected value of RV

2. Mean square value of RV 'X'

$$E\left[X^2\right]=\begin{cases}\int_{-\infty}^{\infty} x^2 f_x(x)dx & X: CRV\\ \sum_i x_i^2 P\{X=x_i\} & X: DRV\end{cases}$$

3. Variance of RV 'X':

$\sigma_x^2=E\left[X^2\right]-(E[X])^2$

= Mean square value –

$\left(\begin{matrix}Mean\ value\\ of\ RV\end{matrix}\right)^2=E\left[x^2\right]=\mu_x^2$

Also, $\sigma_x^2\ge 0: E\left[X^2\right]\ge (E[X])^2$

Variance can also be defined as

$E\left[(X-\bar{X})^2\right]=\sigma_x^2=E\left[X^2\right]-\mu_x^2$

Example 8:

A discrete RV takes values from 1 to 5 with probability shown in the table. A student calculates the mean X as 3.5 and his teacher calculates the variance of X as 1.5. Who is correct?

K	1	2	3	4	5
P {X = K}	0.1	0.2	0.4	0.2	0.1

Solution:

The discrete RV of value 5 has a probability of =

1- {0.1+0.2+0.4+0.2} = 1-0.9 = 0.1

K	1	2	3	4	5
P {X = K}	0.1	0.2	0.4	0.2	0.1

$x_i \to 1$ to 5 here X is the RV

$$\Rightarrow E[X]=\sum_i X_i P\{X=x_i\}=1P\{X=1\}+2P\{X=2\}$$

$$+3P\{X=3\}+4P\{X=4\}+5P\{X=5\}$$

$$=1\times 0.1+2\times 0.2+3\times 0.4+4\times 0.2+5\times 0.1=3$$

So, the student was wrong.

Now, variance:

$$\sigma_x^2=E\left[X^2\right]-(E(x))^2=E\left[X^2\right]-(3)^2=E\left[X^2\right]-9$$

$$E\left[X^2\right] = \sum_i X_1^2 P\left\{X = X_i\right\}$$

$$= (1)^2 P\{X = 1\} + (2)^2 P\{X = 2\} + (3)^2 P\{X = 3\} +$$

$$(4)^2 P\{X = 4\} + (5)^2 P\{X = 5\}$$

$$= 10.2$$

$$\Rightarrow \sigma_x^2 = 10.2 - 9 = 1.2$$

So, the teacher was also wrong.

Example 9:

Let X be a random variable which is uniformly chosen from the set of positive odd number less than 100. Find the expectation of X.

Solution:

Random variable:

$$X = \{1, 3, 5, 7, 9 \ldots 99\}$$

It is a discrete RV (DRV)

$$x_i \rightarrow 1 \text{ to } 99(\text{ odd })$$

DRV:

$$E[X] = \sum_i X_i P\left\{X = X_i\right\}$$

$$= 1P\{X = 1\} + 3P\{X = 3\} + 5P\{X = 5\} + \ldots +$$

$$99P\{X = 99\}$$

As uniformly chosen, so all have the same probability.

AP = 1,3,5,7, ...99 L = 99; a = 1 and d = 2 L = a + (n - 1) d 99 = 1 + (n-1) 2 n = 50 and with probability

$$\Rightarrow E[X] = \frac{1 \times 1}{50} + 3 \times \frac{1}{50} + 5 \times \frac{1}{50} + \ldots 99 \times \frac{1}{50}$$

$$= \frac{1}{50}[1 + 3 + 5 + 7 \ldots + 99] = \frac{1}{50}\left[\frac{50}{2}(1 + 99)\right]$$

$$= \frac{1}{50} \times \frac{50}{2} \times 100 = 50$$

Example 10:

A dice is rolled until 3 is observed for the first time. If no. of times dice rolled is considered as random variable. Calculate the average no. of times dice is rolled.

Solution:

Here a random experiment is given

$$S = \{3, \bar{3}3, \bar{3}\bar{3}3, \bar{3}\bar{3}\bar{3}3, \ldots\}$$

$X \rightarrow$ No. of times dice rolled

$X = \{1, 2, 3, 4 \ldots\} \rightarrow$ R.V.

Average no. of times dice rolled is: E[X]

$$E[X] = \sum_1 x_i P\left\{X = x_i\right\}$$

Previous Years' Question

2. Two random variables X any Y are distributed according to

$$f_{x,y}(x, y) = \begin{cases} (x + y) & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & otherwise \end{cases}$$

The probability is _____.

Solution:

$$X_i - 1 \text{ to } \infty . \in I$$

$$E[X] = 1P\{X = 1\} + 2P\{X = 2\} + 3P\{X = 3\} + \ldots$$

$$x_i - 1 \text{ to } \infty, \in I$$

$$E[X] = 1P\{X = 1\} + 2P\{X = 2\} + 3P\{X = 3\} + \ldots$$

X_i	1	2	3
$P\{X = x_i\}$	1/6	5/6 × 1/6	5/6 × 5/6 × 1/6

$$P\{X = 2\} = P\{\bar{3} \cap 3\} = P\{\bar{3}\}P\{3\} = \frac{5}{6} \times \frac{1}{6}$$

Similarly,

$$P\{X = 3\} = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}$$

Now,

$$E[X] = 1 \times \frac{1}{6} + 2 \times \frac{5}{6} \times \frac{1}{6} + 3 \times \frac{5}{6} \times \frac{1}{6} + 4 \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \ldots$$

$$= \frac{1}{6}\left[1 + 2\left(\frac{5}{6}\right) + 3\left(\frac{5}{6}\right)^2 4\left(\frac{5}{6}\right)^3 + \ldots\right] \text{ if } x = \frac{5}{6}$$

$$= \frac{1}{6}\left(1 + 2x + 3x^2 + 4x^3 + \ldots \infty\right)$$

$$= \frac{1}{6}(1 - x)^{-2} = \frac{1}{6}\left(1 - \frac{5}{6}\right)^{-2}$$

$$= \frac{1}{6}\left[\frac{1}{6}\right]^{-2} = \frac{6^2}{6} = 6$$

Types of Distributions for Continuous Random Variable 'X'

These are the following types of distributions (PDF):

1. Uniform
2. Triangular
3. Gaussian
4. Rayleigh
5. Exponential
6. Laplacian

Uniform Distribution

X: It has a PDF which is uniformly distributed between (a, b).

X is uniform random variable (URV).

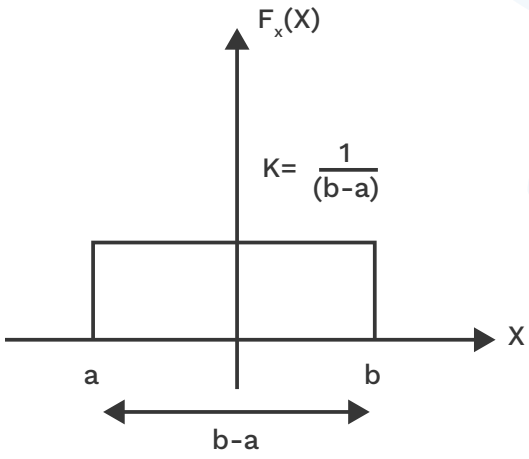
$X \sim \cup [a, b]$

Uniform Distribution

X: It has a PDF which is uniformly distributed between (a, b).

X is uniform random variable (URV)

$\Rightarrow X \sim \cup [a, b]$



$$f_x(x)\begin{cases}\frac{1}{b-a} & a \le x \le b \\ 0 & elsewhere\end{cases}$$

If a < c < b

1. $P\{X = C\} = 0$ (As no impulse in PDF at C.)
2. $E[X] = \int_{-\infty}^{\infty} x f_x(x)dx = \int_a^b x f_x(x)dx = \int_a^b \frac{x}{(b-a)}dx$

$$= \frac{a+b}{2}$$

Previous Years' Question

3. A random variable X takes -0.5 and 0.5 with probabilities $\frac{1}{4}$ and $\frac{3}{4}$, respectively. The noisy observation of X is $Y = X +$ Z, where Z has uniform probability density over the interval (-1, 1). If the MAP rule based detector outputs an

$$\ell = \begin{cases}0.5, & Y<\alpha \\ 0.5, & Y\ge\alpha\end{cases}$$

Then the value of α (accurate to two decimal places) is _____.

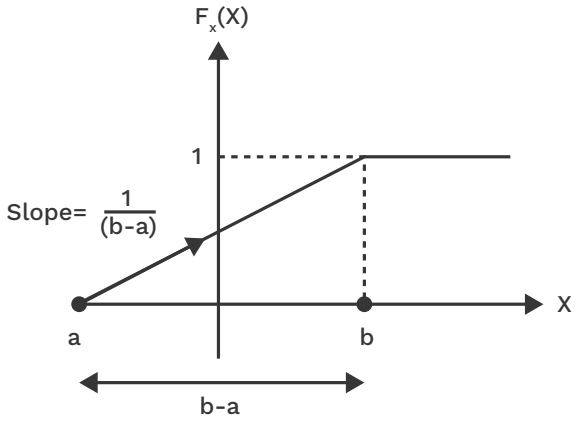
3. $E[X^2] = \int_{-\infty}^{\infty} x^2 f_x(x)dx = \int_a^b x^2 f_x(x)dx = \int_a^b \frac{x^2}{(b-a)}dx$

$$= \frac{a^2 + b^2 + ab}{3}$$

4. $\sigma_x^2 = E[X^2] - (E[X])^2 = \frac{(b-a)^2}{12}$
5. $\sigma_x = \frac{\pm(b-a)}{2\sqrt{3}}$ (Standard deviation)

CDF of uniform distribution:

a) PDF is integrated



Triangular distribution:

X: PDF which is triangular between (a, b)

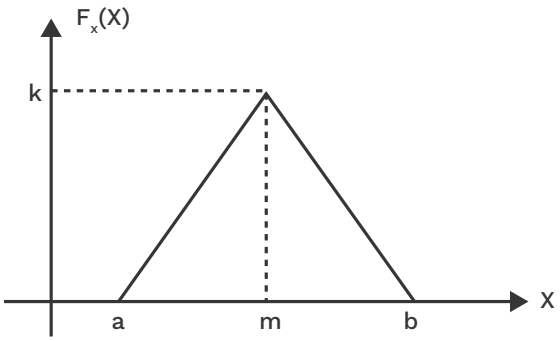
$X \sim \Delta(a, b)$

1. $P\{X = c\} = 0$ (if $a < c < b$)

2. $E[X] = \dfrac{a+m+b}{3}$

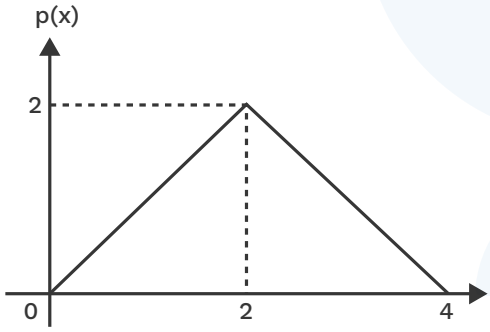
3. $\sigma_x^2 = \left\{\dfrac{a^2+m^2+b^2-am-bm-ab}{18}\right\}$

4. $E\left[X^2\right] \rightarrow$ Here mean square value is always calculated.

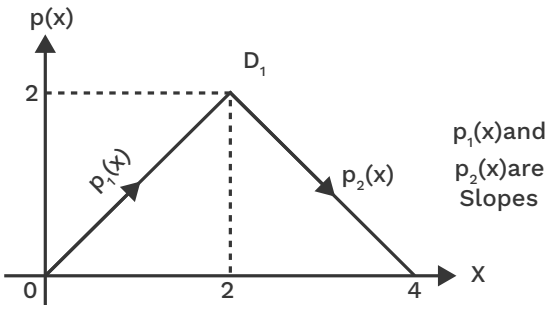


Example 11:

1. Calculate $\int_0^4 xp(x)dx$



Solution:



Area $= \dfrac{1}{2} \times 4 \times 2 = 4$

1. First method (or general method)

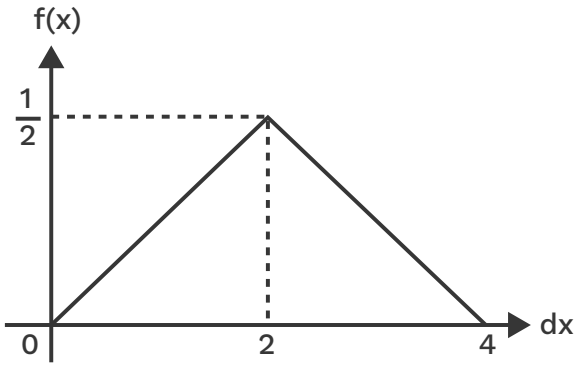
$$\int_0^4 xp(x)dx = \int_0^2 xp_1(x)dx + \int_2^4 xp_2(x)dx$$

Which will take longer to solve.

2. Simple method is by distribution

It can be PDF if its area is 1.

So, we can redraw the distribution as:



Here we divide p(x) by 4.

Now area $= \dfrac{1}{2} \times 4 \times \dfrac{1}{2} = 1$

So, it is a PDF $[p(x) = 4f(x)]$

$$\Rightarrow \int_0^4 xp(x)dx = \int_0^4 x \cdot 4f(x)dx$$

$$\int_0^4 x4.f(x)dx = \underbrace{4\int_0^4 x \cdot f(x)dx}_{PDF}$$

= 4 E[X]

$$= 4\left[\frac{0+2+4}{3}\right]$$

= 4 × 2 = 8

Gaussian Distribution

X: It has a PDF of Gaussian shape to X is called Gaussian Random Variable (GRV)

It can also be defined as:

$X \rightarrow$ GRV or PDF of X is Gaussian or X has Gaussian distribution.

X: GRV or normal RV (NRV)

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{-(x-\mu_2)^2}{2\sigma_x^2}} \quad -\infty < x < \infty$$

$X \sim \left[\mu_x, \sigma_x^2\right]$

1. $\mu_x = E[X] = \int_{-\infty}^{\infty} xf(x)dx$

2. $\sigma_x^2 = E\left[X^2\right] = \mu_x^2$

But we do not use these as integration of $f_x(x)$ is difficult. So, from equation of $f_x(x)$, by comparison, we can find both σ_x^2 and μ_x. The PDF of GRV is an inverted bell shape.

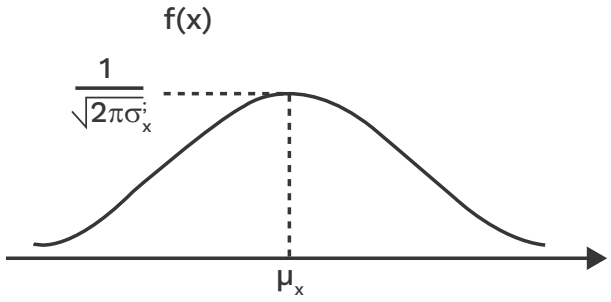
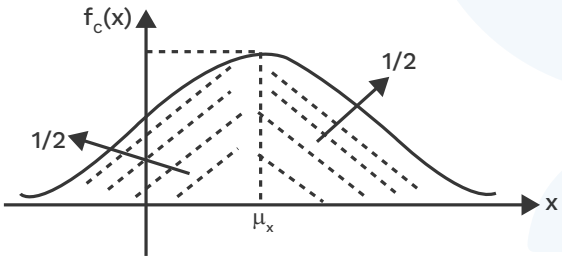


Figure: Gaussian distribution of X

Note: Mean: Divides any PDF into two equal halves.

$$\int_{\mu_x}^{\infty} f_x(x)dx = \int_{-\infty}^{\infty} f_x(x)dx = \frac{1}{2}$$

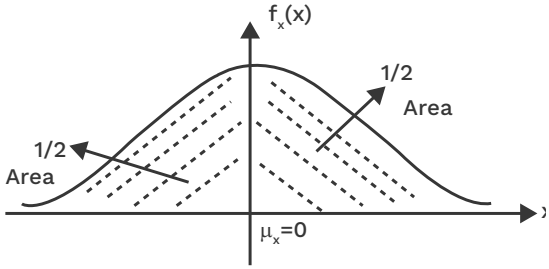


Zero mean GRV:

$X: \mu_x = 0$ and σ_x^2 are the mean and variance

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{-(x-0)^2}{2\sigma_x^2}} = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{-x^2}{2\sigma_x^2}}$$

$$X \sim N\left[0, \sigma_x^2\right] X$$



Zero mean Gaussian random variable PDF is an even function.

Zero mean unit variance GRV:

$X: \mu_x = 0$ and $\sigma_x^2 = 1$ are the mean and variance

$x \sim N[0, 1]$

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{-(x-\mu_x)}{2\sigma_x^2}} = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$$

Area of the PDF is unity.

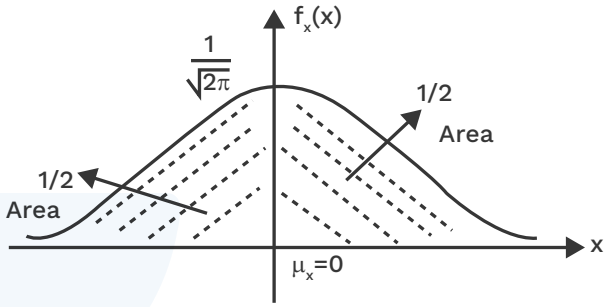


Figure: PDF of zero means and unit variances GRV

The PDF of zero mean unit variance will be symmetric about Y-axis

$$\Rightarrow \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} dx = 1 \ldots (i)$$

Q-function

$$Q(x) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}} dz \ldots (\text{ ii })$$

x: Dummy variable

Whatever variable is in the lower limit, Q is a function of that variable

For example:

$$Q(x) = \int_{x^2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}} dz$$

$$Q\left(\frac{x-2}{3}\right) = \int_{\frac{x-2}{3}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}} dz$$

If we compare above equations (i) and (ii), we can understand that Q-function is the integration of Gaussian distribution's (zero

mean unit variance) PDF for the limit z to ∞, where x is a dummy variable.

So $Q(-\infty) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} dz = 1$

And $Q(\infty) = 0$

As for $x = -\infty; Q(x) = 1$ and for $x = 0; Q(x) = 0$

Hence, it is a decreasing function.

Properties of Q-function:

1. $Q(\infty) = 0$
2. $Q(-\infty) = 1$
3. $Q(\infty) + Q(-\infty) = 1 \Rightarrow Q(x) + Q(-x) = 1$
   (Derivation is not required here)
4. $Q(0) = 0.5$ (From above property)
5. $Q(x) + Q(-x) = 1 \Rightarrow Q(-x) = 1 - Q(-x)$
6. As 'x' increases; $Q(x)$ decreases.

This is an important property

Now,

f(x); the PDF of GRV can be used to find its CDF.

$\Rightarrow F_x(x) = P\{X \le x\} = 1 - P\{X > x\}$

We can find

$$P\{x > x\} = \int_x^{\infty} f_x(x)dx = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-(x-\mu_x)^2}{2\sigma_x^2}} \frac{dx}{\sigma x}$$

We can put $z = \frac{x - \mu_x}{\sigma x}$

When

$$\to x \Rightarrow z = \frac{x - \mu_x}{\sigma_x}$$

$$\to \infty \Rightarrow z \to \infty$$

$$\frac{dx}{\sigma x} = dz$$

$$P[x > x] = \int_{\frac{x-\mu_x}{\sigma_x}} \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}} dz = Q\left[\frac{x - \mu_x}{\sigma x}\right]$$

When X was GRV

We can say, $P[x > a] = Q\left[\frac{a - \mu_x}{\sigma_x}\right]$

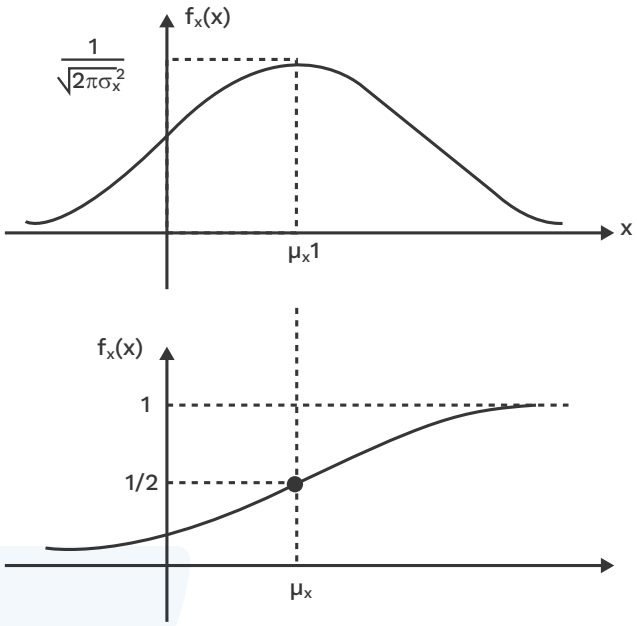


Figure: CDF of GRV (zero mean unit variance)

As, $F_x(x) = 1 - Q\left[\frac{x - \mu_x}{\sigma_x}\right]$

$= 1 - \left[\frac{\mu_x - \mu_x}{\sigma_x}\right]$ (at $x = \mu_x$)

$= 1 - Q(0) = 0.5$ (at $x = \mu_x$)

For X: GRV

$$P[x > x] = Q\left[\frac{x - \mu_x}{\sigma_x}\right]$$

Note:

In this question, it is asked to find probability of GRV that means we need to find Q-function.

[If it is arbitrary then to find probability, we can find it through PDF that means its area].

X: Arbitrary RV	X: GRV
PDF: $f_x(x)$	$\text{PDF}: f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{-(x-\mu_x)^2}{2\sigma_x^2}}$
$\text{CDF}: F_x(x) = \int_{-\infty}^{x} f_x(x)dx$	$\text{CDF}: F_x(x) = 1 - Q\left[\frac{x-\mu_x}{\sigma_x}\right]$
Probability: $P[x > a] = \int_a^{\infty} f_x(x)dx$	Probability: $P[X > a] = Q\left[\frac{a-\mu_x}{\sigma_x}\right]$
(We need to find PDF)	(We can find mean and variance by comparing equation)

Example 12:

PDF of GRV 'X' is given as

$$f_x(x) = \frac{1}{\sqrt{8\pi}} e^{\frac{-(x-3)^2}{8}}$$

Calculate

1. $P\{X \le 3\}$
2. $P\{X = 3\}$
3. $P\{4 < X \le 5\}$

Solution:

By comparing with GRV

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{-\frac{(x-\mu_x)^2}{2\pi\sigma_x^2}}$$

$\mu_x = 3$ and $\sigma_x^2 = 4$

Also $\sigma_x = \pm 2$ but we assume only positive values of σ_x so, $\sigma_x = 2$

$$P\{x \le 3\} = 1 - P[x > 3] = 1 - Q\left[\frac{3 - u_x}{\sigma_x}\right]$$

$$= 1 - Q\left[\frac{3-3}{2}\right] = 1 - Q(0)$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

$P\{x = 3\} = 0$ (As for CRV there are no impulses at points, so it is zero).

$$P\{4 < x \le 5\} = F_x\left(5^+\right) - F_x\left(4^+\right)$$

As, $F_x(a) = 1 - Q\left[\frac{a-u_x}{\sigma_x}\right]$

$$P\{4 < x \le 5\} = 1 - Q\left[\frac{5-3}{2}\right] - 1 + Q\left[\frac{4-3}{2}\right]$$

$$= [Q(0.5) - Q(1)]$$

And this value would be positive.

Example 13:

PDF of GRV is given as

$$f_x(x) = \frac{1}{\sqrt{18\pi}} e^{\frac{-(x-4)^2}{18}}; Q\left(\frac{4}{3}\right) = a$$

Q(2) = b

Calculate

1. $P\{X > 4\}$
2. $P\{X > 0\}$
3. $P\{X > -2\}$
4. $P\{X = 2\}$

Solution:

For GRV, $u_x = 4$ and $\sigma_x = 3$

$$P\{x > 4\} = Q\left[\frac{4-u_x}{\sigma_x}\right] = Q(0) = \frac{1}{2}$$

$$P\{x > 0\} = Q\left[\frac{0-4}{3}\right] = Q\left[\frac{-4}{3}\right] = 1 - Q\left[\frac{4}{3}\right] = 1 - a$$

$$P\{x > -2\} = Q\left[\frac{-2-4}{3}\right] = Q\left[\frac{-6}{3}\right] = Q[-2]$$

$$= 1 - Q[2] = 1 - b$$

$$P\{x = 2\} = 0$$

Rayleigh Distribution

A random variable X has a PDF $f_x(x)$

Where,

$$f_x(x) = \begin{cases} \frac{x}{\sigma_x^2} e^{\frac{-x^2}{2\sigma_x^2}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

And

$$\int_0^\infty \frac{x}{\sigma_x^2} e^{\frac{-x^2}{2\sigma_x^2}} dx = 1$$

Here, the integration can be helpful in solving a few questions.

Exponential Distribution

A random variable X has a PDF $f_x(x)$.

$$f_x(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & x < 0 \end{cases}$$

And

$$\int_0^\infty f_x(x) = 1$$

Example 14:

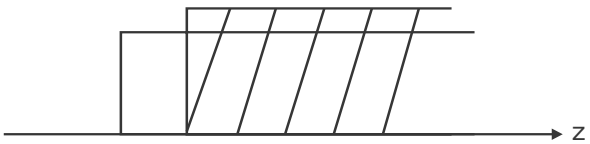
Let Z is an exponential random variable with mean equal to one and CDF of Z is given as:

$$F_x(x) = \begin{cases} 1 - e^{-x} & x > 0 \\ 0 & x < 0 \end{cases}$$

Calculate the probability, $P\left(\frac{Z > 2}{Z > 1}\right)$

Solution:

Suppose $P\left(\frac{Z > 2}{Z > 1}\right) = P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$



As CDF is given:

$$F_z(x) = P(Z \leq x) = 1 - P(Z \geq x)$$

$$\Rightarrow P = P\left(\frac{Z > 2}{Z > 1}\right) = \frac{1 - P(Z \leq 2)}{1 - P(Z \leq 1)}$$

$$= \frac{1 - F_z(2)}{1 - F_x(1)}$$

$$F_z(2) = P(Z \leq 2)$$

$$F_z(1) = P(Z \leq 1)$$

$$F_z(2) = 1 - e^{-2}$$

$$F_z(1) = 1 - e^{-1}$$

$$\Rightarrow P\left(\frac{Z > 2}{Z > 1}\right) = \frac{1 - \left(1 - e^{-2}\right)}{1 - P\left(1 - e^{-1}\right)} = \frac{e^{-2}}{e^{-1}}$$

$$= \frac{1}{e^2} \times e = \frac{1}{e}$$

$$P = \frac{1}{e} = 0.37$$

Laplacian Distribution

For a random variable X, the PDF is $f_x(x)$ –

$$f_x(x) = ae^{-b|x|}$$

The PDF is greater than zero and its integration is unity (depending on values of a and b)

$$\int_{-\infty}^{\infty} f_x(x) = 1,$$

$$\Rightarrow \int_{-\infty}^{\infty} ae^{-b|x|} dx = 1, \text{ this function is even}$$

$$\Rightarrow 2\int_0^{\infty} ae^{-bx} dx = 1 \text{ as } x \text{ is positive so, } |x| = x$$

$$2a\frac{\left[e^{-bx}\right]}{-b} = 1$$

$$\Rightarrow \frac{2a}{-b}[0 - 1] = 1$$

$$4 \Rightarrow \frac{2a}{b} = 1$$

Example 15:

Given $f_x(x) = Me^{-3|x|} + Ne^{-5|x|}$. Find relation between M and N.

Solution:

1. $\int_{-\infty}^{\infty} f_x(x)dx = 1$ (Consider single PDF)

2. $\frac{2\times M}{3}+\frac{2\times N}{5}=1$ (Consider two different PDF)

We will take the second method

$$\frac{2M}{3}+\frac{2N}{5}=1$$

$$\Rightarrow \quad \frac{10M+6N}{15}=1$$

$$\Rightarrow 10M+6N=15$$

$$\left[\int_{-\infty}^{\infty} f_x(x)dx=\int_{-\infty}^{\infty} Me^{-3|x|}dx+\int_{-\infty}^{\infty} Ne^{-5|x|}dx=1\right.$$

Example 16:

Given $f_x(x)=Me^{-3|x|}+Ne^{-2|x|}+Pe^{-2|x|}+Qe^{-2.5|x|}$

Find relation between M,N,P and Q.

Solution:

Similarly to above example,

$$\frac{2M}{3}+\frac{2N}{2}+\frac{2P}{4}+\frac{2Q}{2.5}=1$$

$$\Rightarrow 20M+30M+15P+24Q=30$$

Example 17:

Let X in above example has zero mean, unit variance Gaussian random variable. Calculate the expected value of X.

Solution:

X is zero mean, unit variance GRV

$$X\sim N(0,1),\Rightarrow 20M+3OM+15P+24Q=30$$

$$f_x(x)=\frac{1}{\sqrt{2\pi}}e^{\frac{-x^2}{2}}$$

$$\Rightarrow E[X]=\int_{-\infty}^{\infty} xf_x(x)dx=u_x$$

$$\Rightarrow u_x=0 \text{ (Given in question)}$$

Example 18:

Let X in above example has zero mean, unit variance Gaussian random variable. Calculate the expected value of |X|.

Solution:

From above equation

$$X\sim N[0,1]$$

$$\Rightarrow E[|X|]=\int_{-\infty}^{\infty}|x|f_x(x)dx$$

$$=\int_{-\infty}^{\infty}|x|\times\frac{1}{\sqrt{2\pi}}e^{\frac{-x^2}{2}}dx$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}|x|e^{\frac{-x^2}{2}}dx$$

$$=\frac{2}{\sqrt{2\pi}}\int_{0}^{\infty}|x|e^{\frac{-x^2}{2}}dx \text{ (as even function)}$$

$$=\frac{2}{\sqrt{2\pi}}\int_{0}^{\infty}xe^{\frac{-x^2}{2}}dx \text{ (for } 0\le x\infty\,|\,x\,|=x\text{)}$$

This integration is same Rayleigh distribution. So, the PDF is 1.

$$=\frac{2}{\sqrt{2\pi}}(1)=\sqrt{\frac{2}{\pi}}$$

Conversion or transformation of random variable:

A CRV can be converted to CRV or it can also be converted to DRV. We study only linear transformations.

Case 1.

CRV to CRV (by linear transformations)

Here 'X' is CRV

$Y=aX+b$ is CRV

Here 'X' is RV has $-E[X]=u_x$;

$E\left[X^2\right]$ and $\sigma_x^2=E\left[X^2\right]-u_x^2$

As mean, mean square value and variance

Now,

$Y=aX+b$, so 'Y' RV has

$$E[Y]=E[aX+b]=aE[X]+E[b]$$

$$\Rightarrow E[Y]=aE[X]+b$$

$$\Rightarrow \mu_y=a\mu_x+b$$

Now,

Y = aX+b

$$\Rightarrow Y^2=a^2X^2+b^2+2abX$$

$$\Rightarrow E\left[X^2\right]=a^2E\left[X^2\right]+b^2+2abE[X]$$

$\Rightarrow E\left[Y^2\right]=a^2E\left[X^2\right]+2ab\mu_x+b^2$ that is the mean square value.

And variance,

$$\Rightarrow \sigma_x^2 = E\left[Y^2\right] - (E[Y])^2$$

$$\Rightarrow \sigma_x^2 = a^2 E\left[X^2\right] + 2ab\mu_x + b^2 - a\left(\mu_x + b\right)^2$$

$$= a^2 E\left[X^2\right] + 2ab\mu_x + b^2 - a^2\mu_x^2 - b^2 - 2ab\mu_x$$

$$= a^2\left\{E\left[X^2\right] - \mu_x^3\right\} = a^2\sigma_x^2$$

We need to remember here the mean and variance as they are important. Therefore,

X is a CRV

$Y = aX + b$ is also a CRV

$E[Y] = a\mu_x + b$ and $\sigma y^2 = a^2\sigma_y^2$

Linear transformations do not change the shape of distribution (i.e., PDE)

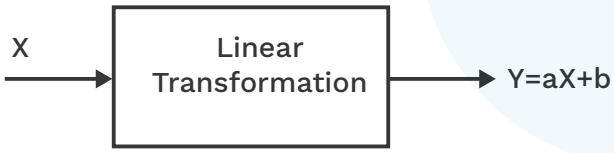


Figure: Block diagram of linear Transformation

But linear transformation changes the statistical parameters.

Now, X URV (uniformly distributed RV between m_1 and m_2)

Or

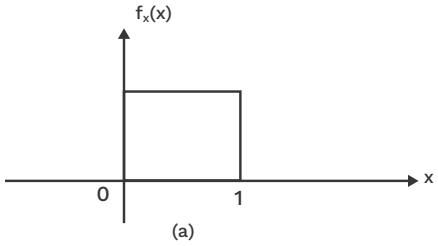
So, $Y = aX + b \sim URV$

$$\Rightarrow Y \sim U\left[am_1 + b, am_2 + b\right]$$

Suppose,

$X \sim U[0, 1]$

Distribution



$\mathbf{Y = 2X; Y \sim \cup[0,2]}$

Distribution

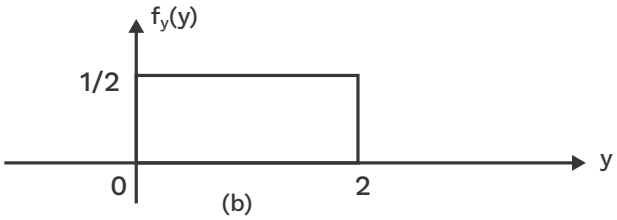
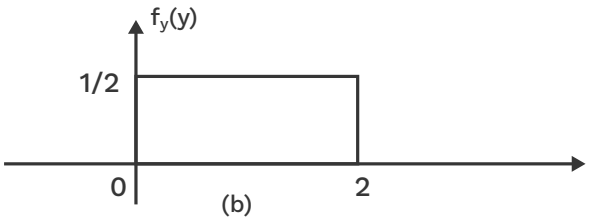


Figure: uniform Distribution of (a) X (b) Y = 2X

$\mathbf{Y \cup [-1, 1]}$



$Y = X + 2$ for X is $[-1, 1]$

So Y is from $[-1+2, 1+2] = [1,3]$

$\Rightarrow Y \sim U[1,3]$

Note: First adjust the width and then height.

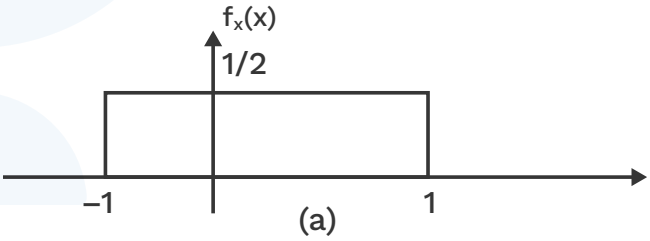


Figure: Uniform Distribution of (a) X (b) Y = X + 2

For Y = X-2, the rate would be $[-1,-2, 1-2]$

$Y \sim U[1,3]$

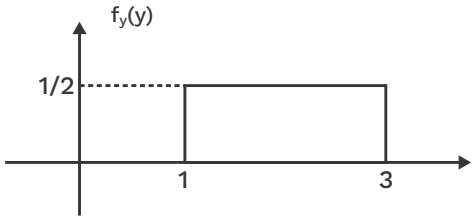


Figure: uniform Distribution of Y = X-2

So, $X \rightarrow f_x(x)$, for $a > 0$

$Y = X + a \rightarrow f_{(x+a)}(x) = f_Y(y)$

$\Rightarrow$ Shift $f_x(x)$ by 'a' unit right wards.

$Y = X - a \rightarrow f_{(x-a)}(x) = f_Y(y)$

$\Rightarrow$ Shift $f_x(x)$ by 'a' unit left wards.

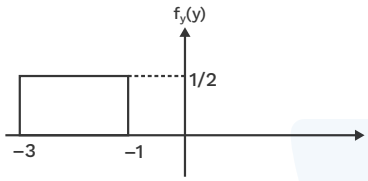
Example 19:

If $X \sim U[0, 1]$. Find

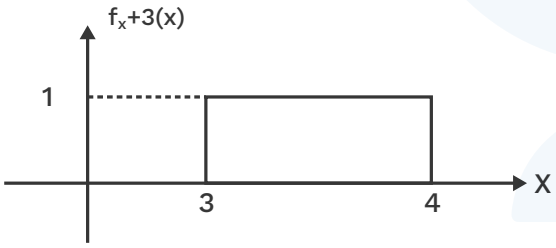
1. Y = X + 3
2. Y = -2X
3. Y = 2X + 5
4. Y = 2X + 5
5. Y = -2X + 3

Solution:

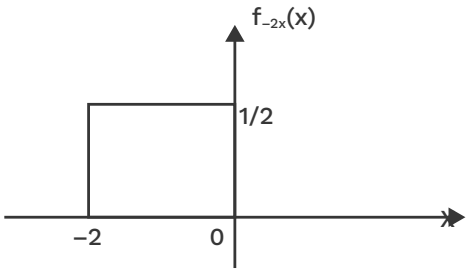
For $X \sim U[0, 1]$, PDF is



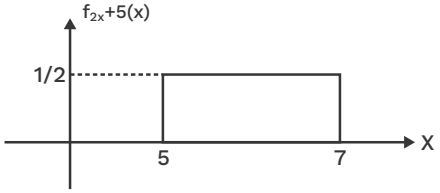
1. Y = X + 3



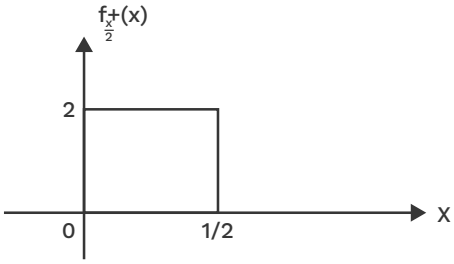
2. Y = -2X



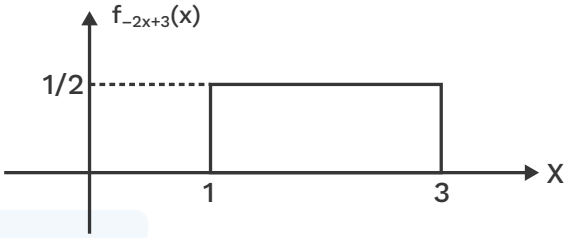
3. Y = 2X + 5



4. $Y = \frac{X}{2}$



5. Y = -2X + 3



X is GRV or $X \sim N\left(\mu_x, \sigma_x^2\right)$

Then Y = aX + b is also GRV.

$\Rightarrow Y \sim N\left(\mu_x, \sigma_x^2\right)$

Where $\mu_y = a\mu_x + b$ and $\sigma_Y^2 = a^2\sigma_x^2$

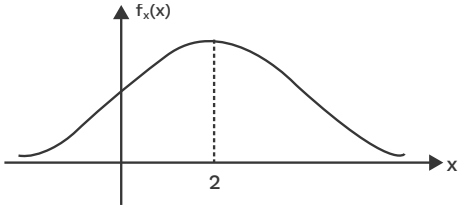
$X \sim N\left(\mu_x, \sigma_x^2\right)$

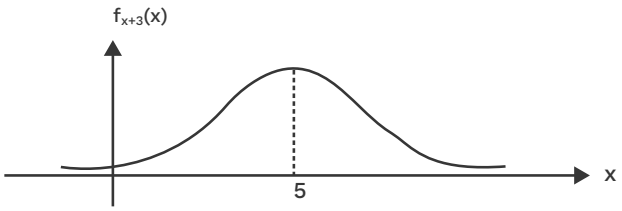
$\Rightarrow Y \sim N\left(a\mu_x + b, a^2\sigma_x^2\right)$

PDF: $\Rightarrow f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{-(x-\mu_x)^2}{2\sigma_x^2}}$

So $\Rightarrow f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{\frac{-(y-\mu_y)^2}{2\sigma_y^2}}$

Where $\Rightarrow \mu_y = a\mu_x + b; \sigma_y^2 = a^2\sigma_x^2$





Fgure: PDF of x and x + 3 for υ_x = 2

If suppose, $Y = 2X + 3$

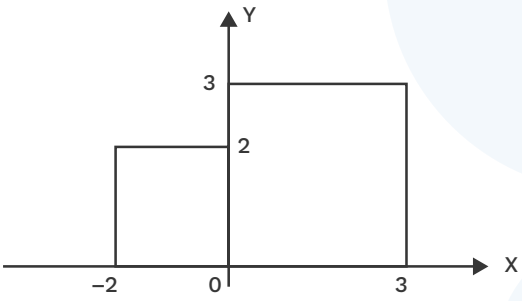
Then $\mu_y = \mathbf{2}\mu_x + 3$ and $\sigma_y^2 = 4\sigma_x^2$

Case 2. CRV to DRV

Let us understand this with help of the following example.

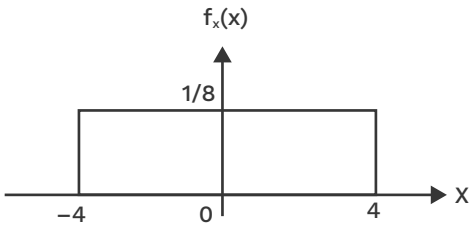
Example 20:

A random variable X is uniformly distributed in the interval -4 to +4. Another random variable Y is related to random variable X as shown in figure obtain PDF of Y.



Solution:

The PDF of X is



X is a continuous RV.

$X : CRV \sim U[-4, 4]$

The figure in question gives a relation between X and Y which is not linear.

For Y at 2, X will be between -2 to 0 and Y at 3, X will be from 0 to 3. Also for Y at 0, X will be either less than -2 or more than -3.

$$\text{So, } Y = \begin{cases} 2 & -2 < X < 0 \\ 3 & 0 < X < 3 \\ 0 & X < -2 \text{ or } X > 3 \end{cases}$$

Y is a discrete random variable.

So, Y: DRV, so its PDF is

$$f_Y(y) = \sum_i P(y_j)\delta(y - y_j) = \sum_j P\{y = y_j\}\delta(y - y_j)$$

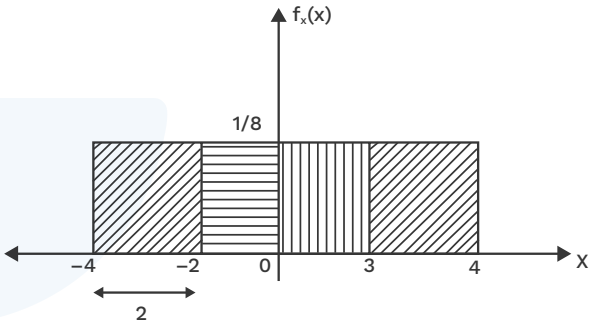
$$= P\{y = 2\}\delta(y - 2) + P\{y = 3\}\delta(y - 3) +$$

$$P\{y = 4\}\delta(y - 0)$$

For $P\{Y = 0\}$

$Y = 0$ for $X < -2$ or $X > 3$

$$P\{Y = 0\} = P\{X < -2\} + P\{X > 3\}$$



(Area under the curve)

$$P\{Y = 0\} = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

For $P\{Y = 2\}, -2 < X < 0$

(Area under the curve)

$$P\{Y = 2\} = P\{-2 < X < 0\} = \frac{1}{4}$$

Lastly $P\{X = 3\}$

$Y = 3$ for $0 < X < 3$

(Area under the curve)

$$P\{Y = 3\} = 1 - \frac{3}{8} - \frac{1}{4} = \frac{3}{8}$$

$$\Rightarrow f_Y(y) = \frac{3}{8}\delta(y) + \frac{1}{4}\delta(y - 2) + \frac{3}{8}\delta(y - 3)$$

Two dimensional or bivariate random variable:

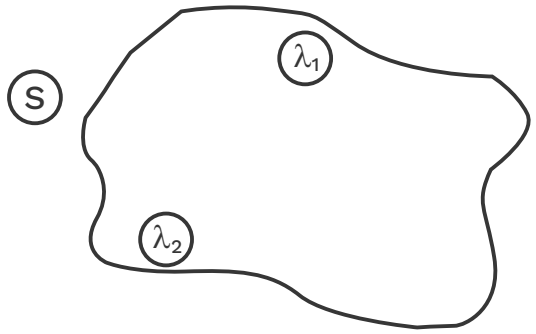


Figure: Sample space 's' with two students λ_1 and λ_2

- $X() \Rightarrow$ Grade and $Y() \Rightarrow$ Height
- Both X and Y have no relation for student λ_1, both functions can be represented by X and Y functions graphically.

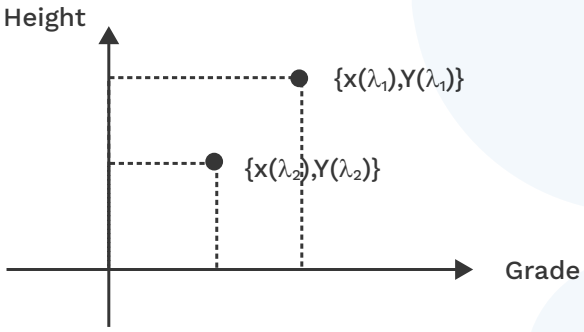


Figure: Dimensional sample point for λ_1 and λ_2

- For 1-dimensional random variable, sample points were mapped on real line which is also 1-dimensional.
- But for 2-dimensional RV, sample points are mapped$\{X(\lambda), Y(\lambda)\}$ or $\{X, Y\}$ where X and Y are two 1-dimensional RV.
- Two random variables (RV) X and Y
- $E[XY] = R_{XY}$ = Correlation between RV, X, and Y.
  $\text{Cov}(XY)$ = Covariance between RV, X, and Y
  $\text{Cov}(XY) = \sigma_{XY}$
  Or
  $\text{Cov}(XY) = E[XY] - E[X]E[Y]$
  $\Rightarrow \sigma_{XY} = R_{XY} - \mu_X\mu_Y$; where R X Y is cross correlation.

Previous Years' Question

4. A fair die with faces {1,2,3,4,5,6} is thrown repeatedly till '3' is observed for the first time. Let X denotes the number of times the die is thrown. The expected value of X is

Note:

1. If two random variables are orthogonal, similar to orthogonality of two vectors $\vec{A}$ and $\vec{B}$ whose dot product becomes zero.
   Here, we take linear product of two signals as zero.
   If there are two signals, and for matrices to be orthogonal we take their linear product, i.e., $[A][B]^T = 0$
   a) For orthogonality of two random variables, their inner product or expectation of X Y, i.e., E[X Y] or R_{X Y} should be zero and if cross correlation between two random variables is zero then they are orthogonal.
   $E[XY] = R_{XY} = 0$
2. If two random variables are uncorrelated, we take covariance as zero and not correlation.
   $\text{Cov}(X.Y) = 0$
   $\text{Cov}(XY) = E[XY] - E[X]E[Y] = 0$
   $\Rightarrow E[XY] - E[X]E[Y] \Rightarrow R_{XY} = \mu_x\mu_y$
3. If two random variables X and Y are independent
   $E\left[X^k Y^r\right] = E\left[X^k\right]E\left[Y^r\right]$
   $K \in I; r \in I$
   We can define the expectation value for any k and r
   For E[X Y] = E[X] E[Y] i.e., k = 1 and r = 1
   We can say that X and Y are uncorrelated.

$\Rightarrow$ If two random variables are uncorrelated then they will always be independent is a false statement.

⇒ If two random variables are independent, then they will always be uncorrelated is true statement.

Therefore,

$E[XY] = R_{XY}$

$\text{CoV}[XY] = E[XY] - E[X]E[Y]$

$= R_{XY} - \mu_X\mu_Y$

$\Rightarrow X, Y$ are orthogonal, then $R_{XY} = 0$ and X, Y uncorrelated or X, Y are independent then $\text{Cov}(XY) = 0$.

Let X and Y are random variables

W = aX + bY

$E(W) = aE[X] + bE[Y] \Rightarrow \mu_w = a\mu_x + b\mu_y$

$W^2 = a^2X^2 + a^2y^2 + 2abxy$

$\Rightarrow E\left(W^2\right) = a^2E\left[X^2\right] + b^2E\left[Y^2\right] + 2abE[XY]$

Or $E\left(W^2\right) = a^2E\left[X^2\right] + b^2E\left[Y^2\right] + 2abR_{XY}$

Case 1.

X, Y are orthogonal

$E\left(W^2\right) = a^2E\left[X^2\right] + b^2E\left[Y^2\right]$

Case 2.

x, y are independent that is

E[X Y] = E[X] E[Y]

$E\left[W^2\right] = a^2E\left[X^2\right] + b^2E\left[Y^2\right] + 2abE[X]E[Y]$

Case 3.

X, Y are uncorrelated same as above, $\text{Cov}(XY)$ is zero so that we can say

$E[XY] = E[X]E[Y]$

Now, for finding the variance

W = aX + bY

Variance: $\sigma_w^2 = \text{E}\left[\text{W}^2\right] - (\text{E}[\text{W}])^2$

$\sigma_w^2 = a^2E\left[X^2\right] + b^2E\left[Y^2\right] + 2ab[XY] -$

$\left(a^2\mu_X^2 + b^2\mu_Y^2 + 2ab\mu_X\mu_Y\right)$

$\sigma_w^2 = a^2E\left[X^2\right]$

$\mu_X^2 + b^2\left\{E\left[Y^2\right] - \mu_Y^2\right\} + 2ab\{E[XY] - E[X]E[Y]\}$

$\Rightarrow \sigma_w^2 = a^2\sigma_Y^2 + b^2\sigma_X^2 + 2ab\,\text{CoV}(XY)$

Example 21:

If W = -2X + 3Y. Find the variance.

Solution:

If $W = -2X + 3Y \Rightarrow W^2 = 4X^2 + 9Y^2 - 12XY$

So, variance is

$\Rightarrow \sigma_w^2 = 4\sigma_Y^2 + 9\sigma_x^2 - 12ab\,\text{cov}(XY)$

Now, we can analyse some cases.

Case 1.

If X and Y are independent, then covariance is zero (as uncorrelated)

$\Rightarrow \sigma_w^2 = a^2\sigma_Y^2 + b^2\sigma_X^2 \quad (\text{ as } \text{Cov}(XY) = 0)$

Case 2.

X and Y are uncorrelated, so this will be same as above.

Case 3.

X and Y are orthogonal,

$\text{Cov}(XY) = R_{XY} - \mu_X\mu_Y$

R_{X Y} = 0

$\Rightarrow \sigma_W^2 = a^2\sigma_X + b^2\sigma_Y^2 - 2ab\mu_X\mu_Y$

Case 4.

X and Y are orthogonal and have zero mean.

w = aX + bY

$\sigma_w^2 = a^2\sigma_x^2 + b^2\sigma_Y^2 + 2ab\,\text{Cov}(XY)$

$\text{cov}(XY) = R_{xY} - \mu_x\mu_Y = 0$

Due to orthogonality R_{x y} = 0 and due to zero mean μ_x &$\mu_y = 0$

$\sigma_w^2 = a^2\sigma_x^2 + b^2\sigma_y^2$

If X_1, X_2, and X_3 are independent random variables, $Z = X_1 + X_2 + X_3$

$Z^2 = X_1^2 + X_2^2 + X_3^2 + 2X_1X_2 + 2X_2X_3 + 2X_3X_1$

$E\left[Z^2\right] = E\left[X_1^2\right] + E\left[X_2^2\right] + E\left[X_3^2\right] + 2R_{x_1x_2} + 2R_{x_2x_3} + 2R_{x_3x_1}$

$\sigma_z^2 = \sigma_{x_1}^2 + \sigma_{X_2}^2 + \sigma_{x_3}^2 + 2\,\text{cov}\left(X_1X_2\right) + 2\,\text{cov}\left(X_2X_3\right)$

$+2\text{cov}(X_3X_1)$

As they are independent, so covariance is zero.

$\sigma_z^2 = \sigma_{x1}^2 + \sigma_{x2}^2 + \sigma_{x3}^2$

If $X_1, X_2, X_3, \ldots X_n$ are mutually independent random variables.

$Z = X_1 + X_2 - X_3 + X_4 \ldots X_n$

$\sigma_z^2 = \sigma_{x1}^2 + \sigma_{x2}^2 + \sigma_{x3}^2 + \ldots \sigma_{xn}^2$

Previous Years' Question

5. Let X_1, X_2, and X_3 be independent and identically distributed random variables with the uniform distribution on [0,1]. The probability $P\{X1$ is the large$\}$ is

- X, Y are independent random variables. Suppose, given $\max(X,Y) < 6 = (X < 6) \cap (Y < 6)$

X	Y	Max (X, Y)	<6
2	3	3	Yes
4	7	7	No
7	4	7	No
8	9	9	no

$P\{\max(X,Y) \le 6\} = P\{(X \le 6) \cap (Y \le 6)\}$

$= P\{(X \le 6)P(Y \le 6)\}$

This can be done by two ways:

$\Rightarrow F_X(6)F_Y(6)$

$\Rightarrow \int_{-\infty}^{6} f_Y(y)dy \int_{-\infty}^{6} f_x(x)dy$

So, $\max(X,Y) \le a = (X \le a) \cap (Y \le a)$

$P\{\max(X,Y) \le a\} = P\{(X \le a) \cap (Y \le a)\}$

$= P\{X \le a\} \cap P\{Y \le a\}$

$\Rightarrow P\{Z > a\} = P[(Z > a] = 1 - P[Z \le a]$

$= 1 - \{P(X \le a)P(Y \le a)\}$

Similarly,

$\min(X,Y) > a = (X > a) \cap P(Y > a)$

$P\{\min(X,Y) > a\} = P\{(X > a) \cap (Y > a)\}$

$= P\{X > a\}P\{Y > a\}$

$\Rightarrow P\{X > a\}P\{Y > a\} = (1 - P\{X \le a\})P(-P\{Y \le a\})$

This can also be done by two ways:

$\Rightarrow (1 - F_X(a))(1 - F_Y(a))$

$\Rightarrow \int_a^{\infty} f_x(x)dx \int_a^{\infty} f_Y(y)dy$, but if given

$P\{\max(X,Y) > 5\}$ we can:

$P\{\max(X,Y) > 5\} = 1 - P\{\max(X,Y) \le 5\}$

So, $P\max(X,Y) \le 5 = (x \le 5) \cap (y \le 5)$

Previous Years' Question

6. A random binary wave y(t) is given by

$$y(t) = \sum_{n=-\infty}^{\infty} x_n p(t - nT - \phi)$$

Where p(t) = u(t) - u(t - T), u(t) is the unit step function and φ is an independent random variable with uniform distribution is [0, T]. The sequence $\{X_n\}$ consists of independent and identically distributed binary valued random variables with $P\{X_n = +1\} = P\{X_n = -1\} = 0.5$ for each n.

The value of the autocorrelation $R_{yy}\left(\frac{3T}{4}\right) = E\left[y(t)y\left(t - \frac{3T}{4}\right)\right]$ equals

Example 22:

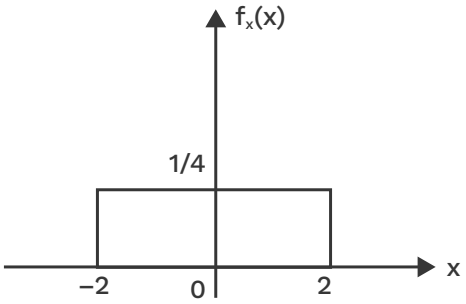
X and Y are two independent random variables uniformly distributed between [-2, 2]

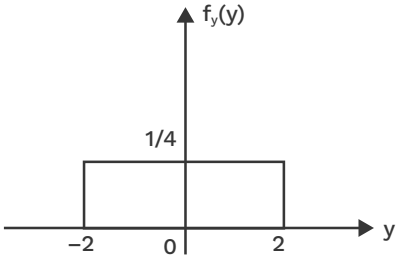
Find

1. $P[\max(X,Y) > 1]$
2. $P[\min(Z,Y) \le 1]$

Solution:

The PDF for X and Y are



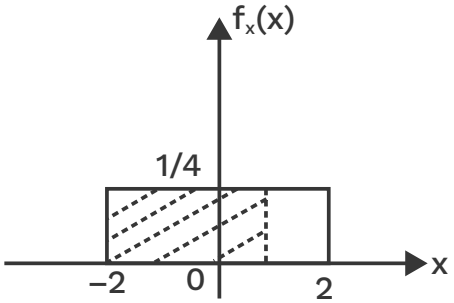


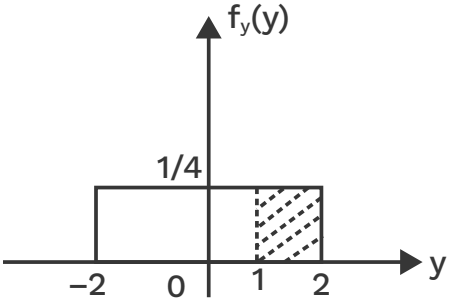
1. $P[\max(X,Y) > 1] = 1 - P\{\max(X,Y) \le 1\}$

$= 1 - P\{(X \le 1) \cap (Y \le 1)\}$

$= 1 - P\{X \le 1\}P\{Y \le 1\}$

$= 1 - \frac{3}{4} \times \frac{3}{4} = 1 - \frac{9}{16} = \frac{7}{16}$





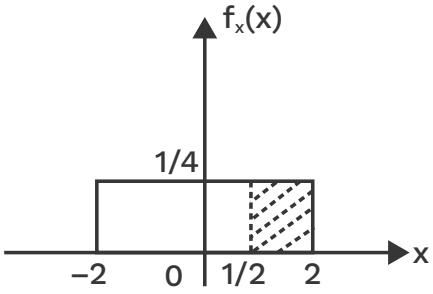
2. $P[\min(2X,Y) \le 1] = 1 - P\{\min(2X,Y) > 1\}$

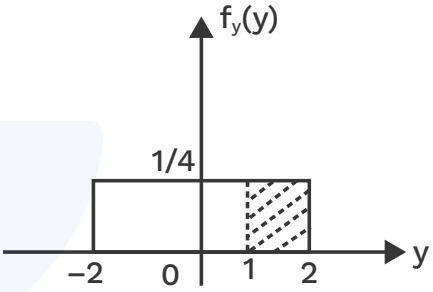
$1 - P\{(2X > 1) \cap (Y > 1)\}$

$= 1 - P\left\{X > \frac{1}{2}\right\} \cap \{Y > 1\}$

$= 1 - P\left(X > \frac{1}{2}\right) P(Y > 1)$

$= 1 - \frac{3}{8} \times \frac{1}{4} = 1 - \frac{3}{32} = \frac{29}{32}$





Previous Years' Question

7. Consider a random process $\sqrt{2}\sin(2\pi t + \phi)$, where the random phase ϕ is uniformly distributed in the interval $[0, 2\pi]$. The auto correlation $E\left[X(t_1)X(t_2)\right]$ is

(A) $\cos\left(2\pi(t_1 + t_2)\right)$

(B) $\sin\left(2\pi(t_1 - t_2)\right)$

(C) $\sin\left(2\pi(t_1 + t_2)\right)$

(D) $\cos\left(2\pi(t_1 - t_2)\right)$

Central Unit Theorem

Suppose X_1 and X_2 are two independent random variables. The PDFs are $f_{x1}(x)$ and $f_{x2}(x)$.

Now, $z = X_1 + X_2$

So, PDF of z will be:

$f_z(z) = f_{x_1}(z) * f_{x_2}(z)$

We take it as function of z.

Whenever two random variables are linearly combined with each other, their PDFs get convolved.

Also, $z = x_1 - x_2 = x_1 + (-x_2)$

$f_z(z) = f_{x1}(z) * f_{-x2}(z)$

We can see that

If z = X_{1}-X_{2}-X_{3}+X_{4}+X_{5}

Where $x_e, i = 1$ to 5 are independent with each other.

$\Rightarrow f_z(z) = f_{x_1}(z) * f_{-x_2}(z) * f_{-x_3}(z) * f_{x_4}(z) * f_{x_5}(z)$

We can see that

Concept 1:

If X and Y are independent random variables then

a) A function of X, i.e., G(X) and Y are independent random variables.

b) X and dual function of Y, i.e., H(Y) independent random variables.

c) Also, $G(X)$ and $H(Y)$ are independent random variables e.g. $X^2, 2X + 5, \sin X, Y^3, 2Y + 7$

Concept 2:

$Z = X_1 + X_2 + X_3 + ... + X_n$ (n: finite)

Hence, $X_1, X_2, X_3 ....X_n$ are independent with each other.

$\Rightarrow f_z(z) = f_{x_1}(z) * f_{x_2}(z) * f_{x_3}(z) ..... * f_{x_n}(z)$

Concept 3:

$X_1, X_2, ...X_n$ are independent random variables (IRV).

$Z = 2X_1 + (3X_2 + 5) + (5X_3 + 7) + ...(-2X_n + 7)$

$f_z(Z) = f_{2x_1}(Z) * f_{3x_2+5}(Z) * f_{5x_3} + 7(Z) * ....f_{-2x_n}(Z)$

Concept 4:

$X_1, X_2, ...X_n$ are IRV general (Non-Gaussian)

$Z = X_1 + X_2 + ...X_n$

$f_z(Z) = f_{x_1}(Z) * f_{x_2}(Z) * ..... * f_{x_n}(Z)$

From here, there are two conclusions for Z:

If n = Finite, then it is general (Non-Gaussian)

n = Infinite, then it is Gaussian

When 'n' no. of Gaussian RV are linearly combined then the resultant RV may or may not be Gaussian.

Suppose X_1, X_2, X_3 are IRV (Non-Gaussian)

$Z = X_1 + X_2 + X_3$

So, Z is also non-Gaussian.

And $f_z(Z) = f_{x_1}(Z) * f_{x_2}(Z) * f_{x_3}(Z)$

Suppose if asked, $P[X_1 + X_2 + X_3 > 1]$, we can

$P[X_1 + X_2 + X_3 > 1] = P[Z > 1] = \int_1^\infty f_z(Z)dz$

Concept 5:

$X_1, X_2 .....X_n$ are RV (Gaussian)

$Z = X_1 + X_2 + ... X_n$

$\Rightarrow f_z(Z) = f_{x_1}(Z) * f_{x_2}(Z) * ... * f_{x_n}(Z)$

From here, the conclusion for Z if n is either finite or infinite, Z is always GRV.

If X_1, X_2, X_3 are RV (GRV)

$Z = X_1 + X_2 + X_3$

$P[X_1 + X_2 + X_3 > 1] = P[Z > 1] = Q\left[\frac{1 - \mu_Z}{\sigma_Z}\right]$

Points To Remember

If linear combination of GRV's are given then central limit theorem is never applied, i.e., $f_z(Z) = f_{x_1}(Z) * f_{x_2}(Z) * f_{x_3}(Z)$ is applicable here. As we can do this by

Q-function.

$Z = X_1 + X_2 + X_3$ has $\mu_z = \mu_{x_1} + \mu_{x_2} + \mu_{x_3}$

And $\sigma_{z^2} = \sigma_{x_1^2} + \sigma_{x_2^2} + \sigma_{x_3^2}$

To summarise

For $P(X_1 + X_2 + X_3 > a)$

Step 1: $Z = X_1 + X_2 + X_3$

Step 2 : $P(Z > a) = 1 - P(Z \le a)$ if Z is a GRV

$$P(Z > a) = Q = \left[\frac{a - \mu_z}{\sigma_z}\right]$$

If Z is not a GRV

$$f_z(Z) = f_{x_1}(Z) * f_{x_2}(Z) * f_{x_3}(Z)$$

Find area here for either $P(Z > a)$ or $1 - P(Z \le a)$

Example 23:

Let U and V be independent random variables with distribution as zero mean GRV. The variance of U is $\frac{1}{4}$ and V is $\frac{1}{9}$. Find the probability that 2U>3V.

Solution:

For U: Zero mean GRV

$$\mu_v = 0$$

$$\sigma_v^2 = \frac{1}{4}$$

And for V: Zero mean GRV

$$\mu_v = 0$$

$$\sigma_v^2 = \frac{1}{9}$$

For $P[2U > 3V] = P[2U - 3V > 0] = P[Z > 0]$ as Z=2U-3V

So, Z is a GRV as well

$$\Rightarrow P[Z > 0] = Q\left[\frac{0 - \mu_z}{\sigma_z}\right]$$

For $z : \mu_z = 2\mu_v - 3\mu_v = 0$

And $\sigma_2^2 = 4\sigma_v^2 + 9\sigma_v^2$

[Both U and V are independent GRV's]

$$Z^2 = 4U^2 + 9V^2 - 12UV$$

As covariance is zero.

$$Z^2 = 4U^2 + 9V^2$$

$$\Rightarrow \sigma_Z^2 = 4 \times \frac{1}{4} + 9 \times \frac{1}{9} = 2$$

$$\Rightarrow \sigma_Z^2 = 2 \Rightarrow \sigma_Z = \sqrt{2}$$

$$\Rightarrow [z > 0] = Q\left[\frac{0 - 0}{\sqrt{2}}\right] = Q(0) = 0.5$$

Example 24:

If X and Y are independent Gaussian random variables, given that $X \sim N(0,4)$ and $Y \sim N(1,2)$. Find $P(X + 2Y > 2)$.

Solution:

For X: GRV

$$\mu_x = 0; \sigma_x^2 = 4$$

For Y: GRV

$$\mu_Y = 1, \sigma_Y^2 = 2$$

For $P[X + 2Y > 2] = P[Z > 2] = Q\left[\frac{2 - \mu_z}{\sigma_z}\right]$

Where, Z = X + 2Y

For Z: GRV

$$\mu_z = \mu_x + 2\mu_y = 0 + 2 \times 1 = 2$$

$$\Rightarrow P[Z > 2] = Q\left[\frac{2 - 2}{\sigma_z}\right] = Q(0) = 0.5$$

Also, $\sigma_z^2 = \sigma_y^2 + 4\sigma_y^2 = 4 + 8 = 12$

$$\sigma_z = \sqrt{12}$$

Example 25:

If X and Y are two independent random variables uniformly distributed as $X \sim [-2,2]$ and $Y \sim [-4,4]$. Find 1. $P[X + Y > 2]$ 2. $P[Y < 2X - 4]$

Solution:

1. $P[X + Y > 2] = P[Z > 2]$

   Where Z = X + Y

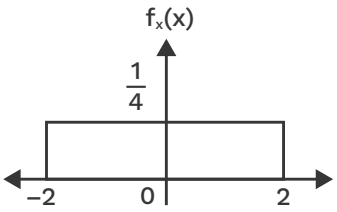
   As both X and Y are uniform random variables, Z is a non-Gaussian random variable. We need to find PDF of Z.

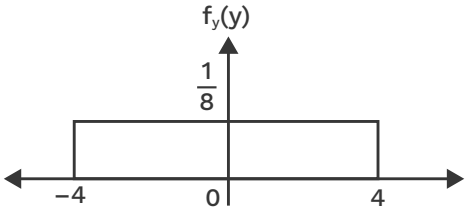
   $$P(Z > 2) = \int_2^{\infty} f_z(z) d_z$$

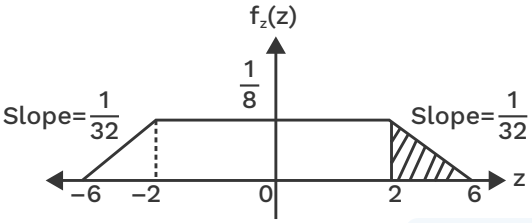
   As Z = X + Y

   We apply the central limit theorem

   $$f_z(z) = f_x(z) * f_Y(z)$$







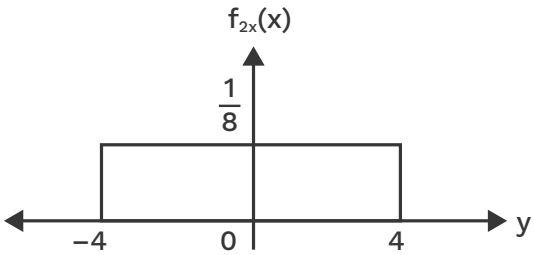
$$\int_2^{\infty} f_z(z)d_z = \frac{1}{2}\times 4\times\frac{1}{8} = \frac{1}{4}$$

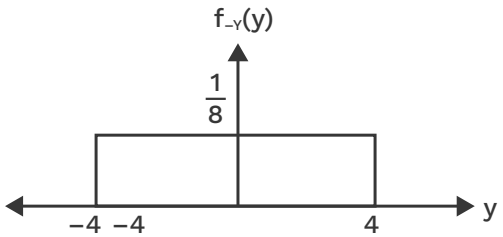
2. $P[Y < 2X - 4] = P[2X - Y > 4] = P[Z > 4]$

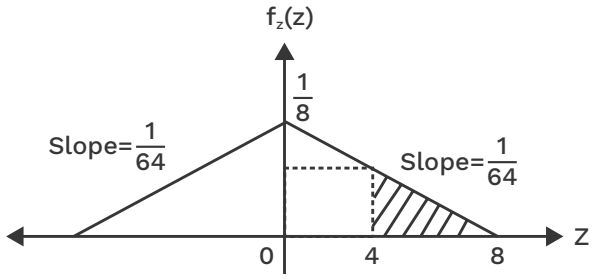
$= \int_4^{\infty} f_z(z)dz$

Z = 2 X - Y; where Z is non-Gaussian. Applying central limit theorem

$f_z(z) = f_X(z) * f_{-Y}(z)$







Example 26:

X_1 and X_2 are independent random variables uniformly distributed between [0,1]. Calculate probability such that one of them is largest.

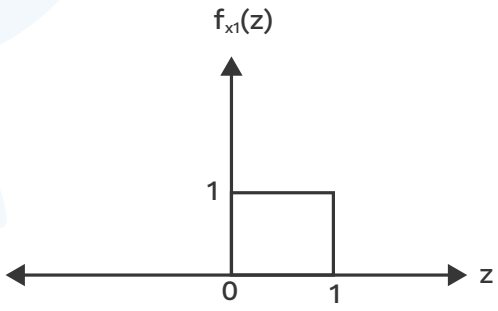
Solution:

$$P\left[X_1 > X_2\right] = P\left[X_1 - X_2 > 0\right] = P[Z > 0]$$

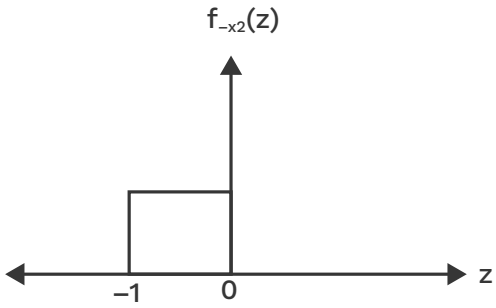
$$= \int_0^{\infty} f_z(z)dz$$

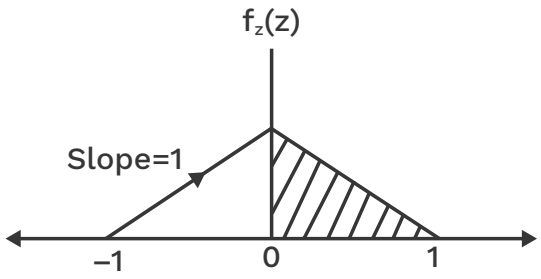
$$Z = X_1 - X_2 = X_1 + \left(-X_2\right)$$

$$f_z(z) = f_{x1}(z) * f_{-x2}(z)$$



$X_1 \sim U[0,1] X_2 \sim U[0,1]$ or $-X_2 \sim U[-1,0]$





Previous Years' Question

8. Let $X \in \{0, 1\}$ and $Y \in \{0, 1\}$ be two independent binary random variables. If P(X = 0) = p and P(Y = 0) = q, then $P(X+Y \geq 1)$ is equal to
   (A) $pq + (1-p)(1-q)$
   (B) pq
   (C) $p(1-q)$
   (D) 1-pq

$$P(Z > 0) = \int_0^{\infty} f_z(z)dz = \frac{1}{2}$$

$X_1, X_2 \ldots\ldots X_n$ are independent random variables, uniformly distributed between [a, b].

[One of them is largest] = P[One of them is smallest]

$$= \frac{1}{n}$$

Example 27:

If $X_1 \sim U[0, 1], X_2 \sim U[0, 1], X_3 \sim U[0, 1]$ and X_1, X_2 and X_3 are independent random variables. Find $P\left[X_1 + 2X_2 > X_3\right]$.

Solution:

$$P\left[X_1 + 2X_2 > X_3\right] = P\left[X_1 + 2X_2 - X_3 > 0\right]$$

$$= P[Z > 0]$$

$$Z = X_1 + 2X_2 - X_3$$

Where Z is non-Gaussian

$$P(Z > 0) = \int_0^{\infty} f_z(z)dz = 1 - \int_{-\infty}^{0} f_z(z)dz$$

$$Z = X_1 + 2X_2 - X_3 = X_1 + 2X_2 + \left(-X_3\right)$$

$$F_z(z) = f_{x_1}(z) * f_{x_2}(z) * f_{-x_3}(z)$$

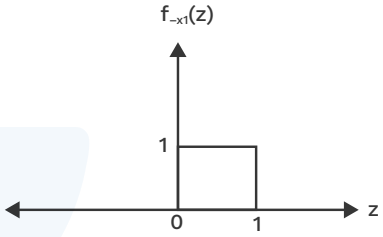
$$x_1 \sim U[0, 1] \Rightarrow x_1 \sim U[0, 1]$$

$$x_2 \sim U[0, 1] \Rightarrow 2x_2 \sim U[0, 2]$$

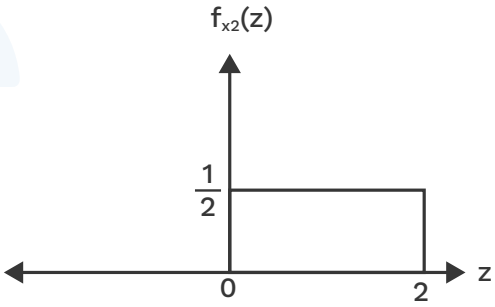
$$x_3 \sim U[0, 1] \Rightarrow -x_3 \sim U[-1, 0]$$

As there are more than two signals, we cannot find conduction by graphical method.

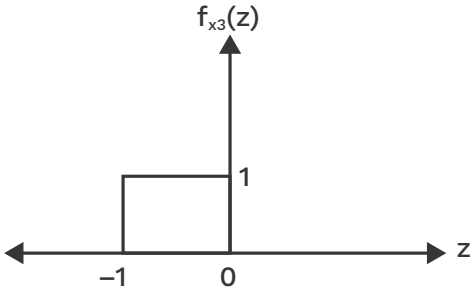
$$f_{x_1}(z) = [U(z) - U(z-1)]$$



$$f_{x_2}(z) = \frac{1}{2}[U(z) - U(z-2)]$$



$$f_{x_3}(z) = [U(z+1) - U(z)]$$



$$[U(z)-U(z-1)]*[U(z+1)-U(z)]*\frac{1}{2}[U(z)-U(z-2)]$$

$$=[r(z+1)-r(z)-r(z)+r(z-1]*\frac{1}{2}[U(z)-U(z-2]$$

$$=\frac{1}{2}[r(z+1)-2r(z)+r(z-1]*[U(z)-U(z-2]$$

$$=\frac{1}{2}[p(z+1)-p(z-1)-2p(z)+2p(z-2)+$$

$$p(z-1)-p(z-3)]$$

$$=\frac{1}{2}[p(z+1)-p(z-1)-2p(z)+2p(z-2)+$$

$$p(z-1)-p(z-3)]$$

$$=\frac{1}{2}p(z+1)-p(z)+p(z-2)-\frac{1}{2}p(z-3)$$

$[U(t-a)\times U(t-b)=r(t-a-b)$ and

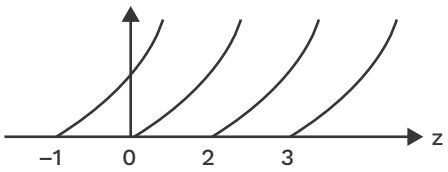
$U(t-a)\times r(t-b)=p(t-a-b)]$

Where $p(t)=\frac{t^2}{2}U(t)$ that is the parabolic function

$$\Rightarrow f_z(z)=\frac{1}{2}p(z+1)-p(z)+p(z-2)-\frac{1}{2}p(z-3)$$

$$P(z>0)=\int_0^{\infty}f_z(z)dz=1-\int_{-\infty}^{0}f_z(z)dz$$

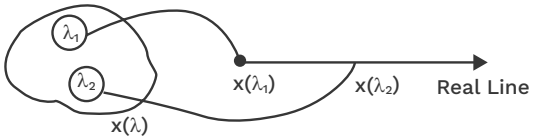
$$=1-\int_{-\infty}^{0}\frac{1}{2}p(z+1)dz=1-\int_{-\infty}^{0}\frac{(z+1)^2}{2}U(z+1)dz$$



$$=1-\frac{1}{2}\int_{-1}^{0}\frac{(z+1)^2}{2}dz=1-\frac{1}{2}\left[\frac{(z+1)^3}{6}\right]_{-1}^{0}$$

$$=1-\frac{1}{12}\left[1^3-0\right]=1-\frac{1}{12}=\frac{11}{12}$$

RANDOM PROCESS

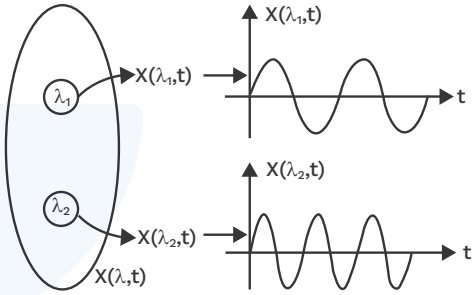


Sample space is the collection of sample points

$$X(\lambda)=\{X(\lambda_1),X(\lambda_2)\}$$

$X(\lambda)$ is the collection of values taken by random variable.

Similarly, from this we can understand random process.

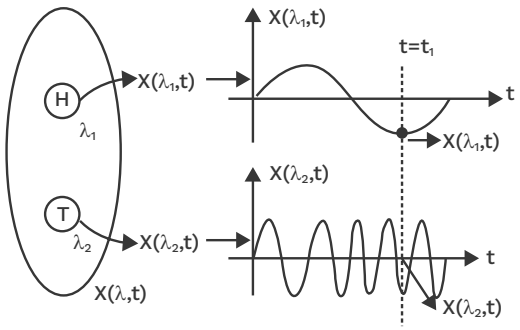


Here, λ_1 and λ_2 are sample points and $x(\lambda,t)$ is a time varying function giving a waveform instead of a value on real line. This is called a sample function. $X(\lambda,t)=\{X(\lambda_1,t),X(\lambda_2,t)\}$ is the collection of sample functions that is random process.

$X(\lambda,t)$ is also called random process/ random signal/stochastic signal.

Suppose,

$S=\{H,T\}$



Where $X(\lambda_1, t)$ and $X(\lambda_2, t)$ are the sample functions and collection of these waveforms is random process that is $X(\lambda, t)$.

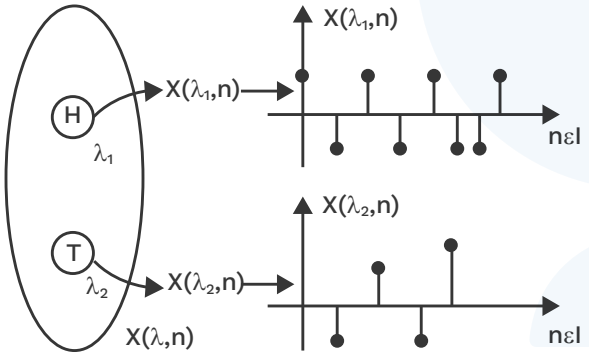
This is also called 'Ensample' and if we observe the random process $X(\lambda, t)$ at a point of line "$t = t_1$", then

$X(\lambda, t) = \{x(\lambda_1, t_1), x(\lambda_2, t_2)\}$ and

at $t = t_1, X(\lambda) = \{x(\lambda_1), x(\lambda_2))$ which is a random variable.

So, if a random process is decreased at a particular instance of time is a random variable and $x(\lambda_1, t_1), x(\lambda_2, t_1)$ are values taken by random variable. $X(\lambda, t)$ is a continuous time random process.

Now, suppose we take $x(\lambda, n)$ is also a random process (discrete).



Here $x(\lambda_1, n)$ and $x(\lambda_2, n)$ are sample functions.

$x(\lambda, n) = \{x(\lambda_1, n), x(\lambda_2, n)\}$ and $x(\lambda, n)$ is the collection of sample functions.

Also, if $n = n_1$, then $x(\lambda, n_1)$ is a random variable and $x(\lambda_1, n_1), x(\lambda_2, n_1)$ are values taken by random variable.

Classification of Random Process

Random Process (RP):

1. Continuous time RP:

   $X(\lambda, t) = X(t)$

   At $t = t_1 \Rightarrow x(t_1)$ is a random variable.

2. Discrete time RP:

   $X(\lambda, n) = X(n)$

   $X(n) = \{x_n\}_{nx}^{\infty}\ n \in i$

   At $n = n_1 \Rightarrow x(n_1)$ is a random variable.

Note:

If X is DRV then its function G(x) is also a DRV and if X is CRV then its function G(x) will also be a CRV.

- A typical example of random process is $X(t) = A\cos(\omega_0 t + \phi)$ where 'ω_0' and 'ϕ' are constants, and 'A' is a random variable and $A\{-1, 1\}$. The value of 'A' varies and suggests that X(t) is a random process.
- For X(t) to be a continuous random process and at t = t_1 this example gives x(t_1) as a discrete random variable as only two values of ' A ' are mentioned.
- Suppose $X(t) = A\cos(\omega_0 t + \phi)$ where 'A' and 'ω_0' are constant and $\phi \sim U[-\pi, \pi]$. As 'ϕ' is uniformly distributed, it is a continuous random variable.
- For X(t), a continuous random process and at t = t_1, the above example gives $X(t_1)$ as a continuous random. Variable as 'ϕ' takes continuous values between $-\pi$ to π.$X(t)$ if $(t, \phi) \rightarrow$ CRTP $\rightarrow t = t_1 \Rightarrow f(t_1, \phi)$ is an RV.
- Similarly, for X(n) which is a discrete time random process can be sampled at $n = n_1$ and the random variable $X(n_1)$ can be continuous or discrete random variable depending on the variable factor in the function is either CRV or DRV.

From above example:

$x(n) = f(x, \phi) \rightarrow$ DTRP $\rightarrow x = n_1 \Rightarrow f(x_1, \phi)$

is an RV and if ϕ is DRV then $X(n_1)$ is DRV and if ϕ is CRV then $X(n_1)$ is CRV.

Joint-Probability Density Function (Joint PDF)

It is denoted by $F_{xY}(x, y)$

Points To Remember

If $Y = g(X)$ then

$$E[X] = \int_{-\infty}^{\infty} x f_x(x)dx$$

$$E[Y] = \int_{-\infty}^{\infty} y f_x(y)dy$$

Statistical Parameter of Random Process

A random process cannt have PDF and CDF, but can be found for a particular time instance.

Suppose X(t) is a random process, for a time $t = t_1$, we have $X(t_1)$, which a random variable with

$$\text{CDF}: F_x(t_1)(x_1) = P\{x(t_1) \le x_1\}$$

$$\text{PDF}: f_x(t_1)(x_1) = \frac{\partial F_x(t_1)(x_1)}{\partial x_1}$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x)dx$$

So, for joint PDF

$$E[XY] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy f_{xY}(x, y)dxdy$$

If random process X(t) is observed at two time instances t_1 and $\Rightarrow$ So, $\overline{4}$ and $\sigma_Y^2 = a$ are two random variables.

Joint CDF: $X \sim N(\mu_x, \sigma_x^2)$ Joint CDF of two random variables observed.

At $t = t_1$ and in a random process or 2nd order joint CDF.

Joint PDF: $\Rightarrow \mu_y = a\mu_x + b; \sigma_y^2 = a^2$ order joint PDF.

If $F_x(t_1)(t_2)(t_3)(x_1, x_2, x_3)$ is given, it is called 3rd order joint PDF.

Mean:

$$E[x(t_0)] = \int_{-\infty}^{\infty} x(t_0) f_x(t_0)(x_0)dx = \mu_x(t_0)$$

Case 1.

CRP: $X(t) = f(t, \phi)$ is deserved at $t = t_0 \Rightarrow X(t_0) = f(t_0, \phi)$ is a random variable (continuous).

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x)dx$$

So, for joint PDF

$$E[XY] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy f_{xY}(x, y)dxdy$$

$$E[X(t_0)] = \int_{-\infty}^{\infty} x(t_0) f_x(t_0)(x_0)dx_0$$

$$E[f(t_0, \phi)] = \int_{-\infty}^{\infty} x(t_0, \phi) f_\phi(\phi)d\phi$$

Case 2.

For DTRP: $X: \text{CRV} \sim U[-$ is observed at $x = x_0 \Rightarrow f(x_0, \phi)$ which CRV as φ is CRV.

$$E[X(x_0)] = E[f(x_0, \phi) = \int_{-\infty}^{\infty} f(x_0, \phi) f_\phi(\phi)d\phi$$

Case 3.

For DTRP: $X(x) = f(x, \phi)$ is observed at x = x_0 $\Rightarrow$ f (x_0, ϕ) which is DRV as φ is DRV.

$$E[X(x_0)] = \sum_i x_{0i} P\{x(x_0) = x_{0i}\}$$

CTRP: Now X(t) is observed at two different time instances $t = t_1$ and $t = t_2$. So, $x(t_1)$ and $x(t_2)$ are random variables.

$$\Rightarrow E[X(t_1) \times (t_2)] = R_{x(t_1)\times(t_2)} = R_{xx}(t_1, t_2)$$

It is called auto correlation, which means correlation between two random variables obtained from same random process.

It can be defined for same instance of time too.

$$\Rightarrow E[X(t_1)Y(t_1)] = R_{X(t_1)Y(t_1)} = R_{XY}(t_1, t_1)$$

Now,

$$\text{cov}(X(t_1)X(t_2)) = E[x(t_1)x(t_2)] -$$

$$E[x(t_1)]E[x(t_2)]$$

$$\text{Cov}(x(t_1)X(t_2)) = R_{xx}(t_1, t_2) - \mu_{x(t_1)}\mu_{x(t_2)}$$

This is called auto covariance.

And

$$\text{Cov}\left(X(t_1)Y(t_2)\right) = E\left[X(t_1)Y(t_2)\right] - E\left[X(t_1)\right]E\left[Y(t_2)\right]$$

$$\text{Cov}\left(X(t_1)X(t_2)\right) = R_{XY}(t_1,t_2) - \mu_{x(t_1)}\mu_{Y(t_2)}$$

This is called cross covariance.

DTRP: Now, X(n) is observed at two different time instance $n = n_1$ and $n = n_2$. So $x(n_1)$ and $X(n_2)$ are random variables.

$\Rightarrow E\left[X(n_1)X(n_2)\right] = R_{xx}(n_1,n_2)$ which is called auto correlation.

Similarly for two random processes $X(n)$ and $Y(n)$ observed at n_1 and n_2 respectively.

$\Rightarrow E\left[X(n_1)y(n_2)\right] = R_{xy}(n_1,n_2)$ which is called cross correlation.

Also, for $X(n)$ at $n = n_1$ and $n = n_2$, we have

$$\text{Cov}\left(x(n_1)x(n_2)\right) = R_{xx}(n_1,n_2) - \mu_{x(n_1)}\mu_{x(n_2)}$$

And for $X(x)$ at $x = x_1$ and $Y(x)$ at $x = x_2$, we have

$$\text{Cov}\left(X(n_1)Y(n_2)\right) = R_{XY}(n_1,n_2) - \mu_{x(n_1)}\mu_{y(n_2)}$$

Those are auto and cross covariance respectively. If $R_{XY}(t_1,t_2) = 0$ for all possible t_1 and t_2. So, $X(t)$ and $Y(t)$ are orthogonal random processes. Similarly,

$$\text{Cov}\left(X(t_1)Y(t_2)\right) = C_{XX}(t_1,t_2) = 0$$

For all possible t_1 and t_2, X(t) and Y(t) are uncorrelated random processes.

Types of random processes:

1. Strict sense stationary random process
2. Wide sense stationary random process
3. Ergodic random process
4. Cyclisation random process

Strict Sense Stationary Random Process (SSSRP):

If a random process X(t) is observed at

$t_1, t_2, t_3 \ldots\ldots\ldots t_k$, we get k random variables

$$\rightarrow X(t_1), X(t_2), X(t_3)\ldots\ldots\ldots X(t_k)$$

$$f_{x(t1),x(t2),x(tk)} \cdots\cdots\cdots x_{x(tk)}(x_1, x_2 \ldots\cdots\cdots \mu_k)\ldots\ldots \quad \text{(iii)}$$

This is k^{th} order joint PDF.

Now, we shift by time 'τ'

$$x(t) \rightarrow x(t_1+\tau), x(t_2+\tau), x(t_3+\tau)\ldots\ldots\ldots x(t_k+\tau)$$

Still there are k random variables.

$$f_x(t+\tau) \rightarrow x(t_1+\tau), x(t_2+\tau)\ldots.$$
$$x(t_k+\tau)(x_1, x_2 \ldots..x_k)\ldots\ldots\ldots \quad \text{(iv)}$$

This is also the kth order joint PDF.

A random process is SSSRP if its kth order joint PDF, CDF, and PMF is independent of time shifting. That is, equation (i) is equal to equation (ii).

For SSSRP, we see the 1st order PDF, i.e., for $X(t_1)$ is:

$$f_x(t_1)^{x_1} = f_x(t_1+\tau)^{x_1}$$

This can only be possible if $f_{x(t)}(x)$ is independent of time (or it should be constant) as well as independent of time shifting.

Now observe 2nd order PDF, i.e., for $X(t_1)$ and $X(t_2)$

$$f_{x(t_1)\times(t_2)}(x_1, x_2) = f_{x(t_1+\tau)\times(t_2+\tau)}(x_1, x_2)$$

Suppose $\tau = -t_1$

$$f_{x(t_1)\times(t_2)}(x_1, x_2) = f_{x(0)x(t_2-t_1)}(x_1, x_2)$$

For both LHS and RHS, the observation time difference for both is $t_2 - t_1$.

So, the 2nd order joint PDF's value depends on time difference '$t_2 - t_1$' (and not on $(a$ or t_2.). Also, it is independent of time shifting.

For SSSRP, we see 3rd order PDF, i.e.,

$$f_{x(t_1)\times(t_2)\times(t_3)}(x_1, x_2, x_3) = f_{x(t_1+\tau)\times(t_2+\tau)\times(t_3+\tau)}(x_1, x_2, x_3)$$

$$= f_{x(0)\times(t_2-t_1)\times(t_3-t_1)}(x_1, x_2, x_3)$$

(If $\tau = -t_1$)

This also does not depend on time instance individually, but not their difference.

Wide Sense Stationary Random Process (WSSRP):

It is a random process which is SSSRP up to 2nd order.

So, all SSSRP is WSSRP but WSSRP can or cannot be SSSRP.

In WSSRP, the 1st order PDF is independent of time.

$$f_{x(t)(x)} = f_{x(t_0)}(x) = f_{x(t_1)}(x) = \ldots\ldots = \text{constant}$$

Similarly, the 2nd order PDF is independent of time instance but dependent upon time difference $(t \sim t_2)$.

$$f_{x(t_1)\times(t_2)}(x_1, x_2) = f_{x(0)\times(t_1-t_2)}(x_1, x_2)$$

For WSSRP X(t) at t = to i.e., X_1, X_2 is a random variable.

$$E\left[X(t_0)\right] = \int x(t_0) f_{x(t_0)}(x_0)\, dx_0 = \text{contant} = \mu_x$$

As $X(t_0)$ and $f_{X(t_0)}$ are constants as it is 1st order PDF.

Also, $E\left[X(t_1)\right] = \mu_x$

Also, $E\left[X(t_2)\right] = \mu_x$

So, for WSSRP has constant mean (at any instant of time)

$\Rightarrow E[X(t)] = \mu_x$ = Mean of RP

So, for WSSRP X(t) as well as the random variables $X(t_1), X(t_2)\ldots$ have WSSRP:

$E[RP] = E[RV] \Rightarrow$ Mean of RP = Mean of RV.

As it is independent of time

X(t) is observed at τduration apart

$$E[X(t)] = E[x(t+\tau)] = \mu_x$$

Also, for WSSRP, we can find

$$E\left[X(t_1)X(t_2)\right] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 x_2 f_{x(0)x(t_2-t_1)}(x_1, x_2)\, dx_1 dx_2$$

$$\Rightarrow R_{xx}(t_1, t_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 x_2 f_{x(0)x(t_2-t_1)}(x_1, x_2)\, dx_1 dx_2$$

$$= R_{xx}(t_2 - t_1)$$

$= f(t_2 - t_1)$, that is function of time difference

$$\Rightarrow E\left[X(t_1)X(t_2)\right] = R_{xx}(t_1, t_2)$$

$$= R_{xx}(t_2 - t_1) \text{ or } R_{xx}(t_1 - t_2)$$

Therefore, in case of WSSRP, the auto correlation function does not depend upon individual time instance rather on the time difference between them.

For CTRP:

'$X(t)$'

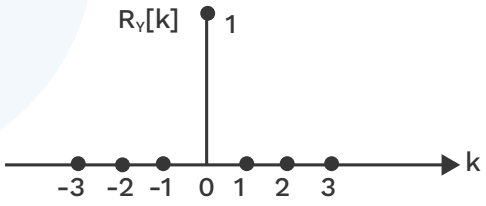
1. $E[x(t)] = \mu_x$ = Constant

Or $E[x(t)] = E[x(t+\tau)] = \mu_x$

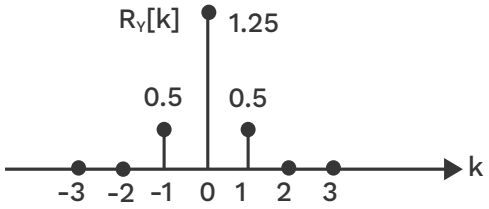
Previous Years' Question

9. $\{X_n\}_{n=-\infty}^{n=\infty}$ is an independent and identically distributed (IID) random process with Xn equally likely to be +1 or −1. $\{X_n\}_{n=-\infty}^{n=\infty}$ is another random process obtained as $Y_n = X_n + 0.5X_{n-1}$. The autocorrelation function of $\{X_n\}_{n=-\infty}^{n=\infty}$, denoted by $R_Y[k]$, is

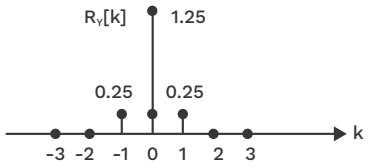
(A)



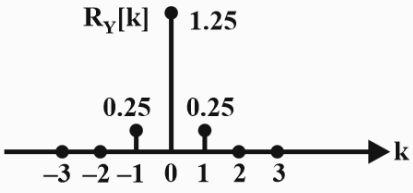
(B)



(D)



(D)



2. $E\left[X(t_1)\times(t_2)\right]=R_{xx}(t_1,t_2)$

We can see (τ is the time difference between observation)

$E[X(t)X(t+\tau)]=R_{xx}(\tau)$

$E[X(t+\tau)X(t)]=R_{xx}(\tau)$

$E[X(t-\tau)X(t)]=R_{xx}(\tau)=R_{xx}(-\tau)$

$E[x(t)x(t-\tau)]=R_{xx}(\tau)$

It is an even function of τ.

3. $E[x(t)\times(t+\tau)]=R_{xx}(\tau)$

If $\tau=0$

$E\left[x^2(t)\right]=R_{xox}(0)=\text{Constant}$

This the mean square value of random process

So, if we observe at $t=t_1$ and $t=t_2$ we get

$E\left[x^2(t_1)\right]=R_{xx}(0)$ and $E\left[x^2(t_2)\right]=R_{xx}(0)$

Therefore, in case of WSSRP,

$E\left[x^2(t)\right]=E\left[x^2(t_1)\right]=R_{xx}(0)=\text{constant}$

If X(t) is energy signal then $E\left[x^2(t)\right]=R_{xx}(0)$ gives energy and if X(t) is a power then $E\left[x^2(t)\right]=R_{xx}(0)$ gives power.

Previous Years' Question

10. A wide sense stationary random process X(t) passes through the LTI system shown in the figure. If the autocorrelation function of X(t) is $R_x(\tau)$, then the auto correlation function $R_Y(\tau)$ of the output Y(t) is equal to

(A) $2R_x(\tau)+R_x(\tau-T_0)+R_x(\tau+T_0)$

(B) $2R_x(\tau)-R_x(\tau-T_0)-R_x(\tau+T_0)$

(C) $2R_x(\tau)+2R_x(\tau-2T_0)$

(D) $2R_x(\tau)-2R_x(\tau-2T_0)$

Rack Your Brain

Write the kth order joint CDF and k^{th} order joint PMF for X(t)

4. $\sigma_x^2(t)=E\left[x^2(t)\right]-(E[x(t)])^2$

$\sigma_x^2(t)=R_{xx}(0)-\mu_x^2=\text{constant}$

So, for time $t=t_1$ and $t=t_2$ we can say that

$\sigma_x^2(t_1)=\sigma_x^2(t_2)=R_{xx}(0)-\mu_x^2$

Therefore,

$\text{Var(RP)}=\text{Var(RV)}$

$\Rightarrow \text{Var}(X(t))=\text{Var}(X(t+\tau))=\text{same}$

$\sigma_x^2(t)=\sigma_x^2(t+\tau)=\text{same}$

5. Let X(t) is power signal

$E\left[x^2(t)\right]=(E[x(t)])^2+\sigma_x^2(t)$

$\Rightarrow R_{xx}(0)=\mu_x^2+\sigma_x^2$

$\Rightarrow$ If $X(t)$ is a power signal then mean square value of RP $X(t), E\left[X^2(t)\right]$, represents the total power of random process $X(t)$ or total power of random variable $X(t_1)$.

$\Rightarrow$ Here, $E[X(t)]^2$ or μ_x^2 is also called DC power of RP $X(t)$ and $\sigma_x^2(t)$ is the AC power of RP X(t).

6. Zero mean WSSRP

It has $E[X(t)]=0$

$\Rightarrow E\left[x^2(t)\right]=\mu_x^2+\sigma_x^2(t)\quad(\mu_x=0)$

$\Rightarrow E\left[x^2(t)\right]=\sigma_x^2(t)$

So, $MSV(X(t))=\text{Var}(X(t))$

$\Rightarrow R_{xx}(0)=\text{Var}(x(t))$

For DTRP: 'X(n)'

1. $E[X(n)]=\mu_x=\text{Constant}$

Or $E[X(n)]=E[X(n+k)]=\mu_x$

$E[X(n)]=E\left[X(n_1)\right]=E\left[X(n_2)\right]=\text{Constant}$

2. $E\left[x(n_1)\times(n_2)\right] = R_{xx}(n_2 \sim n_1)$

From here, $E[x(n)\times(n+k)] = R_{xx}(k)$

$E[x(n)x(n-k)] = R_{xx}(k)$

$E[x(n+k)x(n)] = R_{xx}(k)$

$E[x(n-k)x(n)] = R_{xx}(k)$

3. $E\left[x^2(n)\right] = R_{xx}(0) = E\left[x^2(n_1)\right]$

$= E\left[x^2(n_2)\right] =$ constant

$E\left[x^2(n)\right] = E\left[x^2(n+k)\right]$

$=$ same = constant

4. $\sigma_x^2(n) = \sigma_x^2(n_1) = \sigma_x^2(n_2) =$ constant

$\sigma_x^2(n) = \sigma_x^2(n+k) =$ constant = same

5. Let X(n) is a power signal.

$E\left[x^2(n)\right] = \sigma_x^2(n) + \mu_x^2(n)$

Where $E\left[X^2(n)\right]$ is the total power of DTRP. $[0, 2\pi]$ is the DC power of DTRP and $\sigma_x^2(n)$ is the AC power of DTRP.

6. Zero mean WSSRP; E[X(x)] = 0

$\Rightarrow E\left[x^2(n)\right] = \sigma_x^2(n)$

$R_{xx}(0) = \text{Var}(x(n))$

Covariance of WSSRP

For X(t), a random process observed at time $t = t_1$ and $t = t_2$

$\text{Cov}\left(x(t_1)x(t_2)\right) = E\left[x(t_1)x(t_2)\right] -$

$E\left[x(t_1)\right]E\left[x(t_2)\right]$

$C_{xx}(t_1, t_2) = R_{xx}(t_1, t_2) - \mu_x(t_1)\mu_x(t_2)$

For WSSRP

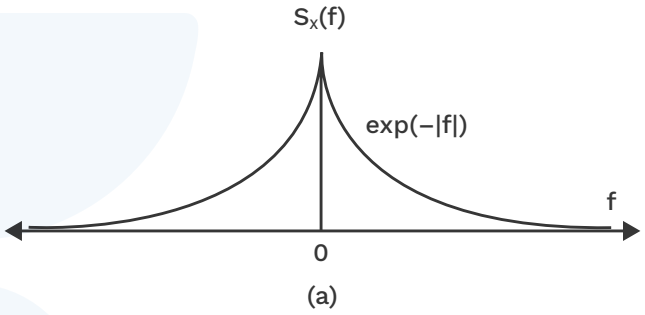
$C_{xx}(t_1 \sim t_2) = R_{xx}(t_1 \sim t_2) - \mu_x^2$

Previous Years' Question

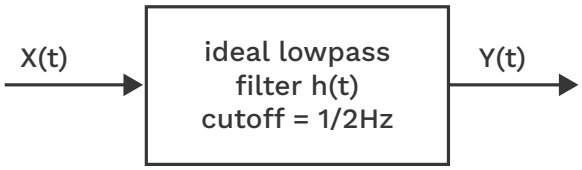
11. Let X(t) be a wide sense stationary random process with the power spectral density. $S_x(f)$ as shown in figure (A), where f is in Hz. The random process X(t) is input to an ideal low pass filter with the frequency response

$$H(f) = \begin{cases} 1, & |f| \le \frac{1}{2} Hz \\ 0, & |f| > \frac{1}{2} Hz \end{cases}$$

as shown in figure (B). The output of the low pass filter is Y(t).



(a)



Let E be the expectation operator and consider the following statements:

I. $E(X(t)) = E(Y(t))$

II. $E\left(X^2(t)\right) = E\left(Y^2(t)\right)$

III. $E\left(Y^2(t)\right) = 2$

Select the correct option:

(A) Only I is true.

(B) Only II and III are true.

(C) Only III and II are true.

(D) Only I and III are true.

If, given

$$\text{Cov}(x(t)x(t+\tau)) = E[x(t)x(t+\tau)] -$$

$$E[X(t)]E[X(t+\tau)]$$

$$C_{xx}(\tau) = R_{xx}(\tau) - \mu_x^2$$

And similarly,

$$\text{Cov}(X(n)x(n+k)) = E[X(n)X(n+k)] -$$

$$E[X(n)]E[x(x+k)]$$

$$C_{xx}(k) = R_{xx}(k) - \mu_x^2$$

Example 28:

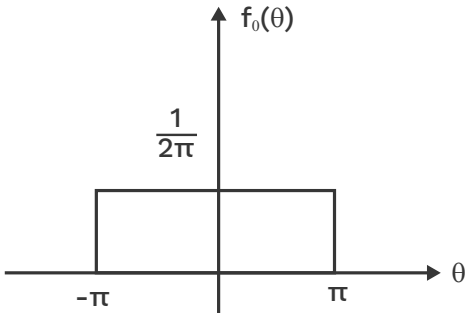
If θ is a uniform random variable and distributed between $-\pi$ to π. Find

1. $E[\theta]$
2. $E[\theta t]$
3. $E[2\theta]$
4. $E[\theta + 2]$
5. E[t]
6. $E\left[\theta t^2\right]$
7. $E[\cos\theta]$

Solution:

θ is URV

$\theta \sim \cup[-\pi, \pi]$



1. $E[(\theta)] = \int_{-\infty}^{\infty} \theta \cdot f_\theta(\theta) d\theta = \int_{-\pi}^{\pi} \theta f_\theta(\theta) d\theta$

   $= \frac{a+b}{2} = \frac{-\pi + \pi}{2} = 0$

2. $E[\theta t] = t\,E[\theta] = 0$
3. $E[2\theta] = 2E[\theta] = 0$
4. $E[\theta + 2] = E[\theta] + E[2] = 0 + 2 = 2$
5. E[t] = t
6. $E\left[\theta t^2\right] = t^2 E[\theta] = 0$
7. $E[\cos\theta] = \int_{-\infty}^{\infty} \cos\theta \cdot f_\theta(\theta) \cdot d\theta = \int_{-\pi}^{\pi} \frac{\cos\theta}{2\pi}$

   $= \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos\theta d\theta = 0$
8. $E\left[t^2 \cos\theta\right] = 0$
9. $E\left[\cos\theta\left(\omega_0 t + \frac{\theta}{2}\right)\right]$; same as

   $E[f(t,\theta)]\theta : CRV$

   So, $E[f(t,\theta)] = \int_{-\infty}^{\infty} f(t,\theta) f_\theta(\theta) d$?

   $= \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos\left(\frac{\theta}{2} + \omega_0 t\right) d\theta$

   $= \frac{1}{2\pi}\int_{-\pi}^{\pi} \left[\sin\left(\frac{\theta}{2} + \omega_0 t\right) \times 2\right]_{-\pi}^{\pi}$

   $= 2 \times \frac{2}{2\pi} \times \sin\left(\frac{\pi}{2} + \omega_0 t\right)$

   $= \frac{2}{\pi}\sin\left(\omega_0 t + \frac{\pi}{2}\right)$

   $= \frac{2}{\pi}\cos\left(\omega_0 t\right)$
10. $E[\cos\theta t] = \int_{-\infty}^{\infty} \cos\theta t f_\theta(\theta) \cdot d\theta = \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos\theta t d\theta$

    $= \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos t\theta d\theta = \frac{1}{2\pi}\left[\frac{\sin t\theta}{t}\right]_{-\pi}^{\pi}$

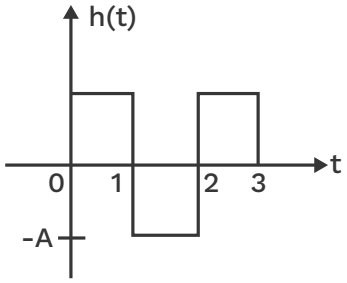
    $= \frac{1}{2\pi}\left[\frac{\sin t\pi - (\sin -\pi t)}{t}\right]$

    $= \frac{\sin \pi t + (\sin \pi t)}{2\pi t} = \frac{2\sin \pi t}{2\pi}$

    $= \sin c(\pi t)$

12. A zero mean white Gaussian noise having power spectral density $\frac{N_0}{2}$ is passed through an LTI filter whose impulse

response h(t) is shown in the figure. The variance of the filtered noise at t = 4 is



(A) $\frac{3}{2}A^2N_0$

(B) $\frac{3}{4}A^2N_0$

(C) A^2N_0

(D) $\frac{3}{4}A^2N_0$

Example 29:

If X(t) is a random process, 'A' and 'ω_0' are constants and φ is uniformly distributed between 0 and 2π. Find the expectation of X(t) when

1. $X(t) = A\cos(\omega_0 t + \phi)$
2. $X(t) = A\cos(2\phi + \omega_0 t)$
3. $X(t) = A\cos(k\phi + \omega_0 t + \alpha)$ $k \in I$ and α = Deterministic

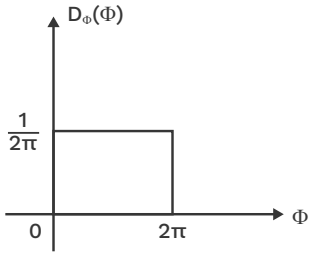
Solution:

Mean of X(t) is

$$E[X(t)] = E\left[A\cos(\omega_0 t + \phi)\right]$$

$$= \int_{-\infty}^{\infty} A\cos(\omega_0 t + \phi) f_{\phi}(\phi) d\phi$$

i. $$= \frac{1}{2\pi}\int_0^{2\pi} A\cos(\phi + \omega_0 t)\, d\phi = 0$$



2.

$$E[X(t)] = E\left[A\cos(2\phi + \omega_0 t)\right] =$$

$$\int_{-\infty}^{\infty} A\cos(2\phi + \omega_0 t) f_{\phi}(\phi) d\phi$$

$$= \frac{1}{2\pi}\int_0^{2\pi} A\cos(2\phi + \omega_0 t)\, d\phi = 0$$

3. $E[X(t)] = E\left[A\cos(k\phi + \omega_0 t + \alpha)\right] = 0$

$\left\{\text{Also } E[X(at+b)] = E\left[A\cos(\omega_0 a\phi + \omega_0 b + \phi)\right] = 0\right\}$

Grey Matter Alert!

For expectation operator:

E[X t] = t E[X]

E[X(t)] will be like that of random variable

E[X(t) h(t)] = h(t) E[X]

Where h(t): Unit impulse response of an LTI system (it is deterministic). E[X(t, θ)] will also be like that of a random variable. (θ = Random variable; t = time) → Implicit function

$E[X(t,\theta)] = tE[\theta]$ → Implicit function

Example 30:

If X(t) is a random process, find the mean square value of X(t) when $X(t) = A\cos(\omega_0 t + \phi)$.

Here 'A' and ω_0 are contains and ϕ is uniformly distributed between 0 and 2π. Also find the auto correlation.

Solution:

The mean square value of X(t) is

$$E\left[X^2(t)\right] = E\left[A^2\cos^2(\omega_0 t + \phi)\right]$$

$$E\left[X^2(t)\right] = E\left[\frac{A^2}{2} + \frac{A^2}{2}\cos^2(2\omega_0 t + 2\phi)\right]$$

$$E\left[X^2(t)\right] = E\left[\frac{A^2}{2}\right] + E\left[\frac{A^2}{2}\cos^2(2\phi + 2\omega_0 t)\right]$$

$$E\left[X^2(t)\right] = \frac{A^2}{2} + 0 = \frac{A^2}{2} =$$ Constant (Independent of time)

The auto correlation of X(t) is

$$E[X(t)X(t+\tau)] = E\left[A\cos\left(\omega_0 t+\phi\right)\right.$$

$$\left.A\cos\left(\omega_0 t+\omega_0 t\tau+\phi\right)\right]$$

$$= A^2E\left[\cos\left(\omega_0 t+\phi\right)\cos\left(\omega_0 t+\phi+\omega_0\tau\right)\right]$$

$$= \frac{A^2}{2}E\left[\cos\left(2\omega_0 t+\omega_0\tau+2\phi\right)+\cos\left(\omega_0\tau\right)\right]$$

$$= \frac{A^2}{2}E\left[\cos\left(2\phi+\omega_0\tau+2\omega_0 t\right)\right]+\frac{A^2}{2}E\left[\cos\left(\omega_0\tau\right)\right]$$

$$= \frac{A^2}{2}\cos\left(\omega_0\tau\right)$$

From examples 2 and 3, we can say that random process $X(t) = A\cos\left(\omega_0 t+\phi\right)$ is a WSSRP.

Now, we can calculate the variance of X(t) in above example.

$$\sigma_x^2(t) = E\left[X^2(t)\right]-(E[X(t)])^2$$

$$\sigma_x^2(t) = E\left[X^2(t)\right] = \frac{A^2}{2}$$

And, the covariance is

$$\text{cov}(X(t)X(t+\tau)) = R_{xx}(\tau)-\mu_x^2$$

$$\Rightarrow C_{xx}(\tau) = R_{xx}(\tau)$$

So, in conclusion, if $X(t) = A\cos\left(\omega_0 t+\phi\right)$ or $X(t) = A\sin\left(\omega_0 t+\phi\right)$, both have zero mean, with

$$= R_{xx}(\tau) = \frac{A^2}{2}\cos\left(\omega_0\tau\right) = \frac{A^2}{2}\cos\left(\omega_0\left(t_1\sim t_2\right)\right)$$

$$= \frac{A^2}{2}\cos\left(\omega_0\left(\left|t_1\sim t_2\right|\right)\right)$$

τ is time difference between time instances.

Example 31:

It $X(t) = \sqrt{2}\cos(4t+\phi)$ where ϕ is uniformly distributed between 0 to 2π. Find $= E\left[X\left(t_1\right)X\left(t_2\right)\right]$.

Solution:

Given $X(t) = \sqrt{2}\cos(4t+\phi)$ and $\phi\sim U[0,2\pi]$

$$= E\left[X\left(t_1\right)\times\left(t_2\right)\right] = R_{xx}\left(t_1\sim t_2\right) = \frac{A^2}{2}\cos\omega_0\tau$$

$$= E[X(t)X(t+\tau)] = \frac{(\sqrt{2})^2}{2}\cos\omega_0\tau$$

$$= \cos 4\left(t_1\sim t_2\right)$$

Or $\cos 4\tau$

Where $\tau = t_1\sim t_2$

Ergodic random process:

If E[X(t)] is the statistical average

$$E[X(t)] = \int_{-\infty}^{\infty}X(t)f_{x(t)}(x)dx$$

And $\bar{X}(t)$ is the time average

$$\bar{x}(t) = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}x(t)dt$$

In ergodic random process, $E[X(t)] = \bar{X}(t)$

Venn Diagram

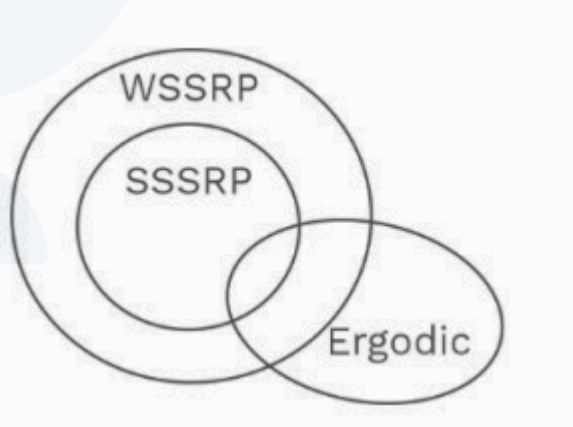
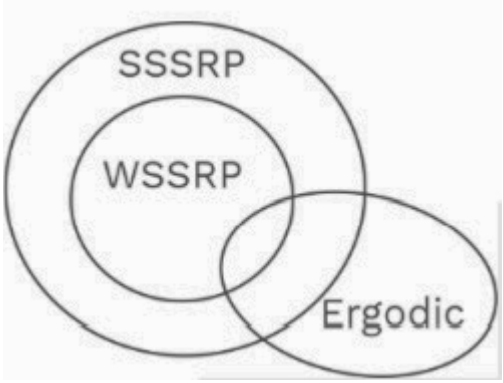


Figure: Venn diagram showing relationship between random processes.



From above figure, we can say that all SSSRP are WSSRP and also if a random process is SSSRP, then it is WSSRP, but can or cannot be ergodic process.

Auto correlation: $E[X(t)X(t+\tau)] = R_{xx}(\tau)$ for WSSRP

$R_{xx}(-\tau) = R_{xx}(\tau)$ (Evan function of time difference)

$$R_{xx}(0) = E\left[X^2(t)\right] = \text{MSV of } X(t)$$

$$\left|R_{xx}(\tau)\right| \le R_{xx}(0) = -R_{xx}(0) \le R_{xx}(\tau) \le R_{xx}(0)$$

$$R_{xx}(\tau)\Big|_{max} = R_{xx}(\tau)\Big|_{\tau=0}$$

X(t) is WSSRP

Periodic (with period T):

$E[X(t)X(t+\tau)] = R_{xx}(\tau)$ is also periodic with time period T.

Non-periodic: $E[X(t)X(t+\tau)] = R_{xx}(\tau)$ is also non-periodic.

For $R_{xx}(\tau)$, if $\tau = 0; = R_{xx}(0) = E\left[X^2(t)\right]$ and if $\tau = \infty; R_{xx}(\infty) = \mu_x^2$ only when X(t) is ergodic.

If a random process is both WSSRP and ergodic,

$$\sigma^2_{x(t)} = R_{xx}(0) - R_{xx}(\infty)$$

For $\Rightarrow F_x($, if $\tau = 0; = R_{xx}(0) = E\left[X^2(t)\right]$ and if $\tau = \infty; R_{xx}(\infty) = \mu_x^2$ only when X(t) is ergodic.

If a random process is both WSSRP and ergodic,

$$\sigma^2_{x(t)} = R_{xx}(0) - R_{xx}(\infty)$$

Example 32:

Let X(t) be an ergodic random process with auto correlation function, $R_{xx} = \frac{50}{2+\tau^2} + 25$

Find

1. E[X(t)]
2. $E[X^2(t)]$
3. E[X(101)]
4. E[X(2) X(4)]

Solution:

The random process X(t) is ergodic

$$\mu_x^2 = R_{xx}(\infty) = 25 \Rightarrow \mu_x = 5 = E[X(t)]$$

$$E[X(t)] = R_{xx}(0) = 25 + 25 = 50$$

E[X(101)] = 5

$$E[X(2)X(4)] = R_{xx}(2) = \frac{50}{2+(2)^2} + 25$$

$$= \frac{50}{6} + 25 = \frac{100}{3}$$

We know that this process was WSSRP as R_{xx} was given as a function of τ and mean came out to be constant.

Example 33:

If X(t) is ergodic random process with ACF

$$R_x(\tau) = \begin{cases} \frac{-25|\tau|}{2} + 100 & |\tau| \le 4 \\ 50 & Elsewhere \end{cases}$$

Find

1. E[X(t)]
2. $E\left[X^2(t)\right]$
3. $E[X(2)X(8)]$
4. $E[X(2)X(4)]$

Solution:

X(t) is an ergodic random process

1. $(E[X(t)])^2 = R_{xx}(\infty) = 50$

   (As $\lim_{\tau\to\infty} R_{xx}(\tau) = 50$)

   $\Rightarrow E[X(t)] = \sqrt{50}$
2. $E\left[X^2(t)\right] = R_{xX}(0) = 100$
3. $E[X(2)X(8)] = R_{xx}(6) = 50$
4. $E[X(2)X(4)] = R_{xx}(2) = 75$

Suppose if $Y(t)X(at+b)$ is a random process we can calculate

a) $\text{Mean} = E[Y(t)] = E[X(at+b)] = \mu_X$

b) $MSV = E\left[Y^2(t)\right] = E\left[X^2(at+b)\right] = R_{xx}(0)$

c) $\sigma_Y^2(t) = E\left[Y^2(t)\right] - (E[Y(t)])^2 = R_{xx}(0) - u_x^2$

d) $R_{xX}(t) = E[Y(t)Y(t+\tau)]$

$= E[X(at+b)X(at+a\tau+b)]$

(Here X(at+b) was WSSRP $= R_{xx}(a\tau)$)

As Y(t) has a constant mean and ACF as a function of τ. So, $Y(t)$ is also a WSSRP.

Independent Identically Distributed Random Process (IIDRP):

X(t) is independent identically distributed.

Suppose, we sample X(t) at 'k' instances

$$X(t) \to X(t_1), X(t_2), \ldots\ldots X(t_k)$$

Whenever random process is independent then $X(t_1), X(t_2), \ldots\ldots X(t_k)$ random variables should be independent of each other.

The statistical parameters of these random variables should be identical, for X(t) to be identical the PDF, CDF, PMF, mean, variance, mean square values should be identical.

For a random process to be WSSRP and ITD:

For WSSRP:

$$E[x(t)] = E\left[X(t_1)\right] - E\left[X(t_2)\right]$$

$$E\left[X(t_1)X(t_2)\right] = E\left[X(t_1)\right]E\left[X(t_2)\right] = \mu_x \cdot \mu_x = \mu_x^2$$

[Also, if $X(t_1)$ and $X(t_2)$ are independent]

If we observe X(t) at τ duration apart

$$E[X(t)] = E[X(t+\tau)] = \mu_x$$

If z = 0

$$R_{xx}(\tau) = E[X(t)X(t+\tau)] = E[X(t)]E[X(t+\tau)] = \mu_X^2$$

Now, for IIDRP:

$$= E\left[X(t_1)X(t_2)\right] = E\left[X(t_1)\right]E\left[X(t_2)\right] = R_{xx}(t_1, t_2)$$

And for DTRP $X(n)$ we can have the same conclusions.

For independent random process:

$$x(n) \to X(n_1), x(n_2), x(n_3) \ldots x(n_k)$$

Random variables will be independent. For identically distributed, the random variables $X(n_1), X(n_2), X(n_3) \ldots . X(n_k)$ will have identical statistical parameters. For WSSRP:

$$E[X(n)] = E\left[X(n_1)\right] = E\left[X(n_2)\right] = \mu_X \text{ and}$$

$$E\left[X(n_1)X(n_2)\right] = E\left[X(n_1)\right]E\left[X(n_2)\right]$$

(As independent)

$$E\left[X(n_1)X(n_1)\right]E\left[X^2(n_1)\right]$$

(As they are not independent)

Now, suppose X(n) is WSSRP and IIDRP and observed at 'l' duration apart

i.e., X(n) and X(n+l)

WSSRP: $E[X(n)] = E[X(n+l)] = \mu_x$

If l = 0

(As they are not independent)

Now, suppose X(n) is WSSRP and IIDRP and observed at 'l' duration

i.e., X(n) and X(n + l)

WSSRP: $E[X(n)] = E[X(n+l)] = \mu_x$

If $l = 0 \quad R_{xx}(l) = E[X(n)X(n+l)] = E[X(n)X(n)]$

$$= E\left[X^2(n)\right]$$

$$= R_{xx}(0)$$

If $l \neq 0$

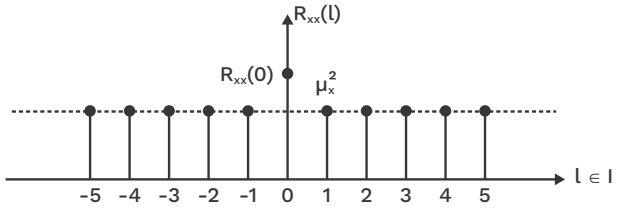
$$R_{xx}(l) = E[X(n)X(n+l)] = \mu_x^2$$

$$R_{xx}(l) = E[X(n)X(n+l)] = E[X(n)X(n)] = E\left[X^2(n)\right]$$

$$= R_{xx}(O)$$

If $l \neq 0$

$$R_{xx}(l) = E[X(n)X(n+l)] = \mu_x^2$$



From above figure:

$$R_{xx}(l) = R_{xx}(0)\delta(l) + \sum_{i=-\infty}^{\infty} \mu_x^2 \delta(l-i)$$

(Where $l \neq 0$ and $i \in I$)

So, it is in the form of discrete impulses.

Density Functions

Density functions enable us to perform the frequency domain analysis of a random process.

Suppose X(t) is a deterministic signal or deterministic process

$$\text{So, } X(t) \xleftrightarrow{\text{F.T.}} X(\omega) \text{ or } X(f)$$

Now, if X(t) is a random process

The density functions are

$X(t) \xrightarrow{\text{Energysignal}}$ Energy Spectral Density (ESD)

Or $G_x\{x\}(f)$ or $G_x\{x\}(\omega)$

And

$X(t) \xrightarrow{\text{Power signal}}$ Power Spectral Density (PSD)

Or $S_{xx}(f)$ or $S_{xx}(\omega)$

Energy Spectral Density Function (ESD)

If X(t) is a stationary random process and is an energy signal,

$$X(t) \xrightarrow{ESD} G_{xx}(\omega) \text{ or } G_{xx}(f)$$

$$X(t) \xrightarrow{F.T.} X(\omega)$$

Or

$$X(t) \xrightarrow{ESD} |X(\omega)|^2 = G_{xx}(\omega)$$

Or

$$X(t) \xrightarrow{ESD} |X(f)|^2 = G_{xx}(f)$$

ESD represents how much energy per unit frequency in frequency domain. The unit of ESD is Joule/Hz.

Fourier transform:

$$G_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau$$

$$\Rightarrow G_{xx}(0) = \int_{-\infty}^{\infty} R_{xx}(\tau) d\tau = 2\int_{0}^{\infty} R_{xx}(\tau) d\tau$$

As $R_{xx}(\tau)$ is an even function

Therefore,

$$G_{xx}(0) = 2 \times \left[\text{ Area of } R_{xx}(\tau) \text{ from } 0 < \tau < \infty\right]$$

Inverse Fourier transform:

$$R_{xx}(\tau) = \begin{cases} \frac{1}{2\pi}\int_{-\infty}^{\infty} G_{xx}(\omega) e^{j\omega\tau} d\omega \\ \int_{-\infty}^{\infty} G_{xx}(f) e^{j2\pi f\tau} df \end{cases}$$

For $\tau = 0$

$$R_{xx}(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} G_{xx}(\omega) d\omega = \frac{\text{Area under } G_{xx}(\omega)}{2\pi}$$

$$\int_{-\infty}^{\infty} G_{xx}(f) df = \text{ Area under } G_{xx}(f)$$

= MSV of X(t)

Where X(t) represents energy signal. So,

$$\Rightarrow E\left[X^2(t)\right] = R_{xx}(0)$$

We can calculate its energy by $R_{xx}(0)$

Power Spectral Density Function (PSD)

If X(t) is a stationary random process and a power signal.

$$X(t) \xrightarrow{PSD} S_{xx}(\omega)$$

Or $X(t) \xrightarrow{PSD} S_{xx}(f)$

PSD is the amount of power per unit frequency range in frequency domain. The unit is Watt Hz.

To calculate the power spectral density, the power signal X(t) has a PSD $S_{xx}(f)$

$$X(t) = S_{xx}(f)$$

First, we convert the power signal X(t) to energy signal $X_T(t)$

$$X_T(t)\left|X_T(f)\right|^2 = \text{ESD(Joule / Hz)}$$

Power signal

$$X(t) = \lim_{T\to\infty} X_T(t) \rightarrow \lim_{T\to\infty} \left|X_T(t)\right|^2$$

$$= PSD watt \text{ / Hz or Joule / sec Hz}$$

Weiner Khinchin theorem:

When X{t} is stationary and a power signal, and observed at τ interval, then

$$E[X(t)X(t+\tau)] = R_{XX}(\tau)$$

Now, $R_{xx}(\tau) \xrightarrow{F.T.} S_{xx}(\omega)$

$$\text{Or}, R_{xxx}(\tau) \xleftrightarrow{F.T} S_{xx}(f)$$

So, according to this theorem, the ACF and PSD form a Fourier transform pair.

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau$$

When ω = 0

$$S_{xx}(0) = \begin{cases} \int_{-\infty}^{\infty} R_{xx}(\tau) d\tau & (\text{ Area under the ACF }) \\ 2\int_{-\infty}^{\infty} R_{xx}(\tau) d\tau & \left[\begin{array}{l} 2x(\text{Area under the} \\ \text{ACF from 0 to } \infty) \end{array}\right] \end{cases}$$

Similarly, $S_{xx}(\omega) \xleftrightarrow{I.F.T.} R_{xx}(\tau)$

or, $S_{xxX}(f) \xleftrightarrow{I.F.T.} R_{xx}(\tau)$

$$E[X(t)X(t+\tau)] = R_{xx}(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{xx}(\omega) e^{j\omega t} d\omega$$

$$\int_{-\infty}^{\infty} S_{xx}(f) e^{j2\pi f\tau} df$$

If τ = 0

$$\text{MSV}[x(t)] = E\left[X^2(t)\right] R_{xx}(0) =$$

$$(i)\frac{1}{2\pi}\int_{-\infty}^{\infty} S_{xx}(\omega) d\omega = \frac{\text{Area under the curve } S_{xx}(\omega)}{2\pi}$$

$$(ii)\int_{-\infty}^{\infty} S_{xx}(f) df = \text{ Area under the curve } S_{xx}(f)$$

So, $R_{xx}(0)$ represents the power of random process $X(t)$.

Methods to calculate power of random process X(t):

1. $E\left[X^2(t)\right] = \text{MSV}(X(t))$

2. $E[X(t)X(t+\tau)] = R_{xx}(\tau) \xrightarrow{z=0} R_{xx}(0) = E\left[X^2(t)\right]$

3.

$$E[X(t)X(t+\tau)] = R_{xx}(\tau) \xrightarrow{F.T.} S_{xx}(\omega) \text{ or } S_{xx}(f)[PSD]$$

$$E\left[X^2(t)\right] = \int_{-\infty}^{\infty} \frac{S_{xx}(\omega)}{2\pi} d\omega$$

$$\int_{-\infty}^{\infty} S_{xx}(f) df$$

In broad sense:

$$E\left[X^2(t)\right] = R_{xx}(0)$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{xx}(\omega) d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\sigma^-} S_{xx}(\omega) d\omega + \frac{1}{2\pi}\int_{0^-}^{0^+} S_{xx}(\omega) d\omega + \frac{1}{2\pi}\int_{0^+}^{\infty} S_{xx}(\omega) d\omega$$

Where $E\left[X^2(t)\right] = R_{xx}(0)$ is the total power of random process X(t) and $\frac{1}{2\pi}\int_{\sigma^-}^{0^+} S_{xx}(\omega) d\omega =$ d.c. power of $X(t)$

And

$$\frac{1}{2\pi}\int_{\sigma^-}^{0^+_{xx}} S_{xx}(\omega) d\omega + \frac{1}{2\pi}\int_{0^+}^{\infty} S_{xx}(\omega) d\omega = \text{AC. power of } X(t)$$

Similarly in frequency domain:

$$E\left[X^2(t)\right] = R_{xx}(0) = \int_{-\infty}^{\infty} S_{xx}(f) df$$

$$= \int_{-\infty}^{0^-} S_{xx}(f) df + \int_{0^-}^{0^+} S_{xx}(f) df + \int_{0^+}^{\infty+} S_{xx}(f) df$$

So, in conclusion, power can be calculated by $-\left[S_{xx}(\omega) \rightarrow \text{Even function when X(t) is real and power signal.}\right.$

Total power:

$$E\left[X^2(t)\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{xx}(\omega) d\omega = \frac{1}{2\pi} \times 2\int_{-\infty}^{\infty} S_{xx}(\omega) d\omega$$

$$\int_{-\infty}^{\infty} S_{xx}(f) df = 2\int_{-\infty}^{\infty} S_{xx}(f) df$$

DC power:

$$(E[X(t)])^2 = \left(\mu_x(t)\right)^2 = \mu_x^2 =$$

$$\frac{1}{2\pi}\int_{0^-}^{0^+} S_{xx}(\omega) d\omega$$

$$\int_{n^-}^{0-} S_{xx}(f) df$$

$S_{xx}(\omega)$ contains impulse at $\omega = 0$ or f = 0, then DC power or mean of random process will be non-zero.

AC power:

$$\sigma_x^2(t) = \sigma_x^2 = \frac{1}{2\pi}\int_{-\infty}^{\sigma} S_{xx}(\omega) d\omega + \frac{1}{2\pi}\int_{0^+}^{\infty} S_{xx}(\omega) d\omega$$

$$= 2 \times \frac{1}{2\pi}\int_{0^+}^{\infty} S_{xx}(\omega) d\omega$$

\int_{-

$$\infty 0^- S_{xx}(f) df + \int_{0^+}^{\infty} S_{xx}(f) df = 2 \times \int_{0^+}^{\infty} S_{xx}(\omega) d\omega$$

Properties of PSD:

When X(t) is real and stationary; X(t) is a power signal then

$$E[X(t)X(t+\tau)] = R_{xx}(\tau)$$

1. $R_{xx}(\tau) \xleftrightarrow{F.T} S_{xx}(\omega)$ or $S_{xx}(f)$
2. $X(t)$ is real $\Rightarrow$ PSD is always even i.e.,
   $S_{xx}(\omega) = S_{xx}(-\omega)$
3. PSD is non-negative quantity
4. As long as X(t) is real, PSD will also be real.

Graph of ACF and PSD should follow the properties then they are valid.

$$ACF(\tau) \rightarrow PSD$$

Energy spectral density of bandpass signal:

Bandpass signals are also known as modulated signal. If X(t) is a baseband signal and X(f) is its Fourier transform

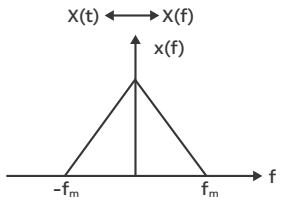


Fig. 4.23(A) Baseband Signal in Frequency Domain

Now

$$X(t)\cos X(t)\cos\omega_c t \longleftrightarrow \frac{X(f+f_c)+X(f-f_c)}{2}$$

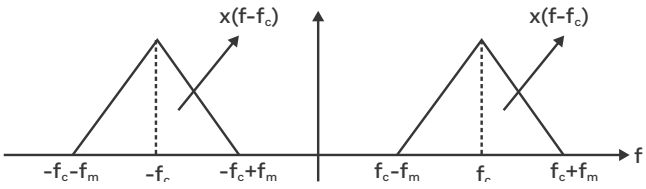


Fig. 4.23(B) Bandpass Signal is Frequency Domain

$X(f+f_e)$ does not overlap with $X(f-f_c)$

Baseband signal:

$$X(t) \leftrightarrow X(t) \rightarrow G_x(t)$$

Bandpass signal:

$$Y(t) = X(t)\cos\omega_c t \leftrightarrow Y(f) \rightarrow G_Y(f) = |Y(f)|^2 \ (ESD)$$

$$Y(t) = \left[\frac{X(f+f_c)+X(f-f_c)}{2}\right]$$

$$|Y(t)| = \left[\frac{A}{2}\left(X(f+f_c)+X(f-f_c)\right)\right]$$

Now,

$$G_Y(f) = |Y(f)|^2$$

$$G_Y(f) = \left|\frac{A}{2}\left\{X(f-f_c)+X(f+f_c)\right\}\right|^2$$

$$= \frac{A^2}{4}\left|X(f+f_c)+X(f+f_c)\right|^2$$

$$G_Y(f) = \frac{A^2}{4}\left\{\begin{array}{l} |f-f_c|^2 + \left|X(f+f_c)\right|^2 + X(f-f_c) \\ X^*(f+f_c) + X^*(f-f_c)X(f+f_c) \end{array}\right.$$

(As the graph doesn't overlap)

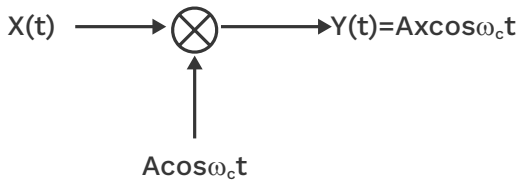
$$G_Y(f) = \frac{A^2}{4}\left\{\left|X(f-f_c)\right|^2 + \left|X(f+f_c)\right|^2\right\}$$

As $G_Y(f) = |X(f)|^2$

$$G_Y(f-f_c) = \left|X(f-f_c)\right|^2$$

$$G_Y(f+f_c) = \left|X(f+f_c)\right|^2$$

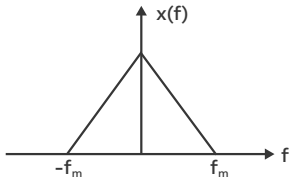
So, $G_y(f) = \frac{A^2}{4}\left\{G_x(f-f_c)+G_x(f+f_c)\right\}$



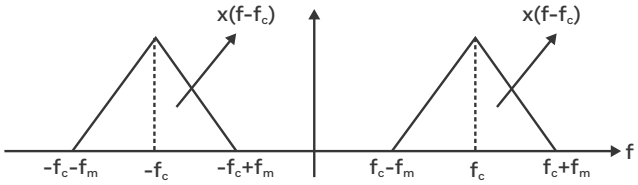
$$Y(t) = AX(t)\cos\omega_c t \overset{ESD}{\Rightarrow} \frac{A^2}{4}\left[G_x X(f-f_c)+G_x(f+f_c)\right]$$

Power Spectral Density of Bandpass Signal

From above we can directly say,



$$X(t)\cos\omega_c t \longleftrightarrow \frac{X(f-f_c)+X(f+f_c)}{2}$$



$X(f+f_c)$ and $X(f-f_c)$ does not overlap

$X(t)$ is a power signal, we will convert it to energy signal $X_T(t)$ to find PSD.

The energy signal XT(t)

$$\lim_{T\to\infty} X_T(t) = X_T(t) \to S_x(f) = \lim_{T\to\infty}\frac{1}{T}\left|X_T(t)\right|^2$$

We know that

$Y(t) = AX(t)\cos\omega_c t$ is the bandpass signal.

So,

$$Y_T(t) \longleftrightarrow Y_T(f) = \frac{A}{2}\left[X_T(f-f_c)+X_T(f+f_c)\right]$$

$$Y_T(t) = AX_T(t)\cos\omega_c t \xleftrightarrow{ESD} \left|Y_T(f)\right|^2$$

$$= \frac{A^2}{4}\left[X_T(f-f_c)+X_T(f+f_c)\right]^2$$

Now,

$$\lim_{T\to\infty} Y_T(t)Y(t) \xleftrightarrow{ESD} \lim_{T\to\infty}\frac{1}{T}\left|Y_T(f)\right|^2$$

$$= \lim_{T\to\infty}\frac{1}{T}\begin{bmatrix}\frac{A^2}{4}\left|x_T(f-f_c)\right|^2+\lim_{T\to\infty}\frac{1}{T}\frac{A^2}{4}\left|x_T(f+f_c)\right|^2\\ +\lim_{T\to\infty}\frac{A^2}{4T}x_T(f-f_c)\,x_T^{-x}(f+f_c)\\ +\lim_{T\to\infty}\frac{A^2}{4T}x^*(f-f_c)\,x_T(f+f_c)\end{bmatrix}$$

The last two terms will be zero as the conjugate of X(t) and X(t) are for away in the spectrum

$$Y_T(t) \xleftrightarrow{PSD} \frac{A^2}{4}\left[\lim_{T\to\infty}\frac{\left|X_T(f-f_c)\right|^2}{T}+\lim_{T\to\infty}\frac{\left|X_T(f+f_c)\right|^2}{T}\right]$$

We know that

$$S_x(f) = \lim_{T\to\infty}\frac{\left|X_T(f)\right|^2}{T}$$

So,

$$S_x(f-f_c) = \lim_{T\to\infty}\frac{\left|X_T(f-f_c)\right|^2}{T}$$

And

$$S_x(f+f_c) = \lim_{T\to\infty}\frac{\left|x_T(f+f_c)\right|^2}{T}$$

So, PSD of Y(t) is

$$Y_T(t) \xleftrightarrow{PSD} \frac{A^2}{4}\left[S_x(f-f_c)+\frac{A^2}{4}S_x(f+f_c)\right]$$

$$X(t) \xrightarrow{PSD} S_x(f)$$

$$Y_T(t) \to \otimes \to Y(t) \xrightarrow{PSD} \frac{A^2}{4}\left[S_x(f-f_c)+S_x(f+f_c)\right]$$

$A\cos\omega_c t$
or
$A\sin\omega_c t$

Chapter Summary

Baseband: X(t): Energy signal $G_X(f)$

$$Y(t) = AX(t)\cos\omega_c t \xleftrightarrow{ESD} \frac{A^2}{4}\begin{bmatrix}G_x(f+f_c)+\\ G_x(f-f_c)\end{bmatrix}$$

Bandpass: Or $AX(t)\cos\omega_c t$

And,

Baseband: X(t): Power signal $\xleftrightarrow{PSD} S_x(f)$

Bandpass: $-Y(t) = AX(t)\cos\omega_c t \xleftrightarrow{PSD} S_Y(f)$

$$AX(t)\sin\omega_c t = \frac{A^2}{4}\left[S_x(f-f_c)+S_x(f+f_c)\right]$$

Previous Years' Question

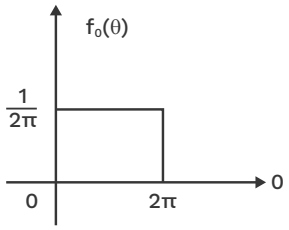
13. Consider a white Gaussian noise process N(t) with two-sided power spectral density $S_N(F) = 0.5 W/Hz$ as input to a filter with impulse response $0.5e^{-t^2/2}$ (where 't' is in seconds) resulting in output Y(t). The power inY(t) in watts is

(A) 0.11 (B) 0.22

(C) 0.33 (D) 0.44

Mixing of a Random Process with Sinusoidal Process

Let X(t) is WSSRP and the process is sinusoidal random process $A\cos(w_c t + \theta)$, where θ is the random variable, uniformly distributed between 0 to 2π.



Random variable, θ and random process X(t) are derived from different sources so they will always be independent.

Now,

$$X(t) \rightarrow \otimes \rightarrow Y(t)$$

$$A\cos(\omega_c t + \theta)$$

$$Y(t) = AX(t)\cos(\omega_c t + \theta)$$

$$R_{rr}(\tau) = E[Y(t)Y(t+\tau)]E\begin{bmatrix} AX(t)\cos(\omega_c t + \theta) \\ AX(t+\tau) \\ \cos(\omega_c t + \omega_c \tau + \theta) \end{bmatrix}$$

$$R_{ry}(\tau) = A^2 E\begin{bmatrix} X(t)X(t+\tau)\cos(\omega_c t + \theta) \\ \cos(\omega_c t + \omega\tau + \theta) \end{bmatrix}$$

Since X and θ are independent

$$R_{\gamma y}(\tau) = A^2 E[X(t)X(t+\tau)]E\begin{bmatrix} \cos(\omega_c t + \theta) \\ \cos(\omega_c t + \omega_c \tau + \theta) \end{bmatrix}$$

$$R_{ry}(\tau) = \frac{A^2}{2} R_{xx}(\tau) E\begin{bmatrix} 2\cos(2\omega_c t + \omega_c \tau + 2\theta) \\ +\cos\omega_c \tau \end{bmatrix}$$

$$R_{ry}(\tau) = \frac{A^2}{2} R_{yx}(\tau) E\begin{bmatrix} \cos(2\theta + 2\omega_c t + \omega_c \tau +) \\ +\cos\omega_c \tau \end{bmatrix}$$

$$R_{ry}(\tau) = \frac{A^2}{2} R_{xx}(\tau)\begin{Bmatrix} E[\cos(2\theta + 2\omega_c t + \omega_c \tau)] \\ +E[\cos\omega_c \tau] \end{Bmatrix}$$

Therefore,

$$R_{ry}(\tau) = \frac{A^2}{2} R_{xx}(\tau)\cos\omega_c \tau \text{ or } \frac{A^2}{2} R_{xx}(\tau)\cos^2 \pi f_c \tau$$

That is, ACF of Y(t)

So,

$$X(t) \xleftrightarrow{ACF} R_{xXX}(\tau) \xleftrightarrow{PSD} S_{xX}(f)$$

And

$$R_{yY}(\tau) = \frac{A^2}{2} R_{xx}(\tau)\cos 2\pi f_c \tau \xleftrightarrow{F.T}$$

$$PSD\,\frac{A^2}{4}\left[S_{xx}(f + f_c) + S_{xx}(f - f_c)\right]$$

Gross power spectral density:

If we have two random processes X(t) and Y(t) then,

$$R_{XY}(\tau) = E[X(t)]Y(t+\tau) \xleftrightarrow{F.T} S_{XY}(f)$$

Here $S_{xy}(f)$ is the cross spectral density

$$S_{XYY}(f) = \int_{-\infty}^{\infty} R_{XY}(\tau)e^{-j2\pi f} = d\tau$$

$$R_{YX}(\tau) = E[X(t)Y(t+\tau)] \xleftrightarrow{F.T} S_{YX}(f)$$

$$S_{YXX}(f) = \int_{-\infty}^{\infty} R_{YXX}(\tau)e^{-j2\pi f_c} d\tau$$

If X(t) and Y(t) will be real then $E[X(t)y(t+\tau)] = R_{XY}(\tau)$ and $E[Y(t)X(t+\tau)] = R_{XY}(\tau)$ are real.

If $R_{xy}(\tau)$ is real, then $R_{xY}(\tau) = R^{*}_{XY}(\tau)$.

Also $R_{XY}(\tau) = R_{XY}(-\tau)$

If we take their Fourier transform

$R_{XY}(\tau) = R^*_{XY}(\tau) \setminus S_{XY}(f) = S^*_{XY}(-f)$

$R_{XY}(\tau) = R_{XY}(-\tau) \leftrightarrow S_{XY}(f) = S_{YX}(-f)$

Also $S_{XY}(f) = S_{YX}(-f)$

Therefore, $S_{xY}(f) = S^*_{yx}(f)$

PSD of combination of two WSSRP

Let $Z(t) = X(t) + Y(t)$

And

$$E[Z(t)Z(t+\tau)] = R_{zz}(\tau)$$
$$= E[(X(t) + Y(t)(X(t+\tau) + Y(t+\tau))]$$

$$R_{zz}(\tau) = E[(X(t)X(t+\tau) + X(t)Y(t+\tau) + Y(t)X(t+\tau) + Y(t)Y(t+\tau)]$$

$$R_{zz}(\tau) = E[(X(t)X(t+\tau)] + E[X(t)Y(t+\tau)] + E[Y(t)X(t+\tau)] + E[Y(t)Y(t+\tau)]$$

$R_{zz}(\tau) = R_{zz}(\tau) + R_{zY}(\tau) + R_{YX}(\tau) + R_{ry}(\tau)$ or

$R_{zz}(\tau) = R_{xX}(\tau) + R_{\gamma Y}(\tau) + R_{XYY}(\tau) + R_{YX}(\tau)$

Now taking Fourier transform on both sides

$S_{zz}(f) = S_{xx}(f) + S_{ry}(f) + S_{yx}(f) + S_{yx}(f)$

Case 1. Assume X(t) and Y(t) are orthogonal signals

$R_{XY}(\tau) = R_{YX}(\tau) = 0$

So, $R_{zz}(\tau) = R_{xx}(\tau) + R_{ry}(\tau)$

After FT $S_{zz}(f) = S_{xx}(f) + S_{\gamma y}(f)$

Case 2. If X(t) and Y(t) are uncorrelated and one of them have zero mean. So,

$$\text{Cov}(x(t)Y(t+\tau)) = 0$$

$$\text{Cov}(x(t)Y(t+\tau)) = E[X(t)Y(t+\tau)] - E[X(t)E[Y(t+\tau)]]$$

$$\Rightarrow 0 = R_{XYY}(\tau) - \mu_x\mu_Y$$

$$\rightarrow R_{XYY}(\tau) = 0 \left(\text{ when } \mu_x = 0 \right)$$

$$\text{Cov}(x(t)X(t+\tau)) = E[Y(t)X(t+\tau)] - E[Y(t)E[X(t+\tau)]]$$

$$\Rightarrow 0 = R_{YX}(\tau) - \mu_Y\mu_x$$

$$\Rightarrow R_{YX}(\tau) = 0 \left(\text{ when } \mu_Y = 0 \right)$$

Therefore,

$R_{zz}(\tau) = R_{xx}(\tau) + R_{YY}(\tau)$

and its Fourier transform

$S_{zz}(f) = S_{xx}(f) + S_{yY}(f)$

Case 3. If X(t) and Y(t) are independent and anyone them has zero mean, X(t) and Y(t) will be uncorrelated

Then,

$R_{zz}(\tau) = R_{xx}(\tau) + R_{yY}(\tau)$

And

$S_{zz}(f) = S_{xx}(f) + S_{YY}(f)$

Similarly, we can extend to

$Z(t) = X_1(t) + X_2(t) + \ldots.. + X_k(t)$

Here are 'k' random processes and if these are mutually orthogonal or mutually uncorrelated (and all of them having zero mean) or mutually independent (and all of them having zero mean) with each other then

$R_{zz}(\tau) = R_{x1}(\tau) + R_{x2}(\tau) + \ldots. + R_{xk}(\tau)$

And

$S_{zz}(f) = S_{x1}(f) + S_{x2}(f) + \ldots. + S_{xk}(f)$

Transmission of WSSRP through LTI System

If x(t), a stationary process is passed through an LTI system

$x(t) \rightarrow \boxed{System} \rightarrow y(t) = Ax^2(t)$ (not WSSRP)

1. Non-linear system

$x(t) \rightarrow \boxed{System} \rightarrow y(t) = Ax(t) + b$ (not WSSRP)

2. Non-linear system

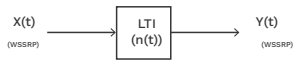
$x(t) \rightarrow \boxed{System} \rightarrow y(t) = a\,x(t) + b(WSSRP)$

3. Linear system

x(t) Passed through System to Get Output y(t)

When X(t) is passed through a system, whenever the system is linear, every time a WSSRP input will generate a WSSRP.

[When system is non-linear, it may or may not generate WSSRP as output]



This is deterministic or real system

$$n(t) \xrightarrow{F.T.} H(f) \quad (WSSRP)$$

$$H(f) = \int_{-\infty}^{\infty} n(t) e^{-j2\pi ft} dt$$

$$H(0) = \int_{-\infty}^{\infty} n(t) dt$$

As the system in above figure is deterministic as well as real, $h(t) = h^*(t)$ and $h(t) = h^*(-t)$

X(t): WSSRP input

$$E[X(t)] = \mu_x$$

$$E[X(t)X(\tau)] = R_{xx}(\tau)$$

For LTI system: $Y(t) = X(t)^* h(t)$

$Y(f) = \int_{-\infty}^{\infty} x(t-\alpha)h(x)d\alpha$ mean of output $Y(f) : E[Y(f)] = E\left[\int_{-\infty}^{\infty} x(t-\alpha)h(x)d\alpha\right]$, here $X(t-\alpha)$ is a random process and $h(\alpha)$ is deterministic.

$$= \int_{-\infty}^{\infty} E[X(t-\alpha)]h(x)d\alpha$$

$$E[X(t)] = \mu_x = E[X(t-\alpha)]$$

$$E[Y(t)] = \int_{-\infty}^{\infty} \mu_x h(\alpha)d\alpha$$

$$= \mu_x \int_{-\infty}^{\infty} h(\alpha)d\alpha$$

$$E[Y(t)] = \mu_x \int_{-\infty}^{\infty} h(t)dt$$

Here both μ_x and $\int_{-\infty}^{\infty} h(t)dt$ are constant. So, $E[Y(t)]$ is constant.

$$E[Y(t)] = \mu_Y = \mu_X \int_{-\infty}^{\infty} h(t)dt$$

$$E[Y(t)] = \mu_Y = \mu_x H(0) \left[H(\omega) \text{ or } H(f) = \int_{-\infty}^{\infty} h(t) \right.$$

when ω or $f = 0$]

$$WSSRP: X(t) \rightarrow \boxed{h(t) \leftrightarrow: H(f)} Y(t) : WSSRP$$

$$E[Y(t)] = \mu_Y$$

$$E[Y(t)] = \mu_x \int^{\infty} h(t)dt = \mu_Y H(0)$$

Therefore, $\mu_Y = \mu_x H(0)$

If X(t) has zero mean, i.e., $\mu_Y = \mu_X H(0) = 0 = 0$

So, in above figure, X(t) and Y(t) is a zero mean WSSRP. Similarly, for auto correlation function

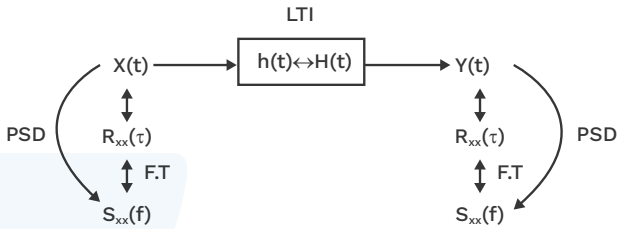


Fig. 4.26 Auto Correlation and PSD of Input and Output of LTI System

Observations

1. $\mu_y = \mu_x H(0)$
2. $R_{ry}(\tau) = R_{xx}(\tau) * h(\tau); S_{yx}(f) = S_{xx}(f)H(f)$
3. $R_{ry}(\tau) = R_{xx}(\tau) * h(-\tau); S_{rx}(f) = S_{xx}(f)H(-f)$
4. $R_{ry}(\tau) = R_{xx}(\tau) * h(\tau) * h(-\tau)$

   $R_{YY}(\tau) = R_{Xx}(\tau) * [h(\tau) * h(-\tau)]$

$$P_{Y(t)} = E\left[Y^2(t)\right] R_{YY}(0)$$

$$= \left[R_{Xx}(\tau) * \int_{-\infty}^{\infty} h(\alpha - \tau) * h(\alpha)d\alpha\right]$$

$\tau = 0$

$$R_{YY}(\tau) = R_{Xx}(\tau) * [h(\tau) * h(-\tau)]$$

Apply Fourier Transform on Both Sides Then

$$S_{YY}(f) = S_{xx}(f)H(f)H(-f)$$

$S_{yy}(f)$ is the PSD of output

We know that h(t) is real: $h(t) = h^*(t)$

$$H(f) = H^*(-f)$$

$$H(-f) = H^*(f)$$

$$S_{YY}(f) = S_{xx}(f)H(f)H^*(f)$$

$$S_{YY}(f) = |H(f)|^2 S_{xx}(f)$$

This is the PSD of the output random process.

$$E\left[X^2(t)\right] = P_x(t) = \left\{\int_{-\infty}^{R_{xx}}(0)\right.$$

$$\Rightarrow E\left[X^2(t)\right] = R_{xx}(0) = \int_{-\infty}^{\infty} S_{xx}(f)df$$

$$\Rightarrow E\left[Y^2(t)\right] = R_{xx}(0) = \left[R_{xx}(\tau) * \int_{-\infty}^{\infty} h(\alpha)h(\alpha-\tau)d\alpha\right]_{\tau=0}$$

$$= \int_{-\infty}^{\infty} S_{YY}(f)df = \int_{-\infty}^{\infty} |H(f)|^2 S_{xx}(f)df$$

Gaussian Random Process

A random process X(t) is a Gaussian random process if the $\left\{X(t_k)\right\}_{k=1}^{l}$ has joint Gaussian PDF.

When $k = 1 \Rightarrow \underset{[GRP]}{X(t)} \rightarrow \underset{[GRP]}{X(t_1)} = Z$

$$f_z(z) = \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{\frac{(z-\mu_z)^2}{2\pi\sigma_z^2}}$$

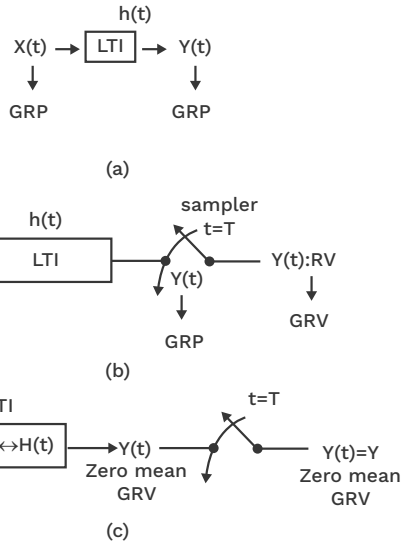
When k = 2

$$X(t) \rightarrow X(t_1)$$

Here, $f_{x(t_1)x(t_2)}(x_1, x_2) \rightarrow$ jointly Gaussian 2nd order PDF

Properties of Gaussian random process:

1. If a Gaussian RP is given, it may or may not be stationary.
2. If a Gaussian RP is WSSRP then it will also become SSSRP.
3. If any two samples of Gaussian random process are uncorrelated, they will always become independent.
4. If a Gaussian RP is passed through an LTI system, the output random process will also be Gaussian random process.



Example 34:

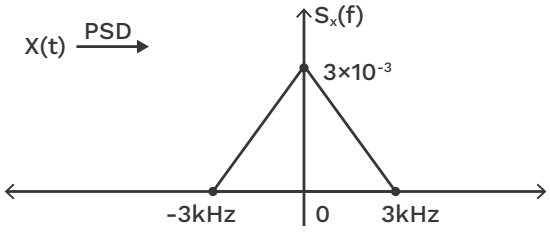
A real band-limited random process X(t) has two-sided power spectral density

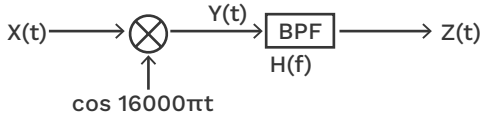
$$S_x(f) = \begin{cases} 10^{-6}(3000 - |f|) \text{ watt / Hz for } |f| \le 3\text{kHz} \\ 0 \qquad\qquad\qquad\qquad otherwise \end{cases}$$

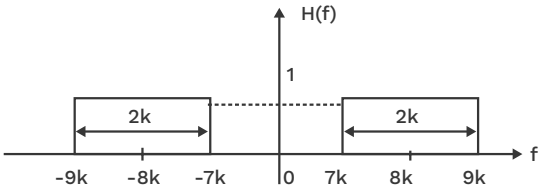
Where f is the frequency expressed in Hz. The signal X(t) modulates a carrier $\cos 16000\pi t$ and the resultant signal is passed through an ideal band pass filter of unity gain with centre frequency of 8 kHz and bandwidth of 2 kHz. The output power (in watts) is _____.

Solution:

Given, PSD of X(t) is



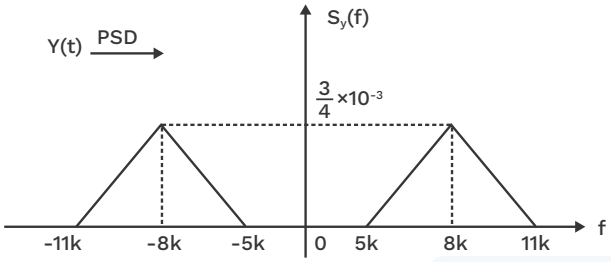




$$Y(t) = X(t)\cos 16000\pi t$$

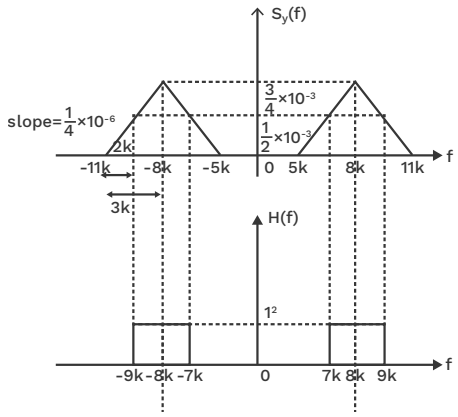
$$S_Y(f) = \left\{\frac{S_X(f+8k) + S_X(f-8k)}{4}\right\}$$

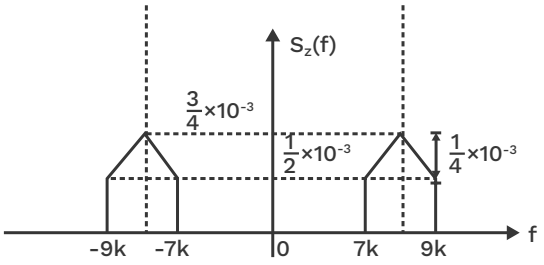
Now, PSD of Y(t) is



$$Y(t) \rightarrow \boxed{H(f)} \rightarrow z(t)$$

$$S_z(f) = |H(f)|^2 \; S_Y(f)$$





$$P_{z(t)} = \int_{-\infty}^{\infty} S_z(f)df = 2\int_0^{\infty} S_z(f)df$$

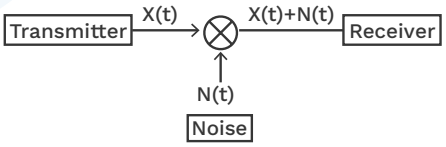
$$= 2 \times (\text{ Area of rectangle+Area of triangle})$$

$$= 2\left[\frac{1}{2} \times \frac{1}{4} \times 10^{-3} \times 2 \times 10^{+3} + \frac{1}{2} \times 10^{-3} \times 2 \times 10^{+3}\right]$$

$$= 2\left[\frac{1}{4} \times 1\right] = 2(1.25) = 2.5 \text{ watt}$$

Noise

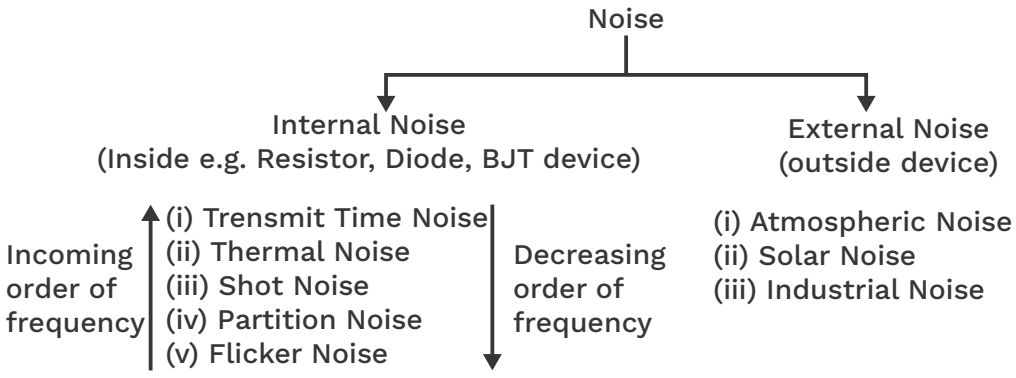
Any unwanted signal which disturbs the transmission and reception processing of signal in the communication system.



Here in above figure, the noise $N(t)$ is an additive noise.

N(t) → Noise is a 'random signal' or N(t) is a random process.

As we compute $N(t)$ its statistical parameter and not its exact variation with time.



Shot Noise

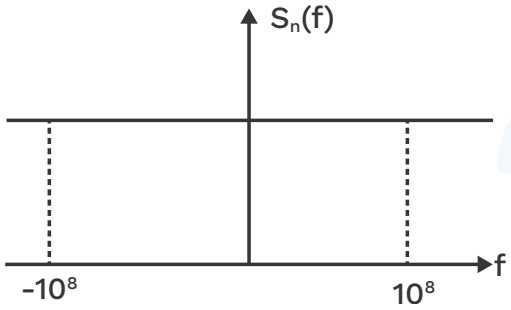
To analyse shot noise mathematically, we can model it as random process. Shot noise arises in electronic devices due to discrete nature of current flow.

In solid state devices, random diffusion of charge carriers, and random generation and recombination of charge carrier active devices and due to the fact that electric current is not continuous, but discrete flow of electrons.

In semiconductor devices, shot noise is due to the carriers and random regeneration and recombination of electron hole pairs.

Properties of shot noise

Shot noise can be mathematically modelled as a random process. It is a zero-mean Gaussian RP. The PSD of shot noise is constant up to 100 MHz. Let shot noise be n(t).



Thermal Noise

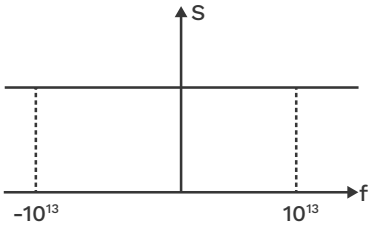
Thermal noise is also known as Johnson noise or resistor noise. In any object with electrical resistance, the thermal fluctuations of the electrons in the object will generate noise due to the random motion of free charge particles (electrons) in the conducting media.

This agitation of free electrons moving randomly gives rise to noise voltage that appears between the resistor terminals.

Properties of thermal noise:

1. Thermal noise as random process.
2. Thermal noise is also zero mean Gaussian random process.
3. PSD of thermal noise is constant up to 10^{13} Hz.

   If n(t): Thermal noise



4. This frequency range 10^{13} Hz covers almost all the practical applications.
5. PSD of thermal noise is independent of frequency.

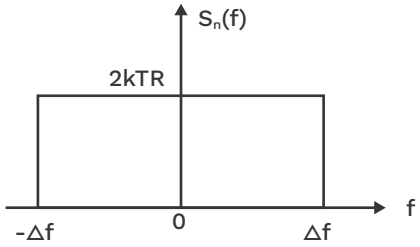
PSD of thermal noise:

$$S_{\gamma}(f) = 2kTR$$

k: Boltzmann constant $= 1.38 \times 10^{-23}$ J / K

T: Ambient temperature in Kelvin

R: Resistor value



$$P_{n(t)} = \int_{kf}^{\Delta f} S_n(f)df = (2kTR)(2\Delta f) = 4kTR\Delta f$$

Δt is considered as the bandwidth over which noise power is being measured.

$$\text{MSV}\left(V_{n(t)}\right) = V^2(t) = 4kTR\Delta f$$

$$V_{n(t)} = rms\left(V_{n(t)}\right) = \sqrt{4kTR(\Delta f)}$$

Shot noise + Thermal noise = White noise

White light contains all possible light. Similarly, PSD of white noise is constant and covering almost all practical frequency.

White noise:

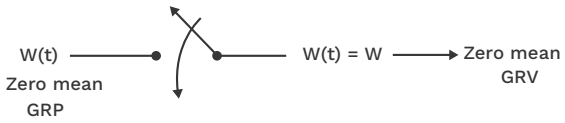
White noise is a random process denoted as W(t). It is a zero mean Gaussian random process.

E[W(t)] = 0

Shot noise and thermal noise are examples of white noise.

$$E\left[N^2(t)\right] = \sigma^2_{W(t)} + (E[W(t)])^2$$

$$\text{MSV}(W(t)) = \text{Var}(W(t)) = \text{Power of white noise}$$



$$f_w(w) = \frac{1}{\sqrt{2\pi\sigma_w^2}} e^{\frac{-w^2}{2\pi\sigma_w^2}}$$

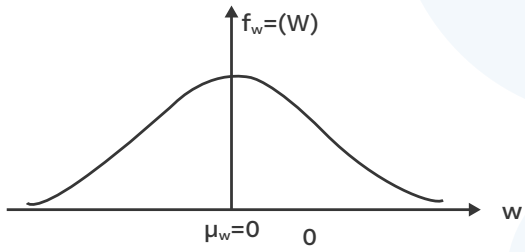


Figure: PDF white noise

White noise is also known as thermal noise, Gaussian noise, Johnson noise, resistor noise.

The PSD of white noise is flat or PSD is independent of frequency.

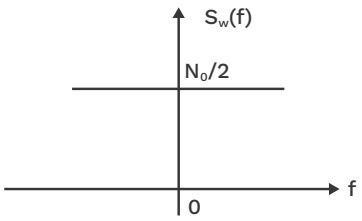


Figure: PDF white noise

$$S_w(f) = \frac{N_0}{2}\frac{(watt)}{Hz} \quad -\infty < f < \infty$$

$N_0 / 2$ or $\eta / 2$ is the PSD of white noise.

From above figure, we can calculate the DC and AC Power.

$$(E[W(t)])^2 = \int_{0^-}^{0^+} S_w(f)df = 0 \Rightarrow E[W(t)] = 0$$

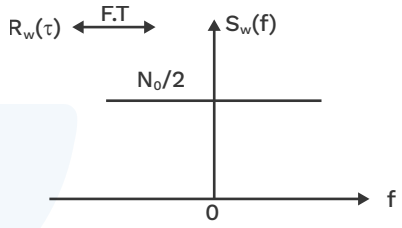
$$(E[W(t)])^2 = 2\int_{0^-}^{0^+} S_w(f)df = \infty$$

$$\therefore \sigma^2_{w(t)} = \infty$$

From Weiner Khinchin Theorem

$$R_W(\tau) \xleftrightarrow{F.T.} S_w(f)$$

(ACF of white Noise) (PSD of white Noise)

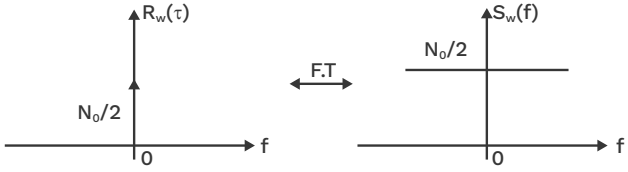


$$x(t) = \delta(\tau) \longleftrightarrow 1 = X(f)$$

$$x(t) = \frac{N_0}{2}\delta(\tau) \longleftrightarrow \frac{N_0}{2} = X(f)$$

$$W(t) \longleftrightarrow W \longleftrightarrow W(t)W(t+\tau)] = R_w(\tau)$$

$$= \frac{N_0}{2}\delta(\tau)$$



$$E\left[W^2(t)\right] = R_w(0) = 2\int_0^\infty S_w(f)$$

Which we found as infinite.

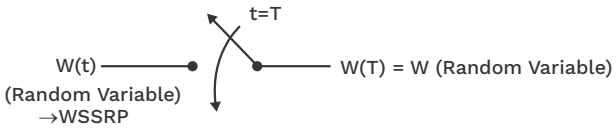
So, power inside white noise is infinite

W(t): Zero mean GRP

E[W(t)] = 0 = Constant

$$\Rightarrow R_w(\tau) = \frac{N_0}{2}(\tau)$$

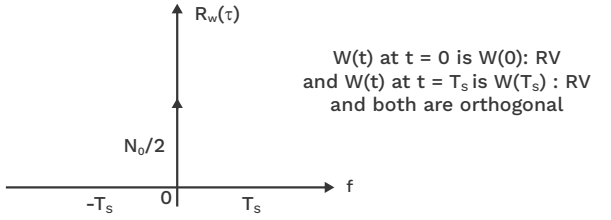
So, this random process is also WSSRP.



So,

$$E[W(t)] = E[W] = 0; E\left[W^2(t)\right] = E\left[W^2\right] = 8$$

$$W(t) \to E[W(t)W(t+\tau)] = R_W(\tau) \xleftrightarrow{F.T.} S_W(f)$$



$$E\left[W(0)W\left(T_s\right)\right] = R\left(T_s\right) = 0 \quad (E[XY]) = R_{XY} = 0$$

Any two samples of GRP are always orthogonal and $E[W(t)] = E[W(0)] = E\left[W\left(T_s\right)\right] = 0$

So,

$$\text{Cov}\left(W(0)W\left(T_s\right) = E\left[W(0)W\left(T_s\right)\right] - E[W(0)]EW\left[\left(T_s\right)\right]\right.$$

$$\text{Cov}\left(W(0)W\left(T_s\right) = R_w\left(T_s\right) - 0\right.$$

$$\Rightarrow \text{Cov}\left(W(0)W\left(T_s\right) = 0\right. \quad \text{(As two GRP are orthogonal)}$$

Any two sample of white noise will always be uncorrelated.

Since, W(t) is a GRP if uncorrelated then independent also.

Transmission of white noise through LTI system

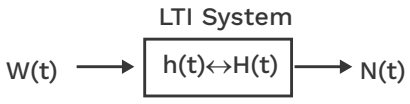


Figure: White Noise passed through an LTI System

When white noise is passed through LTI system, the output is coloured noise. When zero mean GRP is passed through LTI system it gives zero mean GRP. Similarly, zero mean LOSS RP gives zero mean LOSS RP.

$$\therefore E[W(t)] = 0 \underset{\text{System}}{LT} E[N(t)] = 0$$

And

$$E\left[W^2(t)\right] = \sigma^2_{W(t)} 0 \overset{LT1}{\underset{\text{System}}{\longrightarrow}} E[N(t)] = \sigma^2_{N(t)}$$

For above figure, the input has ACF as $R_w(\tau)$ and transforming, we get $S_w(f)$, i.e., PSD of $W(t)$ and similarly for the output $N(t)$, we have the ACF as $R_N(\tau)$ and transforming it, we get $S_N(f)$, i.e., PSD of $N(t)$.

$$E[N(t)N(t+\tau)] = R_N(\tau) = \left[R_w(\tau) * \int_{-\infty}^{\infty} h(\tau-\alpha)d\alpha\right]$$

$$= R_N(\tau) = \frac{N_0}{2}\delta(\tau) \star \left[\int_{-\infty}^{\infty} h(\alpha)(\tau-\alpha)d\alpha\right]$$

$$= \mathrm{R_N}(\tau) = \frac{\mathrm{N_0}}{2}\int_{-\infty}^{\infty} h(\alpha)(\tau-\alpha)d\alpha$$

$$E\left[N^2(t)\right] = R_N(0) = \frac{N_0}{2}\int_{-\infty}^{\infty} h(\alpha)d\alpha$$

$$h(t): \int_{-\infty}^{\infty} h^2(\alpha)d\alpha = \text{ Energy of } h(t)$$

$$P_{N(t)} = E\left[N^2(t)\right] = \sigma^2_{N(t)} = R_N(0) = \frac{N_0}{2}\int_{-\infty}^{\infty} h(\alpha)d\alpha$$

$$= \frac{N_0}{2} \times \text{ Energy of } h(t)$$

$$(PSD)_{0/p} = |H(f)|^2 \ (PSD)_{i/P}$$

$$S_N(f) = |H(f)|^2 \ S_w(f)$$

$$= \frac{N_0}{2} \times \text{ Energy of } h(t)$$

$$(PSD)_{0/p} = |H(f)|^2 \ (PSD)_{i/p}$$

$$S_N(f) = |H(f)|^2 \ S_w(f)$$

$$E\left[N^2(t)\right] = P_{N(t)} = \sigma^2_{N(t)} = \int_{-\infty}^{\infty} S_N(f)df$$

$$= \int_{-\infty}^{\infty} |H(f)|^2 \ S_w(f)df$$

$$= \int_{-\infty}^{\infty} \frac{N_0}{2} |H(f)|^2 \ df$$

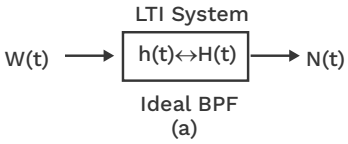
$$E\left[N^2(t)\right] = P_{N(t)} = \sigma^2_{N(t)} = \frac{N_0}{2}\int_{-\infty}^{\infty} |H(f)|^2 \ df$$

Power of N(t):

$$E\left[N^2(t)\right] = \sigma^2_{N(t)} = \begin{cases} R_{xx}(0) = \frac{N_0}{2}\int_{-\infty}^{\infty} h^2(\alpha)dx = \frac{N_0}{2} \times \text{ Energy of } h(t) \\ \int_{-\infty}^{\infty} S_N(f)df = \frac{N_0}{2}\int_{-\infty}^{\infty} |H(f)|^2\, df = \frac{N_0}{2} \times \text{ Energy of } h(t) \text{(in frequency domain)} \end{cases}$$

Transmission of white noise through ideal LPF:

In this case, the LTI system is the ideal Low pass Filter (LPF)



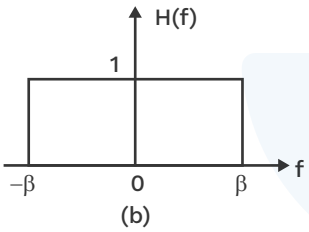


Figure: (a) W(t) through LPF (b) Characteristics of LPF

N(t) = W(t) * h(t)

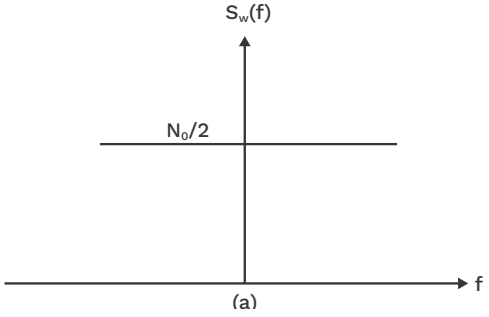
The ACF of input is $R_w(\tau)$ and PSD is $S_w(f)$, while ACF of output is $R_N(\tau)$ and PSD is $S_N(f)$.

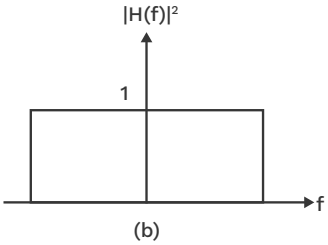
$$S_N(f) = |H(f)|^2\, S_w(f)$$

$$S_w(f) = \frac{N_0}{2} \qquad -\infty < f < \infty$$

$$H(f) = 1 \qquad -\beta < f < \beta$$

$$|H(f)|^2 = 1 \qquad -\beta < f < \beta$$





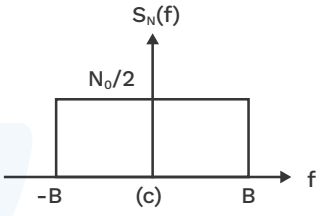
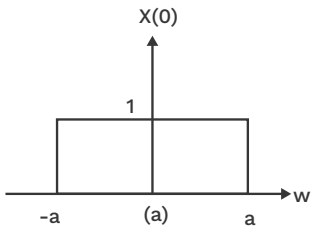


Figure: PSD of output (Graphically)

ACF of N(t): $R_N(\tau) \longrightarrow S_N(f)$

$$\delta(t) \xrightarrow{\frac{\sin at}{\pi t}}$$



$$R_N(\tau) = \frac{N_0}{2}\frac{\sin(2\pi B\tau)}{\pi\tau} \longleftrightarrow$$

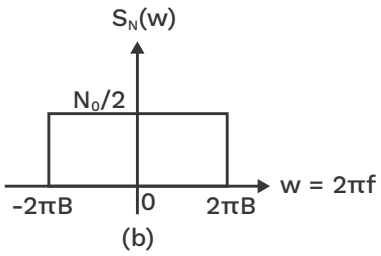
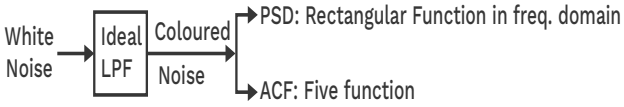


Figure: ACF and PDF of output N(t)



Let 2 B τ = R

$[\text{sinc}(x) : x = k$ where $K \in I$ and $k \neq 0]$

$\tau = \frac{k}{2B} = \frac{1}{2B}, \frac{2}{2B}, \frac{3}{2B} \ldots\ldots\ldots\ldots$

τ : Samples of N(t) random process duration apart. For all these cases $R_N(\tau) = 0$, so samples are orthogonal.

$$E\left[N(O)N\left(\frac{k}{2B}\right)\right] = R_N\left(\frac{k}{2B}\right) = 0$$

$$\text{Cov}\left[N(O)N\left(\frac{k}{2B}\right)\right] = R_N\left(\frac{k}{2B}\right) = 0$$

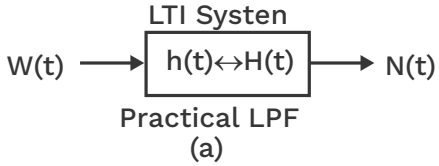
$$\text{Cov}\left[N(O)N\left(\frac{k}{2B}\right)\right] = 0$$

[Since $E[N(0)]$ and $E\left[N\left(\frac{k}{2B}\right)\right] = 0$]

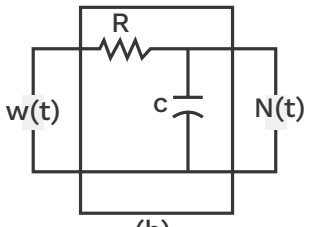
$$\left(\text{sin}\,ceR_N\left(\frac{k}{2B}\right) = 0\right)$$

Samples $\tau = \frac{k}{2B}$ duration apart will be uncorrelated and N(t) is GRP. So, it is dependent also.

Transmission of white noise through practical LPF:



(a)



(b)

$$\Rightarrow H(B) = \frac{1}{HsRC} \quad \Rightarrow H(j\omega) = \frac{1}{1 + j\omega RC}$$

$$\text{or } H(f) = \frac{1}{1 + j2\pi fRC}$$

$$S_N(f) = |H(f)|^2 \; S_N(f)$$

$$S_w(f) = \frac{N_0}{2} \quad -\infty < f < \infty$$

$$H(f) = \frac{1}{1 + j2\pi f}$$

$$|H(f)| = \frac{1}{\sqrt{1 + 4n^2f^2}} \Rightarrow |H(f)|^2 = \frac{1}{1 + 4\pi^2f^2}$$

Therefore,

$$S_N(f) = \frac{N_0 / 2}{1 + 4\pi^2f^2K^2C^2}$$

Taking inverse F.T., we get $R_N(\tau)$

$$\left[As,\ e^{-a|e|}_{(\text{ for } a>0)} \longleftrightarrow \frac{2a}{a^2 + \omega^2} - \frac{2a}{a^4 + 4\pi^2t^2}\right]$$

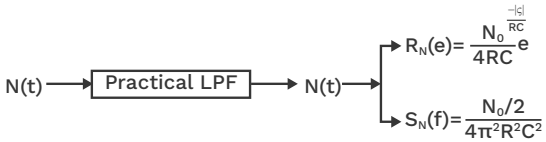
$$R_N(\tau) \longleftrightarrow S_N(f) = \frac{N_0}{2}\left[\frac{1/\,R^2C^2}{\frac{1}{R^2C^2} + 4\pi^2f^2}\right]$$

$$= \frac{N_0}{4RC}\left[\frac{R/\,RC}{\left(\frac{1}{RC}\right)^2 + 4\pi^2f^2}\right]$$

Solving its inverse Fourier transform, we get

$$\frac{N_0}{4RC}e^{-\frac{1}{RC}|\xi|}$$

$$\Rightarrow R_N(\tau) = \frac{N_0}{4RC}e^{-\frac{|\tau|}{RC}}$$



$$R_N(f) = E\left[N^2(t)\right] = \sigma^2_{N(t)} = R_N(0) = \frac{N_0}{4RC}$$

$$\Rightarrow \int_{-\infty}^{\infty} S_N(f)df = \int_{-\infty}^{\infty} \frac{N_0/2}{1+4\pi^2t^2R^2C^2}df = \frac{N_0}{4RC}$$

Narrow band noise:

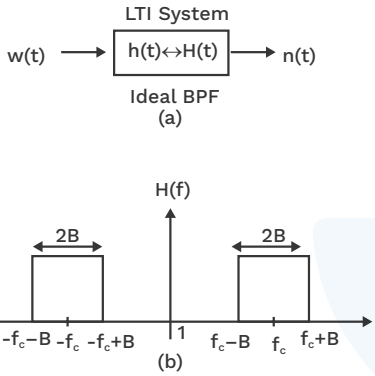
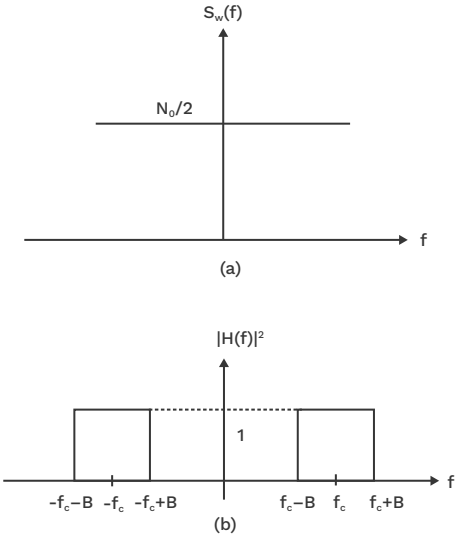
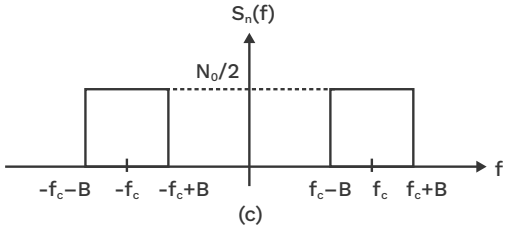


Figure: (a) Noise through ideal BPF
(b) Response of ideal BPF

$$S_N(f) = |H(f)|^2 \, S_w(f)$$

$\Rightarrow S_w(f) = N_0/2$ and characteristics of $H(f)$ is known





$S_n(f)$ represents narrow band spectrum.

Where $E[\theta t] = tE$, $E[2\theta] = 2E$ and $\frac{f_H}{f_L} \approx 1$

As PSD of output noise is narrow band, hence n(t) is narrow band noise or bandpass noise.

PSD of n(t) will have bandpass nature.

Mathematically, a narrow band noise can be represented as

$$n(t) = n_1(t)\cos\omega_c t - n_Q(t)\sin\omega_c t$$

(Based on narrow band signal)

$n_1(t)$: In-phase compound of $n(t)$

$n_Q(t)$: Quadrature phase compound of $n(t)$

$$n_1(t) = n(t)\cos\omega_c t + \hat{n}(t)\sin\omega_c t$$

$$n_Q(t) = \hat{n}(t)\cos\omega_c t - n(t)\sin\omega_c t$$

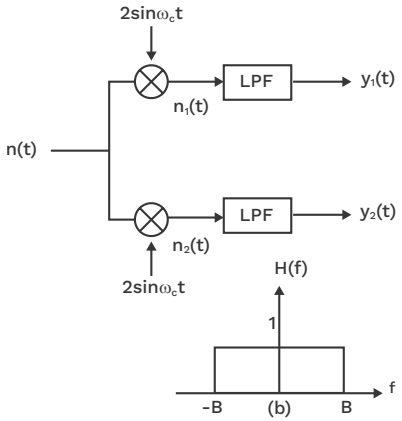


Figure: Block diagram of signal n(t)

$$n_1(t) = 2n(t)\cos\omega_c t = n_1(t)2\cos^2\omega_c t - n_Q(t)\sin 2\omega_c t$$

$$= n_1(t) + n_1(t)\cos 2\omega_c t - n_Q(t)\sin 2\omega_c t$$

When passed through LPF, $n_1(f)$ and $n_Q(t)$ are low pass signals and n(t) is bandpass signal.

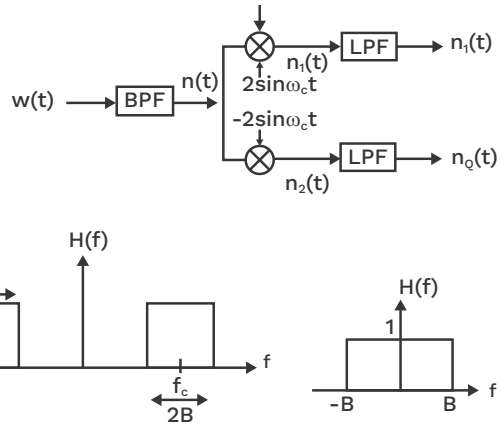


Figure: (a) Output of $n_1(t)$ from LPF

Now, $n(t) = n(t)\left(-2\sin\omega_c t\right)$

$= n_1(t)\sin 2\omega_c t + n_Q(t) - n_Q(t)\cos 2\omega_c t$

$n_2(t) = n_Q(t) - n_1(t)\sin 2\omega_c t - n_Q(t)\cos\omega_c t$

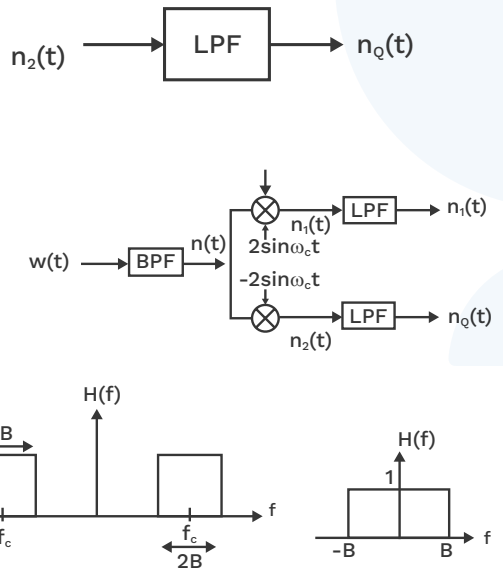


Figure: (a) Output of $n_1(t)$ from LPF

Here, $S_w(f) = \frac{N_0}{2}$

$S_n(f) = \frac{N_0}{2}|H(f)|^2$

$= \frac{a^{\prime} + b^{\prime} + ab}{3}$

$n(t)\left(2\cos\omega_c t\right) \xleftrightarrow{PSD} \frac{(2)^2}{4}\left[S_n\left(f + f_c\right)S_n\left(f - f_c\right)\right]$

$= S_{n_2}(f)$

$n_1(t)\,n_2(t) \overset{PSD}{\left[S_n\left(f + f_c\right) + S_n\left(f - f_c\right)\right]} = S_{n_1}(f)$

$S_{nD}\left[S_n\left(f + f_c\right) + S_n\left(f - f_c\right)\right] = S_{n_2}(f)$

$S_{n_1}(f) = S_n\left(f + f_L\right) + S_n\left(f - f_c\right) - B \le f \le \left|^2 S_1(f)\right.$

$S_{nQ}(f) = |H(f)|^2, S_{n2}(f)$

$S_{nQ}(f) = S_n\left(f + f_c\right) + S_n\left(f - f_c\right) - B \le f \le B$쩔 B

We know,

$n(t) = n_1(t)\cos\omega_c t - n_Q(t)\sin\omega_c t$

$n(t) \leftrightarrow S_n(f) = |H(f)|^2\ S_w(f)$ from BPF

$n_1(t) \xrightarrow{PSD} S_n\left(f + f_c\right) + S_n\left(f - f_c\right) = S_{nI}(f)\,|f| \le B$

$n_Q(t) \xrightarrow{PSD} S_n\left(f + f_c\right) + S_n\left(f - f_c\right) = S_{nQ}(f)\,|f| \le B$

PSD of both in phase and quadrature phase will be same

$$S_{nI}(f) = S_{nQ}(f) = \begin{cases} S_n\left(f - f_c\right) + S_n\left(f - f_c\right) & |f| \le B \\ 0 & elsewhere \end{cases}$$

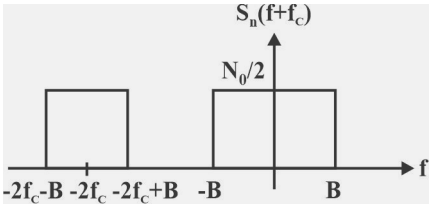
Where B: Band limiting frequency of LPF or half of the BW of BPF.

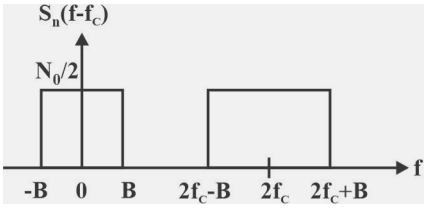
Now, $E[n(t)] = 0$

$E\left[n^2(t)\right] = \sigma n^2_{(t)} = \int_{-\infty}^{\infty} S_n(f)df$

$= \frac{N_0}{2} \times 4B = 2N_0 B$

Spectrum of $n_1(f)$ or $S_{nQ}(f)$





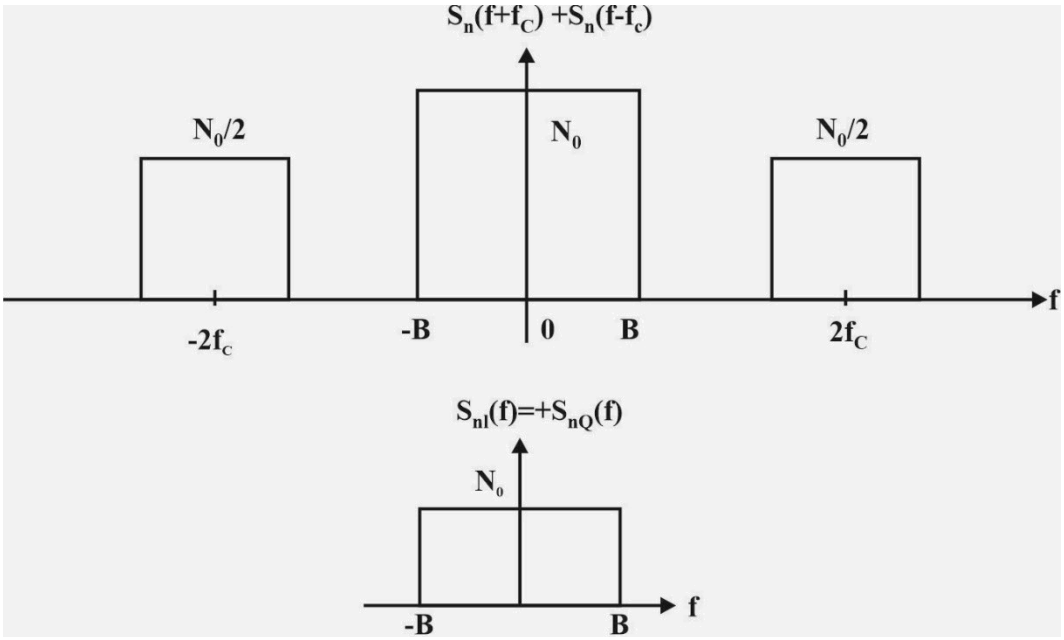


Figure: Spectrum of power beats of noise

$$E\left[n_1^2(t)\right] = E\left[n_Q^2(t) = \int_{-\infty}^{\infty} S_{nl}(f)af\right]$$

$$= N_0(2B)$$

$$= E\left[n^2(t)\right]$$

Properties:

PSD of n(t) will represent bandpass or narrowband spectrum.

$$S_{nl}(f) = S_{nQ}(f) = S_n\left(f + f_c\right) + S_n\left(f - f_c\right) \mid f \mid \leq B$$

PSD of $n_1(t)$ and $n_Q(t)$ will represent low pass or baseband spectrum. Since ω(t) is zero mean, so $n(t), n_1(t)$ and $n_e(t)$ are zero mean.

Since $\omega(f)$ is zero mean GRP then $n(t), n_1(t), n_Q(t)$ are zero mean GRP.

$$E\left[n^2(t)\right] = E\left[n_1^2(t)\right] = E\left[n_Q^2(t)\right]$$

$$\sigma n_{(t)}^2 = \sigma n_{1(f)}^2 = \sigma n_{Q(t)}^2$$

$n(t) = n_1(t)\cos\omega_c t - n_Q(t)\sin\omega_c t$ envelope of $n(t) = \sqrt{n_1^2(t) + n_Q^2(t)}$

(Narrowband noise = Rayleigh distribution)

Exercise

1. Let X(t) be a wide-sense stationary (WSS) random process with power spectral density $S_x(f)$. If Y(t) is the process defined as $Y(t) = X(2t-1)$, the power spectral density $S_Y(f)$ is

[Set-03]

(A) $S_Y(f) = \frac{1}{2}S_x\left(\frac{f}{2}\right)e^{-j\pi f}$

(B) $S_Y(f) = \frac{1}{2}S_x\left(\frac{f}{2}\right)e^{-j\pi f/2}$

(C) $S_Y(f) = \frac{1}{2}S_x\left(\frac{f}{2}\right)$

(D) $S_Y(f) = \frac{1}{2}S_x\left(\frac{f}{2}\right)e^{-j2\pi f}$

Solution: (C)

2. Let $X \in \{0, 1\}$ and $Y \in (0, 1)$ be two independent binary random variable. If

P(X=0) = p and P(Y=0)=q, then $P(X+Y \geq 1)$ is equal to

[Set-02]

(A) pq + (1 - p) (1 - q)

(B) pq

(C) p(1 - q)

(D) 1 – pq

Solution: (D)

3. The variance of the random variable X with probability density function $f(x)=\frac{1}{2}|x|e^{-|x|}$ is _____.

[Set-03]

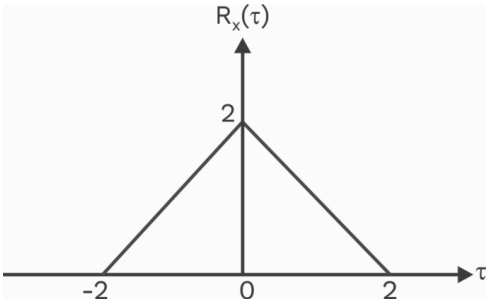
Solution: 6

4. Consider the random process $X(t)=U+Vt$, where U is zero mean Gaussian random variable and V is a random variable uniformly distributed between 0 and 2. Assume that U and V are statistically independent. The mean value of the random process at t = 2 is _____.

[Set-02]

Solution: 2

5. The autocorrelation function $R_x(\tau)$ of a wide-sense stationary random process X(t) is shown in the figure.



The average power of X(t) is _____.

Solution: 2

Solutions

1.

Given: X(t) is a wide-sense stationary process.

So, its autocorrelation function is given by,

$$R_{xx}(\tau)=E\left[X(t)X(t-\tau)\right]$$

From Wiener-Khinchin relationship, autocorrelation function and power spectral density are Fourier transform pair,

$$R_{xx}(\tau)\xrightarrow{F.T.}S_{xx}(f)$$

Autocorrelation function of random process Y(t) is given by,

$$R_{YY}(\tau)=E\left[Y(t)Y(t-\tau)\right]$$

$$R_{YY}(\tau)=E\left[X(2t-1)X(2t-1-2\tau)\right]$$

$$R_{YY}(\tau)=R_{XX}(2\tau)$$

From Wiener-Khinchin relationship,

$$S_{YY}(f)=FT\left[R_{XX}(2\tau)\right]$$

(From scaling property of Fourier transform)

$$S_{YY}(f)=\frac{1}{2}S_{XX}\left(\frac{f}{2}\right)$$

Hence, the correct option is (c).

2.

Given: P(X = 0) = p

So, P(X = 1) = 1 – p

P(Y = 0) = q and P(Y = 1) = 1 – q

Let, Z = X + Y

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	2

So,

$$P(Z\geq 1)=P\{X=0 \text{ and } Y=1\}+P\{X=1 \text{ and } Y=0\}+P\{X=1 \text{ and } Y=1\}$$

$$P(Z \geq 1) = 1 - P\{X = 0 \text{ and } Y = 0\}$$
$$= 1 - P[X = 0]P[Y = 0]$$

Since, X and Y are independent

$$P(Z \geq 1) = 1 - pq$$

Hence, the correct option is (d).

3. Solution:

Given: Probability density function,

$$f(x) = \frac{1}{2}|x|e^{-|x|}$$

Variance is given by,

$$\sigma_X^2 = E[X^2] - \{E[X]\}^2 \quad \text{...(i)}$$

Mean value is given by,

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^{\infty} x\frac{1}{2}|x|e^{-|x|}dx$$

Since, it is an odd function,

$$E[X] = 0$$

Mean square value is given by,

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)dx$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 \frac{1}{2}|x|e^{-|x|}dx$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 e^{-x}dx$$

$$E[X^2] = \left[x^3\left(\frac{e^{-x}}{-1}\right) - 3x^2\left(\frac{e^{-x}}{(-1)^2}\right) + 6x\left(\frac{e^{-x}}{(-1)^3}\right) - 6\left(\frac{e^{-x}}{(-1)^4}\right) \right]_{-\infty}^{\infty} = 6$$

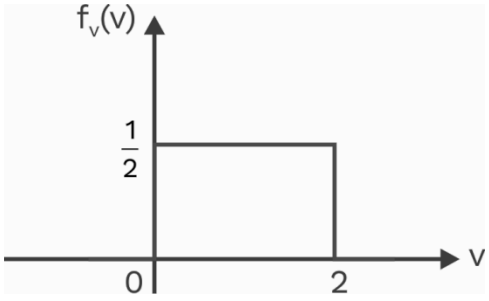
From equation (i),

$$\sigma_x^2 = 6 - (0)^2 = 6$$

Hence, the variance of the random variable X is 6.

4. Solution:

$$X(t) = U + Vt, E[U] = 0, V = UDF[0,2]$$



$$f_v(v) = \begin{cases} \frac{1}{2}, & 0 < v < 2 \\ 0, & Otherwise \end{cases}$$

Mean value of V is given by,

$$E[V] = \frac{a+b}{2} = \frac{0+2}{2} = 1$$

$$E[X(t)] = E[U] + E[Vt]$$

At t = 2 sec,

$$E[X(t=2)] = E[U] + E[2V]$$

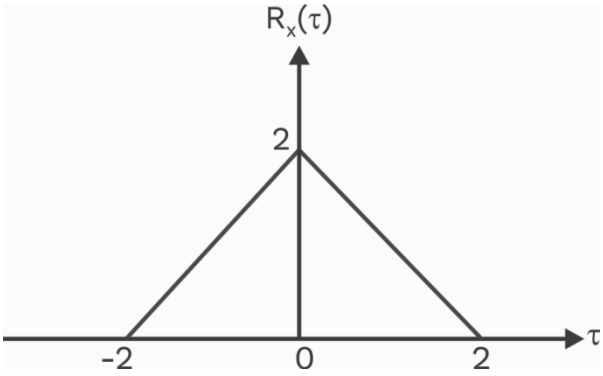
$$E[X(t=2)] = E[U] + 2E[V]$$

$$E[X(t=2)] = 0 + 2 \times 1 = 2$$

Hence, the mean value of the random process at t = 2 is 2.

5. Solution:

Given: Autocorrelation function $R_x(\tau)$ of wide sense stationary random process X(t) is shown in the figure.



Here. $R_x(0) = 2$

Average power of X(t) is given as mean square value of X(t), i.e.,

$$P_x = E\left[X^2(t)\right] = E\left[X(t).X(t)\right] \quad \text{...(i)}$$

Autocorrelation function of X(t) is given by,

$$R_x(\tau) = E\left[X(t).X(t+\tau)\right]$$

$$R_x(0) = E\left[X(t).X(t+0)\right]$$

$$R_x(0) = E\left[X(t).X(t)\right] \quad \text{...(ii)}$$

From equations (i) and (ii),

$$P_x = R_x(0) = 2W$$

Hence, the average power of X(t) is 2 W.